NOTE: DOD-STD-2003-4 has been redesignated as a standard practice. The cover page has been changed for Administrative reasons. There are no other changes to this Document.

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DOD-STD-2003-4(SH)
24 June 1987
SUPERSEDING
NAVSEA S9300-AW-EDG-010/EPISM
(INCLUDING NAVSEA DWG. NO.
803-5001027) AND NAVSEC NO.
9000-S6202-73980

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

ELECTRIC PLANT INSTALLATION
STANDARD METHODS FOR
SURFACE SHIPS AND SUBMARINES
(CABLEWAYS)

SECTION 4 OF 5 SECTIONS

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DOD-STD-2003-4(NAVY)
24 June 1987

SECTION 4

CABLEWAYS

DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Electric Plant Installation Standard Methods For Surface Ships and Submarines

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. The criteria contained herein for the installation of the electrical plant on ships of the United States Navy supersede the data contained in Sections 1 through 5 of NAVSHIPS Drawing 9000-S6202-73980, NAVSEA Drawing No. 803-5001027 and NAVSEA PUBLICATION S9300-AW-EDG-010/EPISM.

2. This standard disseminates up-to-date information detailing Requirements for Standard Installation Methods Employed for Submarine and Surface Ship Electrical Distribution Systems.

3. These criteria apply to work on a specific ship or ships only when invoked by the Ship Specifications or similar contractual documents.

4. Although these criteria are primarily for application to new construction, their use may be considered in the conversion or alteration of existing ships. In such cases the degree of applicability of these criteria will be specified by the activity preparing the instructions for the work.

5. Considering the magnitude of this standard, along with the changing requirements imposed on the Electric Plant, it is inevitable that changes will be required to up-date these criteria. Therefore, as comments arise they should be forwarded to Naval Sea Systems Command (NAVSEA) 55Z3 to keep this standard as current as possible through subsequent revisions. Revisions will be accomplished by the issuance of additional or revised figures to be inserted in the basic standard sections. Document Improvement Proposal Form DD 1426 attached. Superseded pages may be retained for reference if so desired.

6. This standard is available in a 8-1/2 X 11 hard copy, in microfilm aperture cards, or in microfiche. It is available in 8-1/2 X 11 hard copy from Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Microfilm aperture card or microfiche are available from Commanding Officer, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801. All revisions on microfilm aperture cards, or on microfiche are automatically distributed to a previously approved distribution list. (Tel: (207) 439-1000, Ext. 1718, Autovon 684-1718). Activities having a requirement to be placed on the distribution or for additional copies should forward these requests to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101. Aperture cards have been distributed to those activities presently on the distribution for NAVSEA Standard and Type Drawing microfilm aperture card sets. Microfiche has been distributed to all active ships.

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1. SCOPE

1.1 Purpose. The purpose of DOD-STD-2003-4(NAVY) is to disseminate up-to-date information for cableways on surface ships and submarines.

1.1.1 Application. These installation standards shall be used by all installing activities. These standards do not identify ship or type, but do establish minimum standards of acceptance for NAVSEA ships. It is the responsibility of the user activity to determine which standard satisfies their requirements. It does not authorize relaxation of any requirement specifically invoked by new construction, conversion, overhaul, or refurbishment contracts. In instances where deviated design requirements (for example, ship type, ship class, and so forth) conflict with the requirements of this standard, the requirements of this standard shall govern. Any deviation for electric plant installation identified in this standard shall be submitted to NAVSEA 56Z2 for resolution.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915	- Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MIL-I-3064	- Insulation, Electrical, Plastic-Sealer.
MIL-T-16366	- Terminals, Electrical Lug and Conductor Splices, Crimp Style.
MIL-S-24149	- Studs, Arc Welding, and Arc Shields (Ferrules), General Specification for.
MIL-C-24640	- Cable, Electrical, Lightweight For Shipboard Use, General Specification for.
MIL-C-24643	- Cable and Cord, Electrical, Low Smoke, For Shipboard Use, General Specification for.

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278	- Welding and Casting Standard.

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2.1.2 Other Government publications and documents. The following other Government publications and documents form a part of this standard to the extent specified herein.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
S9407-AB-HBK-010 - Electromagnetic Shielding Practices, Shipboard.
0978-LP-003-9000 - Safety and Damage Control Facilities Liquid.
0900-LP-016-6081 - Quality Assurance Standards for Submarine Antenna and Mast Assemblies.

DOCUMENTS

General Specifications for Ships, Section 075.
DDS 304-2 - Electric Cable Ratings and Characteristics.
DDS 305-1 - Designation and Marking of Electric System.

(Copies of specifications, standards, publications, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS

3.1 Cableway tier. Cableway tier is one row of cable installed in a cableway.

3.2 Double banking. Double banking is two or more rows of cable installed in a single tier of a cableway.

4. GENERAL REQUIREMENTS

4.1 Location of cable runs. Main fore-and-aft cableways shall be located port and starboard, high and low in ship; lower cableways shall be through machinery spaces and corresponding platform decks, upper cableways under the main deck. This is to provide a quadrangular pattern to allow maximum athwartship and vertical separation of cables to loads requiring two sources of power for systems requiring alternate reliability circuits. For example, a ship service power feeder in the lower starboard cableway should have its complementary emergency or alternate feeder in the upper port cableway.

4.1.1 Vertical cable runs. Vertical cable runs shall be organized on the basis of the fore-and-aft or athwartship separation.

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4.1.2 Emergency diesel generator seawater booster pumps. Cables supplying emergency diesel generator seawater booster pumps shall be run in the same compartment with, and as close as practicable to, but shall not be attached to the water piping connecting the pump to the diesel engine.

4.1.3 Interior communications and weapons control. The main runs of interior communications and weapons control system cables shall follow cable runs of lighting and power installations. Interior communication and weapons control cables for circuits having several interrelated main and interrelated auxiliary circuits shall be run so that main circuits including normal power supply will be in one group and auxiliary circuits including alternate power supply will be in another group.

4.1.4 Control from more than one location. Where equipment is controlled from more than one location, the control cabling from each location to the equipment shall be installed in separate wireways.

4.1.5 Weather exposure. Cables shall not be exposed to the weather except where they penetrate a deck or bulkhead to supply equipment. Cables exposed to the weather shall be kept to a minimum. Where possible, cables to equipment on masts, staffs, macks, and yardarms shall be installed within masts, staffs, macks, and yardarms.

4.1.6 Circuit trunking. Location and arrangement of cable runs shall provide circuit trunking avoiding physical interferences with piping and ventilation ducts. Spaces considered harmful to cable or subject to explosion hazards shall be avoided. Cableways shall be arranged so that ventilation will be adequate for maintaining the minimum practicable temperature of cables within the cableway. Cableways shall be located so the cables will not be disturbed by disassembly or removal of machinery.

4.1.7 Armored cable. Armored cable is required to be used on all nuclear ships for propulsion plant and reactor compartments and is desirable in all other areas unless technically prohibited. The use of armored cable on non-nuclear ships is optional and to be determined by the building shipyard, except armored cables shall not be installed in weather locations due to EMC considerations.

4.1.8 Deadended cable. Cables installed through error or rendered useless as a result of modification shall be removed.

4.1.9 Gun mounts. Cables to gun mounts and directors shall be routed from the deck below through the center column in accordance with NAVSEA standard methods. Watertight integrity shall be maintained. Cable shall be located and installed to avoid the possibility of chafing. Non-flexing cables shall terminate in submersible or watertight changeover connection boxes at the point of connection to the flexible cables. Length of flexible cables between changeover connection boxes and the slack cable enclosure shall be kept to a minimum. The slack cable enclosure shall be entered through stuffing tubes. In compartments containing hydraulic systems, the cable installation shall not impair the airtightness or watertightness of decks and bulkheads forming the boundaries of the compartment.

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4.1.10 Rotating missile launchers. Cable connections to rotating missile launchers shall comply with drawings furnished by NAVORD.

4.1.11 Ballistic structures. Attachment of cables and supports to ballistic structures shall be in accordance with NAVSEA standard drawings and the following:

(a) First preference shall be given to routing cables on the inboard or after bulkheads in the forward half of the ship, and on the inboard or forward bulkheads in the after half of the ship.
(b) Second preference shall be given to routing cables on channels, or in cable racks on angles, attached to overhead deck beams.

4.1.12 Bulkheads and overheads. Cables shall not be routed on the outboard or forward bulkheads in the forward half of the ship, and on the outboard or after bulkheads in the after half of the ship, when the plating is 50 pounds per square foot (lb/ft^2) or heavier. This routing shall also be avoided for lighter plating unless no other route is feasible. The installation of cables on the overhead and on bulkheads shall comply with NAVSEA standard methods. Cables for vital systems such as interior communication and weapons control systems shall not be secured to the overhead, or to shell planking, to shell plating or to ballistic bulkheads.

4.1.13 Magnetic minesweeping generators. Cable runs between magnetic minesweeping generators and minesweeping cable terminal boxes shall consist of four single-conductor cables arranged to form a quad. The cable centers of the quad shall form a square with alternate positive (+) and negative (-) polarity. The quad shall be twisted, having a right hand lay of 360 degrees each 10 feet. The outer jacket of adjacent cables in each quad shall be in contact with each other throughout the run. In the event that cables forming the quad pass through bulkhead stuffing tubes, separation of the outer jacket of adjacent cables shall not exceed 3 inches at the bulkhead and 1/2 inch at a distance of 1-1/2 feet from the bulkhead. The same maximum spacing limits shall apply at termination. Where space permits, a single community type stuffing tube shall be used for the cables of a quad when penetrating a watertight bulkhead. Details of the arrangement shall be in accordance with the standard plan for community type stuffing tubes on wood hull minesweepers.

4.1.14 Sonar system cables. Sonar system cables and other low frequency system cables shall be installed in accordance with the requirements of NAVSEA S9407-AB-HBK-010.

4.1.15 Hypergolic missile stowage areas. Installation and enclosure requirements for electric equipment located in hypergolic missile stowage area shall be in accordance with NAVSEA 0978-LP-003-9000.

4.2 Spare cable space allowance. In the organization of principal cableways, spare cable space of approximately 20 percent of that to be occupied by the final cable installation (as known at time of delivery of the ship) shall be reserved on cableways and in cable penetration areas for future cable installations. The additional cable space may consist of unused hangers or a combination of unused hangers and space available on used hangers, assuming

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that for future addition of cable, double banking will be allowed. During the design phase, the contractor shall provide cableway space in excess of the spare 20 percent in order to accommodate cables added as a result of developments occurring during the construction period. Through horizontal cable runs in aircraft carriers' hanger spaces will not be permitted. Through vertical runs such as those from the second deck to the gallery or flight deck levels shall be grouped to reduce the number of protective casings required and shall be protected from fire by NAVSEA standard methods.

5. DETAILED REQUIREMENTS

5.1 Cable slack. Cable slack requirements are as follows:

(a) Cables shall be installed so that sufficient slack exists to allow for deflection of bulkheads.
(b) Sag between hangers shall be uniform for each row of cables so that clearance between rows will be the same throughout the cable run.
(c) Where cables spread out to enter bulkhead stuffing tubes, bends shall have liberal sweep to provide as much flexibility as practicable.
(d) Cables having only a minimum spread where they pass through bulkhead stuffing tubes shall have enough slack to give them the same flexibility as other cables in the group.
(e) Cables from switchboards or other electric equipment shall enter cableways in a curve of sufficient radius to prevent transmission of stresses to the equipment during severe cableway deflection.
(f) Cables crossing expansion joints, such as those under the flight deck and gallery deck of aircraft carriers, shall have slack allowance at such points at least equal to the maximum movement of the expansion joints.

5.2 Cable bend radius. Cable bend radius shall be in accordance with NAVSEA'S Cable Comparison Handbook. The following general requirements are also applicable:

(a) Bends at terminal entrances, where stuffing tubes are required, shall be made through angle stuffing tubes.
(b) Straight stuffing tubes may be used in place of angle stuffing tubes if the bend in the cable is not less than the minimum bend radius.
(c) Measurement point for minimum radius of bend is that surface of the cable jacket which is on the innermost portion of the cable bend.

5.3 Protection of cable.

5.3.1 Battle damage. Protection afforded by the ship structure shall be used to the greatest extent practicable. Cables shall not be run on the exterior of deck houses or similar structures above the main deck (including the island structure of aircraft carriers) except where necessary because of the location of the equipment served.

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5.3.2 Mechanical damage. Cables subject to mechanical damage because of their proximity to areas frequented by personnel shall be protected by metal casings. Cableways in areas where their misuse as steps or hand holds would cause damage shall be protected. Protective plates shall be installed over the cableways in all passages where cables might be stepped on. At hatch openings and in trunks where objects are raised or lowered, cables shall be protected by steel casings.

5.3.2.1 Auxiliary ship installation. Auxiliary (cargo) ship cable installations shall be routed outside cargo spaces wherever practicable. Where routing through cargo spaces is unavoidable, cables shall be protected from mechanical damage, including damage due to shifting of cargo.

5.3.2.2 Riser boxes. Riser boxes, or multiple cable penetrators with mechanical protection, shall be installed for three or more cables passing through a deck in a group. Where fewer than three cables pass through a deck, kickpipes shall be installed. Kickpipes installed in the vicinity of hydraulic equipment, where the possibility of oil leakage exists, shall be welded in place. Details of kickpipe installations and the installation of top bracing, where required, shall conform to this standard.

5.3.2.3 Oiler weather deck. Cableways installed on the weather deck of oilers shall be protected by a substantial open bottom steel enclosure. Cableways installed alongside or under weather deck catwalks shall be completely enclosed in a steel enclosure, the bottom section of which shall consist of a removable perforated steel plate. Enclosures shall be constructed so as to permit periodic inspection and maintenance of cables and hangers.

5.3.3 Excessive heat. Installation of cables in locations subject to excessive heat shall be avoided. Where required, heat insulating barriers or heat resistant types of cables shall be installed. Cable shall not be installed adjacent to machinery, piping or other surfaces having an exposed surface temperature greater than 70 degrees Celsius (°C). Cables shall not be run over boilers in the upper portions of firerooms or in other locations where they may be surrounded by hot air. Cables shall not be run in locations where they will be exposed to hot stack gases unless they are of heat resistant types protected by methods as shown in this standard.

5.3.4 Excessive moisture. Cables shall not be installed in locations where they may be subjected to excessive moisture. Where cable routing near firemains, water, steam, oil or other piping is unavoidable, dripproof shields shall be provided for protection. Where cables must be installed in spaces subject to flooding, they shall be installed as high as practicable within the space. In such spaces, cable installation shall be arranged so as to permit the cables to be painted.

5.3.5 Cables installed through submerged spaces and voids. Cables that would normally be submerged, except for cables in the sonar dome, shall be enclosed in a single pipe in a manner similar to that for cables passing through tanks. Cable runs in voids and other dead air spaces shall be avoided. If it is not practicable to avoid cable runs in such spaces, cables shall be installed

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only in those spaces which are not provided with a means for flooding, and then only when temperature calculations show that the spaces have sufficient radiation area to maintain an ambient temperature at or below the temperature in which the cable was selected to operate. Cables shall be supported clear of decks and bulkheads to avoid condensate which might form on such surfaces.

5.3.6 Cable installations in hazardous locations. Cable runs shall be avoided through hazardous spaces such as the following (except such cable runs that terminate in or cannot be routed around these spaces):

(a) Compartments where hazardous materials are stowed, such as explosive signalling apparatus, ready service ammunition, chemical warfare materials, photographic materials, film, paint, compressed oxygen, chlorine and combustible gases.
(b) Diesel, JP-5, and fuel tanks and stowage compartments (stowage in separate tanks or containers).
(c) Gasoline hazard areas.
(d) Hangar spaces.
(e) LOX (liquid oxygen) handling and transfer spaces.
(f) Magazines and other spaces where exposed powder is handled or stored.
(g) Missile magazines.
(h) Warhead, depth charge, mine charge, aerial bomb and cased explosives magazines.

Cables within hazardous spaces shall be protected against mechanical damage. This protection shall be in the form of nontight metal guards (expanded or solid) of sufficient strength to provide the required protection. Clearance shall be provided between the protective guards and the cables to provide ample ventilation. Protective guards shall be provided with drainage holes where required. Cables protected by the ship structure or by permanently installed equipment are considered adequately protected. Only cables having conductor insulation of silicone rubber (except for interior communication and weapons control cables totally within the compartment) shall be used. Through cables shall be of unbroken length within compartments. Cables terminating at lighting fixtures or other equipment shall be of broken lengths. Separate cables to each lighting fixture are not required; a single cable may be run between fixtures where more than one fixture is installed in a space.

5.3.7 Channel rubber. Channel rubber shall be used with banding straps in cableways to protect the outer jacket of cables from being cut and coaxial cables from being deformed by the banding strap. For cableways that have unarmored cable in contact with banding straps, channel rubber is required only in the following instances:

(a) For each banding strap of a vertical cableway.
(b) For each banding strap at a cableway bend, including breakout bends.
(c) For rubber jacket and coaxial cables in contact with the banding strap. These cables, however, should be run in the middle of the cableway for increased mechanical protection.
(d) For submarine external cableways.

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Where hose clamp cable retention devices are used, channel rubber is not required, except when coaxial cable would otherwise be in contact with the hose clamp cable retention device. For cableways that have only armored cables in contact with the banding strap, channel rubber is not required.

5.4 Installation of propulsion system cables. Propulsion system cables shall be installed in cableways separate from those used for other electrical system cables. Direct current or alternating current propulsion system cables shall be arranged and installed in such manner as to minimize heating of adjacent structural material by induced currents. Alternating current propulsion cables installed on insulating hangers shall be grounded at approximately the midpoint of the cable run. Propulsion system cables shall not be installed in bilges. When cables are located near dripping water, or oil, they shall be protected by drip covers. These protective covers shall be spaced away from the cables so as not to restrict ventilation. Cables entering propulsion system equipment from above or from the sides shall enter through stuffing tubes and shall be braced and secured to prevent dislodgment under vibration, shock and magnetic stresses. The ends of propulsion system cables 9,000 circular mils and larger shall be sealed with solderless waterseal type lug terminals in accordance with MIL-T-16366.

5.5 Cable handling and stowage. All cables except flexing service and silicone rubber insulated type of MIL-C-915, MIL-C-24643 and MIL-C-24640 shall be given the following special handling at temperatures below 35 degrees Fahrenheit (°F):

(a) If the compartment in which the cable is to be installed cannot be heated, the cable shall first be stored in an ambient temperature of at least 50°F but not above 120°F until it is warm enough to be completely installed before it cools to 35°F.

(b) If cable must be installed when its temperature is 35°F or lower, extra care will be required. The radius of bends shall be not shorter than the minimum values in DDS 304-2. Before bending to the final radius, that portion of the cable comprising the bend shall be warmed thoroughly by a portable warm air blower.

(c) Cable shall be stored in a dry place which is not subject to accidental flooding, protected from the weather, and subjected to a minimum variation of temperature.

(d) Portable cables for aircraft servicing shall be stowed in bins. In hangar bay areas, they shall be stowed on cable racks.

(e) Cable that has been in storage for prolonged periods may be installed provided a visual inspection shows that it has sustained no mechanical damage that would impair the watertight integrity of its outer sheath.

(f) Cable armor showing signs of corrosion shall be cleaned and a coat of zinc-chromate primer applied to all corroded areas.

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5.6 Cable tags. All permanently installed cables shall be tagged to each point of connection, and on both sides of decks and bulkheads except as follows:

(a) Where through cable runs within a compartment are direct (such as a vertical run between decks), a single tag will suffice.
(b) For cables with both points of connection within a compartment and which can be readily traced, a single tag will suffice.
(c) Where compartments are subdivided by internal bulkheads or where machinery or installed equipment makes tracking of cable runs difficult, additional tags shall be provided.
(d) For multiple cable penetrations of decks and bulkheads (main cableways), individual cable tags can be omitted, and in lieu thereof, an identification plate shall be installed adjacent to the cableway penetration area showing each cable designation in the order of location in the penetration area.

Cable tags shall be in accordance with DDS 305-1. Capital letters shall be used on cable tags; height of all letters and numbers shall be not less than 3/16 inch, and letters and numbers shall be embossed to at least 1/64 inch above the surface.

5.7 Cable hangers and supports. Cable hangers and supports shall meet the following requirements.

(a) MIL-S-901 shock.
(b) MIL-STD-167-1 vibration.
(c) Minimum weight.
(d) Minimum maintenance on cable supporting systems.
(e) Facilitate cabling procedures.
(f) Convenience of servicing system.
(g) Fireproof except in specific areas such as wooden ships.
(h) Material used is non-toxic under all conditions.
(i) Free of sharp protrusions, corners, items, and so forth, dangerous to personnel.
(j) Shall be designed to permit repair with local market items. It is not intended that extensive back-up stocks be supplied for item by item replacement.
(k) Compatible with and non-damaging to cables.

5.7.1 Insulation spacer block. The use of insulation spacer blocks is mandatory for propulsion system cables in order to prevent bunching.

5.7.2 Shock design. All cable supporting arrangements shall be tested for resistance to shock and vibration. Deformation to supporting members and retaining members is satisfactory providing cables are prevented from becoming whips or missiles and retainers, racks and supports do not become missiles. Shock and vibration tests shall be conducted at an approved laboratory. No mechanical damage shall occur to cable specimens, and insulation resistance shall comply with cable specifications before and after the testing. Copies of the shock test, with details of any damage and clear photographs of the damage to cable supports or cables, certified by the Government inspector witnessing the test shall be warded to NAVSEA, Code 56Z24. Copies shall also be furnished to purchasing activities upon request. These tests shall be conducted by each manufacturer

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for each type hanger supplied. Retesting shall be required when design, manufacturing process, or material is changed. Testing of a hanger with lesser tiers or lesser dimensions and spacing shall not be extended or exempt from testing for other hangers. The cable supports when tested shall be loaded with maximum design cable load. Material is not satisfactory unless it has successfully passed these tests. Shock tests shall be in accordance with MIL-S-901, class I. Cable supporting devices shall be arranged for testing as follows:

5.7.2.1 Multiple cable test racks (15 inches free length of cable support). If multiple cable racks are being tested for ability of cable support member only to withstand shock and vibration as fully loaded, a minimum of three supporting stations with 32-inch, 21-inch or 16-inch spacing shall be provided. The maximum number of the largest cables that can be accommodated on these racks shall be applied. Cable overhang at each end shall be 16 inches or one-half the distance between supports. If tiers of multiple cable racks and supporting devices for attachments to ships' structure are to be tested for capability to withstand shock and vibration, a duplicate of the assembly proposed for shipboard installation with a minimum of three supporting stations with 32-inch, 21-inch or 16-inch spacing shall be provided. The maximum number of the largest cables that can be accommodated on these racks shall be applied. Cable overhang at each end shall be 16 inches.

5.7.2.2 Single cable supports. All single cable supports shall be tested. These shall be arranged using a minimum of three supporting stations spaced 32 inches, 21 inches or 16 inches apart. Cable overhang shall be 16 inches. Maximum size cables shall be mounted with these single cable supports. Two, three or four cables supported as shown on figure 4C5, or by use of retainer bars or straps, shall be arranged for tests in a manner similar to that for single cables. A single tier of cables shall be arranged as described for multiple cable racks.

5.7.3 Installation.

5.7.3.1 Spacing. Spacing of steel and aluminum cable supports shall be 32 inches except as follows:

(a) The spacing of hangers shall be 21 inches to reduce cable sag in cable racks when headroom clearance under the cable racks before installation of cables is 6 feet 7 inches or less (6 feet 4 inches minimum).
(b) Hangers for overhead multiple tier aluminum cableways, when welded to aluminum decks, shall be spaced 16 inches and the number of tiers shall be limited to three.
(c) The spacing of hanger supports for single cables or single tier of cables shall be 32 inches but may be adjusted between 32 inches and 16 inches as necessary to suit particular ship conditions.

5.7.3.2 Cable racks. A maximum of five single tiers of cables may be supported from an overhead in one cable rack. Aluminum tiers shall be limited to three. A maximum of two tiers of cables may be supported from bulkheads in

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one cable rack. Eight tiers of double banked cables may be supported in main cableways in machinery spaces and boiler rooms, when approved by NAVSEA. Backing plates are not required for cable support except that when bulkheads are 1/16-inch thick or less and cable run is over 4 inch width, backing washers shall be used. Multiple tiers of cable shall not exceed two on bulkheads except in congested locations, in which case 3 tiers may be used on request and subject to NAVSEA approval. Cable support angles shall be secured to ship structure by 180 degree weld except that in the following cases, the weld shall be 360 degrees:

(a) Where the number of tiers of 15-inch cable rows in wireway exceeds two.
(b) When welding to ballistic structure.
(c) When the welds are in wet spaces such as weather decks or shower rooms.

5.7.3.2.1 Support components. Cable support components shall be of the following materials:

Nonmagnetic vessels

Inboard	Outboard
Hangers - aluminum	Hangers - aluminum
Supports - aluminum	Supports - aluminum
Ferrules - CRES or aluminum	Ferrules - CRES
Bolts and nuts - CRES or aluminum	Bolts and nuts - CRES
Banding - aluminum	Banding - CRES

Other vessels

Inboard	Outboard
Hangers - aluminum or steel 1/	Hangers - aluminum or steel 1/
Supports - aluminum or steel	Supports - aluminum or steel
Bolts and nuts - steel-zinc plated	Bolts and nuts - CRES
Banding - steel-zinc plated	Banding - CRES

1/ If aluminum hangers or supports are being utilized, a steel hanger or support or both shall be installed in lieu of an aluminum hanger or support at least every 6 feet.

Appropriate method details specified as submarine only may be utilized for surface ship applications, particularly for weather deck, wet/humid areas, and hull applications. Aluminum cable supports may be attached to steel deck and bulkhead where suitable design and interface is approved. Cable hangers shall be bracketed away from deck and bulkhead surfaces in food handling and food storage spaces to permit spraying for insect control.

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and steel decks and bulkheads (submarine ships only).

5.7.3.3 Double banking of cable. Not more than one row of cables shall be installed on a cable hanger. Where space is limited, and if specific approval has been obtained, double banking of cables on hangers will be permitted provided the clearance between the top of the upper layer of cables and the hanger above shall be at least 3/8 inch.

5.7.3.4 Cable retention. Devices for cable retention meeting the requirements of 5.7 and 5.7.2 are required on horizontal cable supports in multiple racks. Retention of cables on supports can be accomplished by the use of retainers such as contour straps, soft iron flat bars bent over the cables, semi-contour straps or angle iron retainers. Non-toxic strips or channel material shall be used with semi-contour straps, bars and angle retainers to reduce cable damage, distortion, and chafing. On cable runs in which the cable is vertical, cable retention is required at every hanger.

(a) Cable retention spacing shall be as specified in 5.7.3.1.
(b) Cable straps shall be omitted on horizontal cable runs except as follows:

(1) Where the hanger has no side brackets.
(2) At those locations where the cable runs change direction or pass through beams or bulkheads.
(3) Where four consecutive hangers would not require straps. In this case, a minimum of one strap shall be installed on every fourth hanger.

5.7.3.5 Plastic sealer. After the cables are properly secured, plastic sealer electrical insulation, MIL-I-3064, type HF, shall be used to seal the space around the cable as follows:

(a) In cable clamps and bushings entering the top of an electrical enclosure.
(b) In bushings, collars or nipples used for passing cables through lightight and fumetight bulkheads.

5.7.3.6 Welding. Welding shall be in accordance with MIL-STD-278.

5.7.3.7 Threaded fastener. Threaded fasteners shall be in accordance with the General Specifications for Ships, Sections 075; for submarine antenna and mast material, see NAVSHIPS 0900-LP-016-6081. Unless otherwise specified, threaded fasteners of normally stocked lengths shall be used. Male threads on threaded fasteners, after being installed and tightened, shall protrude at least one thread beyond the top of the nut or plastic locking element. Excessive protrusion shall be avoided, particularly where necessary clearances, accessibility, and safety are important. Where practicable, the number of threads protruding shall not exceed five; however, in no case shall thread protrusion exceed ten threads. Washers shall not be used for the sole purpose of lessening thread protrusion.

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5.7.3.8 Collar studs. Collar studs shall be in accordance with MIL-S-24149.

5.7.4 Deadended cable. Cables installed through error or rendered useless as a result of modifications shall be removed. Vacated and unused penetrations shall be sealed by methods which satisfy the tightness requirements of the structure penetrated.

6. NOTES

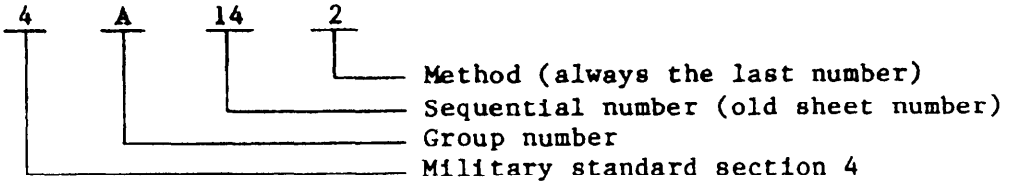
6.1 Intended use. This section specifies the requirements for Submarine and Surface Ship Cableways and Protection Methods to be employed both on surface ships and submarines. Standard Methods identified for electric plant installation are intended for new construction only.

6.2 Designation of Electric Plant Installation Standard Methods figures. The Electric Plant Installation Standard Methods Military Standard (DOD-STD-2003-4) contains drawings that depict Standard Methods that are applicable for general electric plant installation on both surface ships and submarines. Each drawing has been assigned a figure number. The methods shown on the figures are grouped together providing similar functions. These groups are:

DOD-STD-2003-4 (Cableways) Group A. Cableways (Submarines)
B. Cableways (Surface Ships)
C. Cableways (General)
D. Cable Protection

The methods shown on the figures are identified by the following alphanumeric designation system:

METHOD 4A142



Thus, method 4A142 identifies method 2, sequential number 14 in group A of DOD-STD-2003-4.

6.3 Subject term (key word) listing.

Cableways (submarines)
Cableways (surface ships)
Cableways (general)
Cable protection

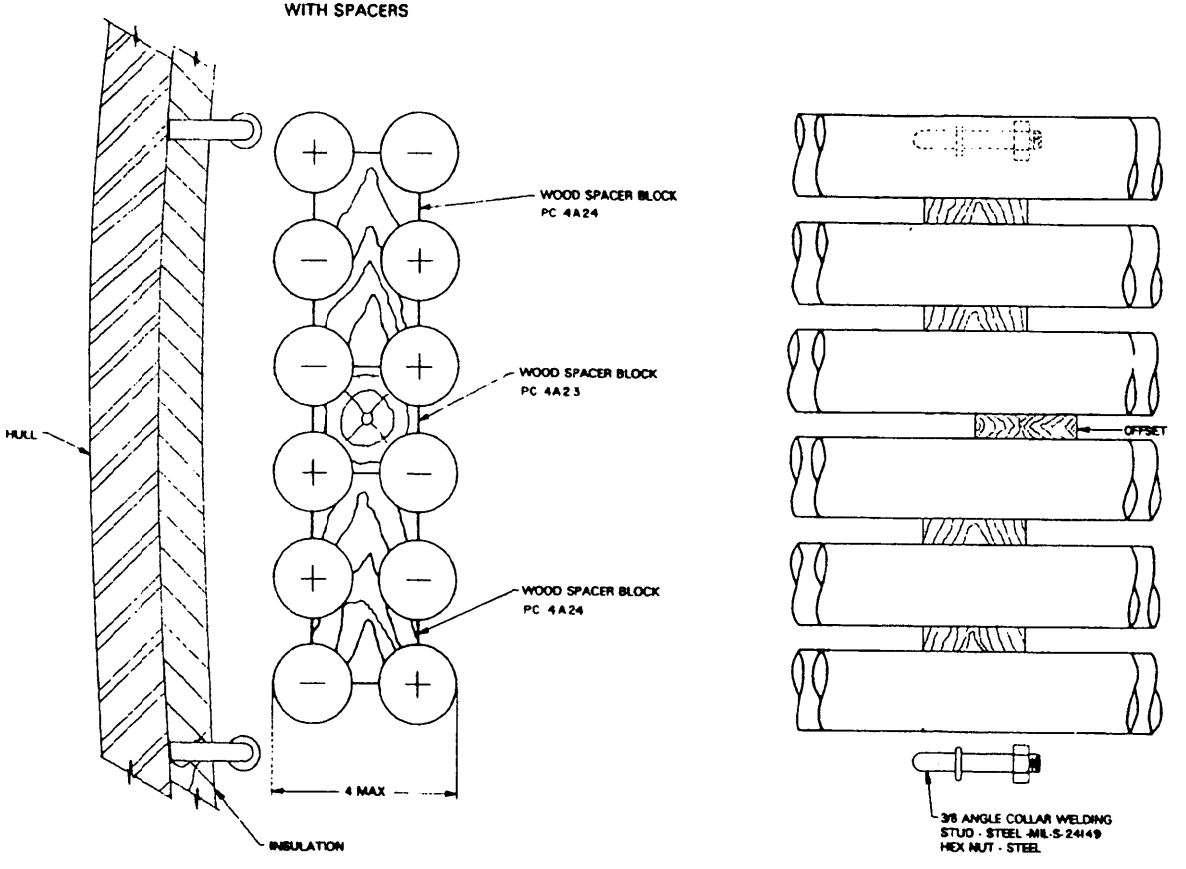
Preparing activity:
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(Project GDRQ-N066-4)

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NOTES:

1. HANGERS WITH 16" CENTER SPACING IS THE MAXIMUM SIZE THAT CAN WITHSTAND HI SHOCKS WHERE THE NUMBER OF CABLES EXCEED THE MAXIMUM SIZE OF SUPPORT SHOWN. ADDITIONAL WIREWAYS SHALL BE PROVIDED.
2. METHODS SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATION WHERE APPLICABLE.
3. THIS FIGURE SUPERSEDES SHEET 4A1 OF DWG 803-5001027 AND SECTION 5, SHEET 35, OF DRAWING, NAVSEC No. 9000-S6202-73980.

4 A 11
BANK OF 12 CABLES
ARRANGED IN DOUBLE VERTICAL TIERS
WITH SPACERS



SH 132317204

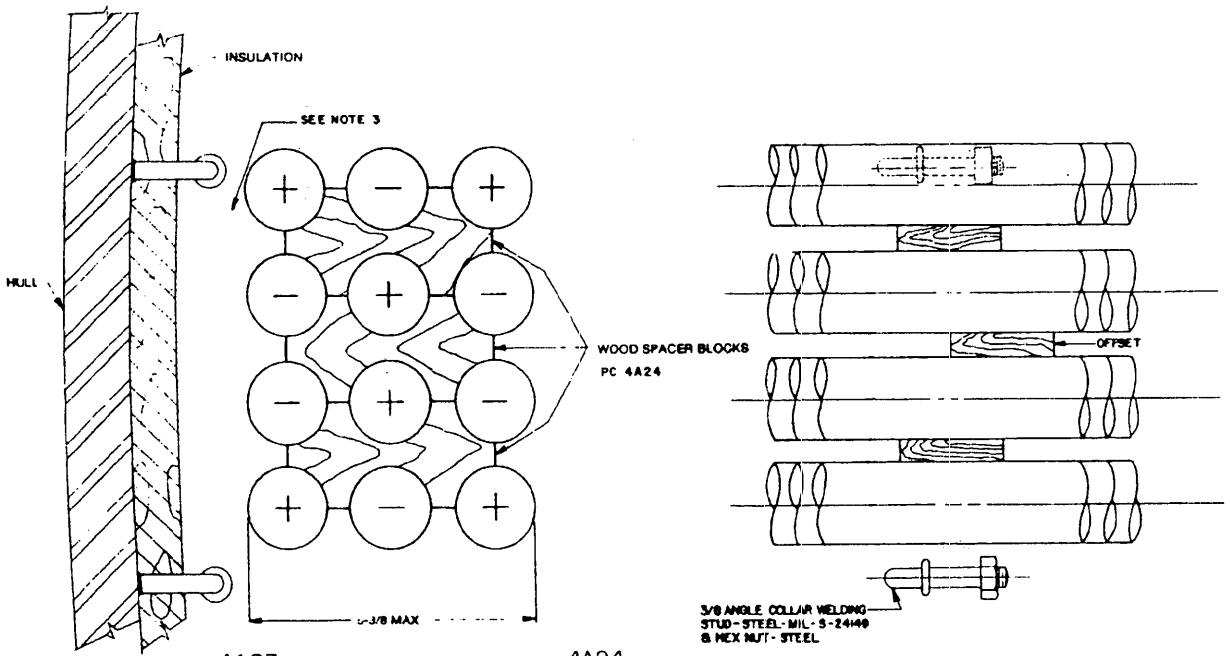
FIGURE 4A1. Main cableway for battery cables (submarines).

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NOTES:

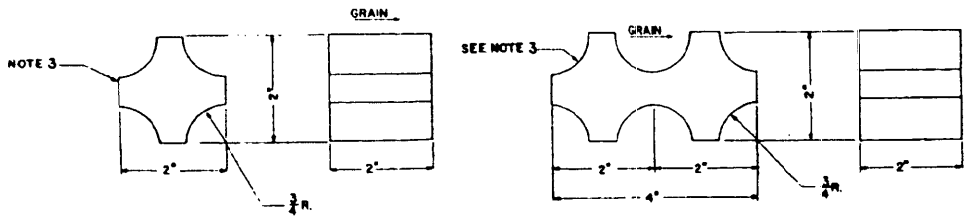
1. THE METHOD SHOWN ON THIS FIGURE MAY BE USED FOR GENERAL INSTALLATION WHERE APPLICABLE.
2. HANGERS WITH 16" CENTER TO CENTER SPACING IS THE MAXIMUM THAT CAN WITHSTAND HI SHOCKS WHERE THE NUMBER OF CABLES EXCEEDS THE MAXIMUM SIZE OF SUPPORT SHOWN, ADDITIONAL WIREWAYS SHALL BE PROVIDED
3. MATERIAL - MAPLE FINISHED WITH TWO COATS OF VARNISH. SPEC. TT-V-121
4. THIS FIGURE SUPERSEDES SHEET 4A2 OF DWG. 803-5001027 AND SECTION 1 SHEET 54, AND SECTION 6, SHEET 36 OF DRAWING, NAVSEC NO.9000-S6202-73980

4A21
BANK OF 12 CABLES
ARRANGED IN TRIPLE VERTICAL TIERS
WITH SPACER BLOCKS



4A23
WOOD SPACER BLOCK
FOR FOUR CABLES

4A24
WOOD SPACER BLOCKS
FOR SIX CABLES



SH 132317205

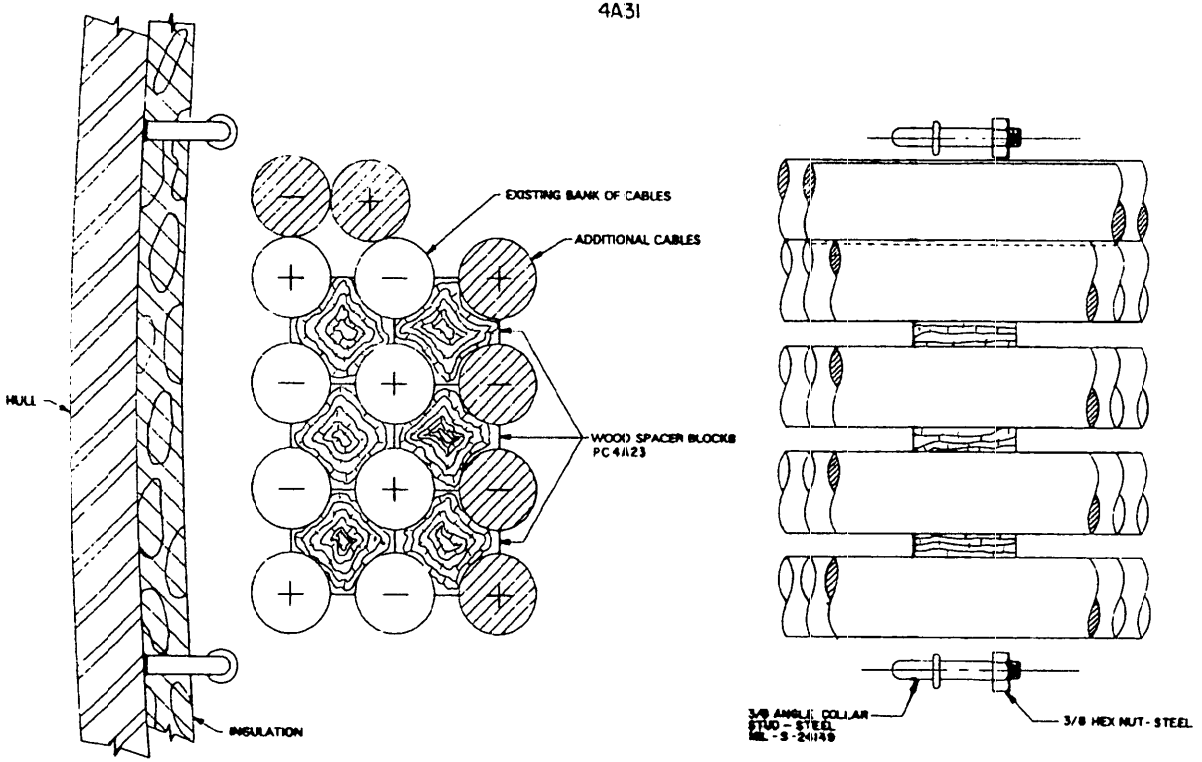
FIGURE 4A2. Main cableway for battery cables (submarines).

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NOTES:

1. HANGERS WITH 16" CENTER TO CENTER SPACING IS THE MAXIMUM SIZE THAT CAN WITHSTAND HI SHOCK WHERE THE NUMBER OF CABLES EXCEED THE MAXIMUM SIZE OF SUPPORT, ADDITIONAL WIREWAYS SHALL BE PROVIDED.
2. PROCEDURES SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATION
3. THIS FIGURE SUPERSEDES SHEET 4A3 OF DWG 803-5001027 AND SECTION 5, SHEET 37, OF DRAWING, NAVSEC NO. 9000-S6202-73980

4A31



SH 132317206

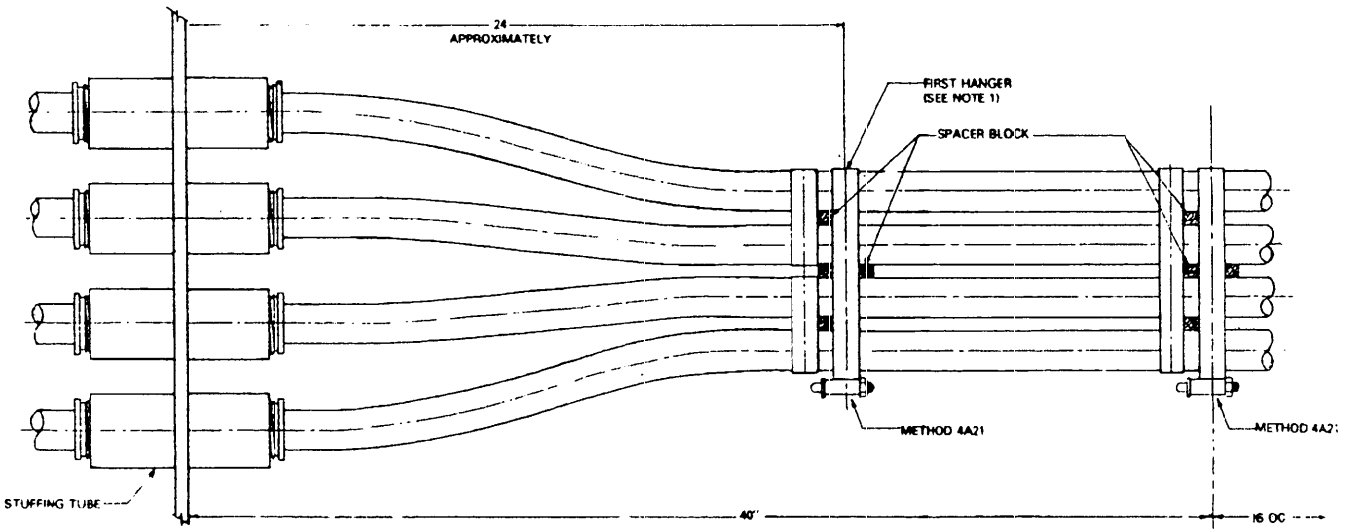
Figure 4A3. Addition of cable to existing battery cableway (submarines).

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NOTES:

1. THE FIRST HANGER NEXT TO THE BULKHEAD SHALL BE LOCATED TO SUIT CABLE BENDS AT PENETRATION BUT SHALL NOT EXCEED 32".
2. THIS FIGURE SUPERSEDES SHEET 4A4 OF DRAWING 803-5001027 AND SECTION 5, SHEET 40, OF DRAWING NAVSEC NO. 9000-98202-73980.

4A4(1



SH 132-17207

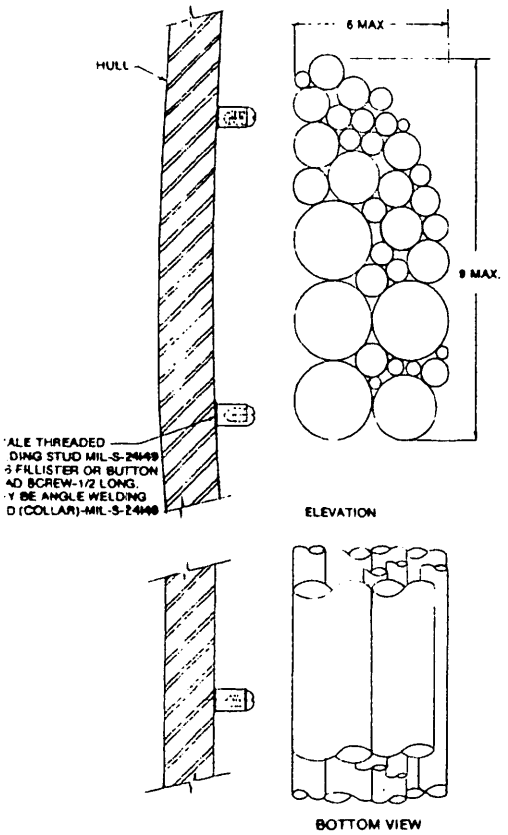
FIGURE 4A4. Cableway adjacent to bulkhead penetration (submarines).

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NOTES:

1. HANGERS WITH A 16" CENTER TO CENTER SPACING IS THE MAXIMUM SIZE THAT CAN WITHSTAND HI SHOCK WHERE THE NUMBER OF CABLES EXCEEDS THE MAXIMUM SIZE OF SUPPORT SHOWN. ADDITIONAL WIREWAYS SHALL BE PROVIDED.
2. THIS FIGURE SUPERSEDES SHEET 4A5 OF DRAWING 803-500 1027 AND SECTION 5, SHEET 38, OF DRAWING, NAVSEC NO. 9000-S6202-73960.

4A51
CABLE BANKS
SUPPORTED BY WELDING STUDS ON HULL
WITHOUT INSULATION



SH 132317208

4A52
CABLE BANKS
SUPPORTED BY WELDING STUDS ON HULL
WITH INSULATION

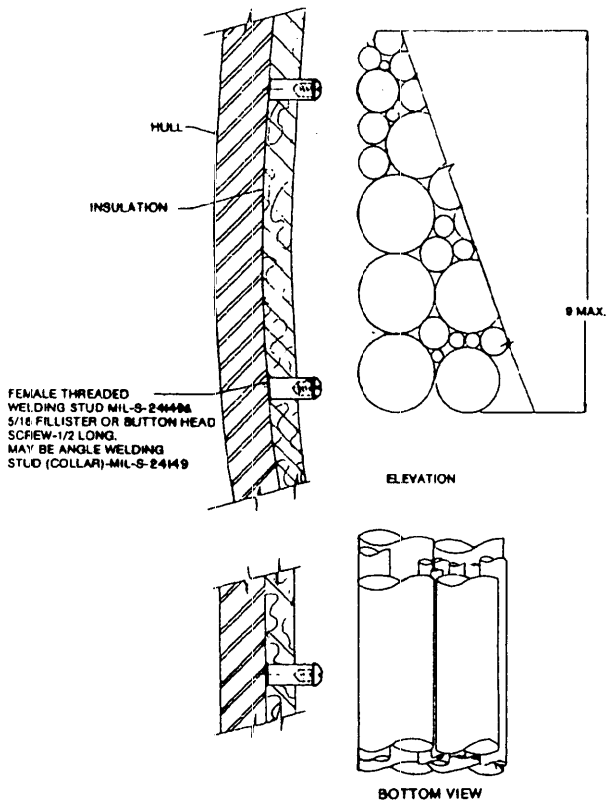
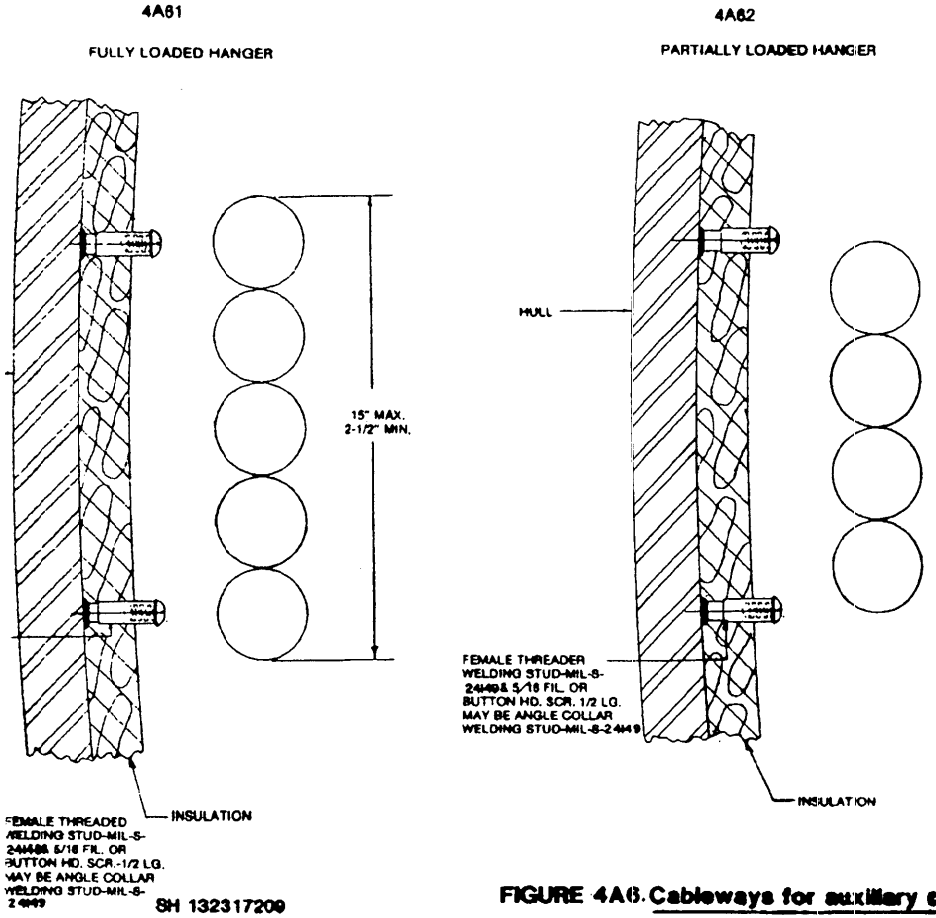


FIGURE 4A5. Cableway for auxiliary cables (submarines).

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NOTES:

1. METHODS SHOWN ON THIS FIGURE MAY BE USED FOR GENERAL INSTALLATION WHERE APPLICABLE.
2. CABLES SHALL BE ARRANGED SO THAT THE LARGER CABLES ARE IN THE CENTER OF THE HANGER IF POSSIBLE.
3. THIS FIGURE SUPERSEDES SHEET 4A6 OF DRAWING 803-5001027 AND SECTION 5, SHEET 39, OF DRAWING, NAVSEC NO. 8000-S6202-73960.



SH 132317209

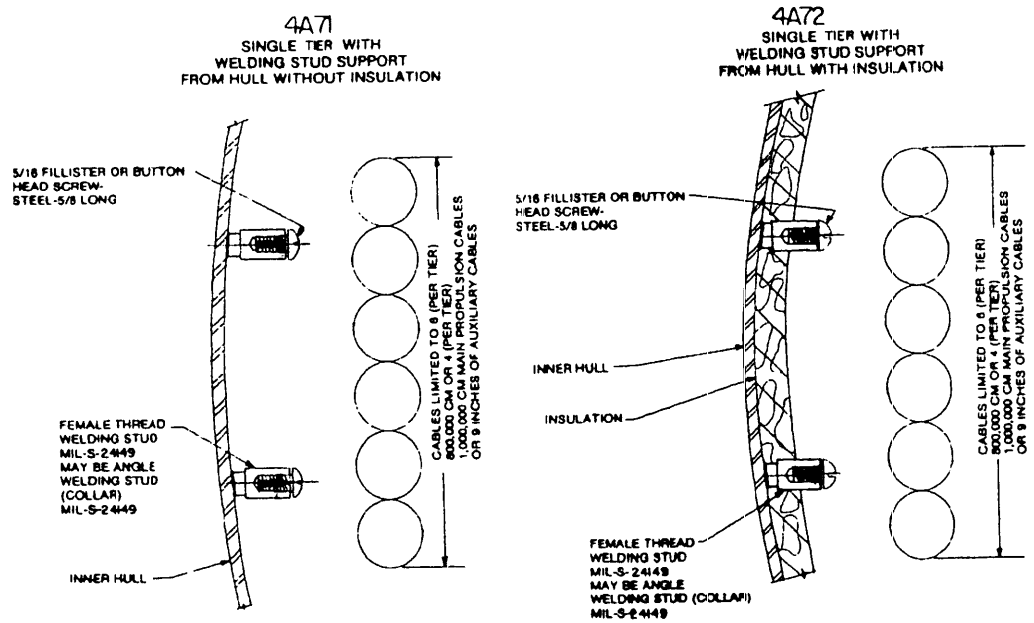
FIGURE 4A63. Cableways for auxiliary cables (submarines only).

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NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4A7 OF DRAWING 803-6001027 AND SECTION 5, SHEET 1, OF DRAWING NAVSEC NO. 9000-S6202-73980



SH 132317210

FIGURE 4A7. Single row cableway (submarine).

NOTES:
1. THIS METHOD IS FOR VERTICAL CABLE RUNS ONLY.
2. THIS FIGURE SUPERSEDES SHEET 4A8 OF DRAWING 803-5001027 AND SECTION 6, SHEET 41, OF DRAWING NAVSEC NO. 8000-S6702-73980.

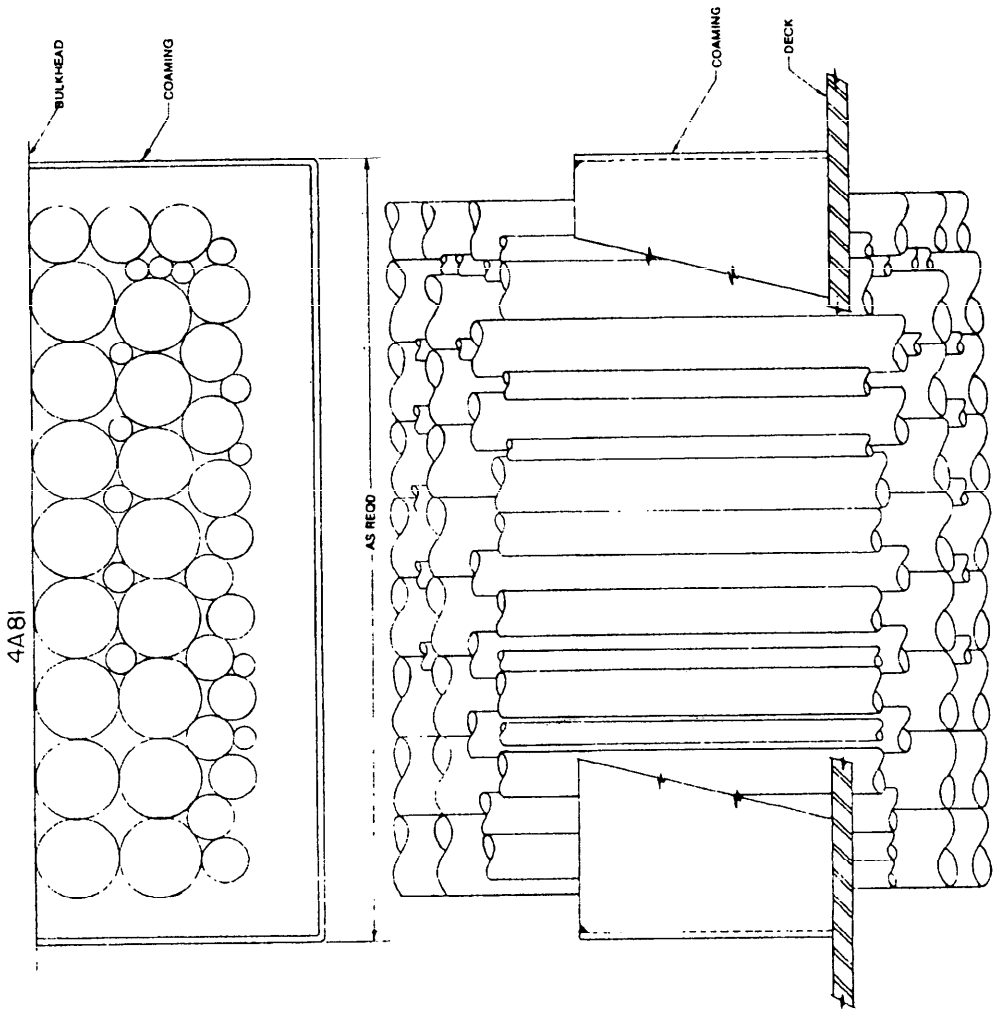


FIGURE 4A8. Cableway through coamings and hatchways (submarines).

SH 132317211

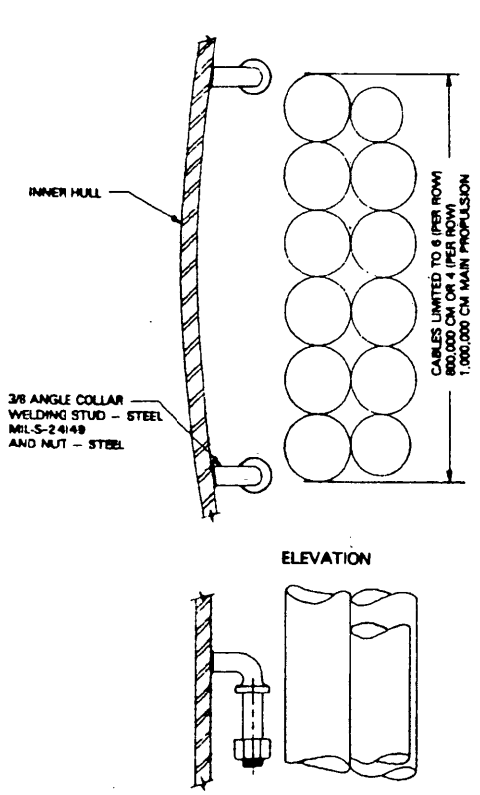
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NOTES:
1. HANGERS WITH 16" CENTER TO CENTER SPACING IS THE MAXIMUM SIZE THAT CAN WITHSTAND HI SHOCK WHERE NUMBER OF CABLES EXCEEDS THE MAXIMUM SIZE OF SUPPORT SHOWN. ADDITIONAL WIREWAYS SHALL BE PROVIDED.
2. THIS FIGURE SUPERSEDES SHEET 4A10 OF DRAWING 803-5001027 AND SECTION 5, SHEET 2, OF DRAWING, NAVSEC NO. 9000-S6202-73910.

4A101
CABLE BANKS
SUPPORTED BY WELDING STUDS
ON INNER HULL WITHOUT INSULATION



SH 132317213

4A102
CABLE BANKS
SUPPORTED BY WELDING STUDS
ON INNER HULL WITH INSULATION

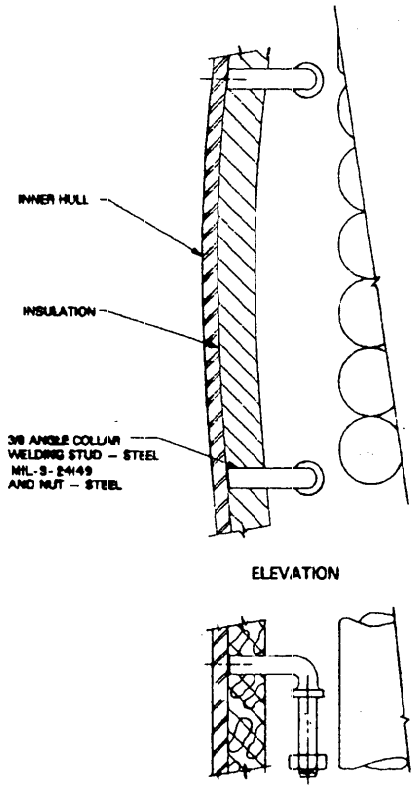
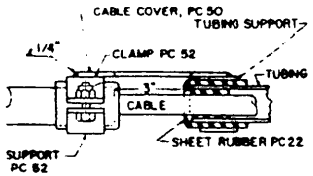


FIGURE 4A10. Cableway for double bank of main propulsion cables (submarines).

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RUBBER IS TO BE INSERTED INTO TUBING (2" MIN) & FOLDED BACK 2" OVER TUBING. TUBING IS NOT TO BE FLARED, REMOVE ALL SHARP EDGES. THE RUBBER IS TO BE HELD IN PLACE WITH TUBING SUPPORT CLAMP. A PIECE OF 1/4"X2" FLATBAR IS TO BE WELDED TO THE TOP HALF OF THE CLAMP. THE THK OF THE RUBBER & SIZE OF CLAMP DEPENDS ON THE SIZE OF TUBING & CABLES INST. THERE IS TO BE NO NUTS LEFT ON CLAMPS

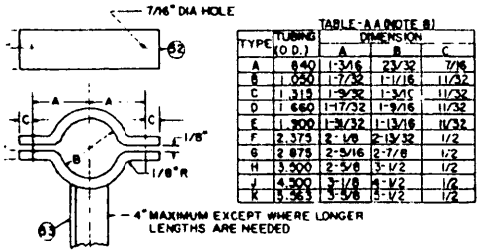
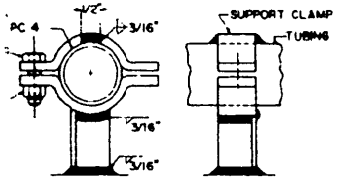


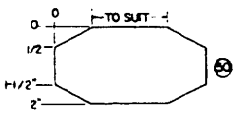
TABLE-A A (NOTE 8)

TYPE	TUBING (O.D.)	DIMENSION A	B	C
A	.640	1-3/16	23/32	7/16
B	1.050	1-7/32	1-1/16	11/32
C	1.315	1-9/32	1-3/16	11/32
D	1.660	1-17/32	1-9/16	11/32
E	1.900	1-3/32	1-13/16	11/32
F	2.375	2-1/8	2-13/32	1/2
G	2.875	2-5/16	2-7/8	1/2
H	3.500	2-5/8	3-1/2	1/2
J	4.500	3-1/8	4-1/2	1/2
K	5.563	3-5/8	5-1/2	1/2

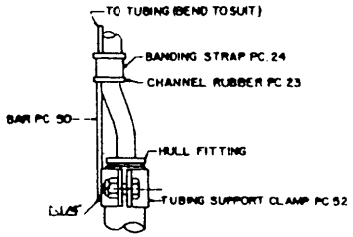


TUBING SUPPORT CLAMP ASSY
SEE TABLE AA

DETAIL 1A
SEE NOTE 7, 8 & 10

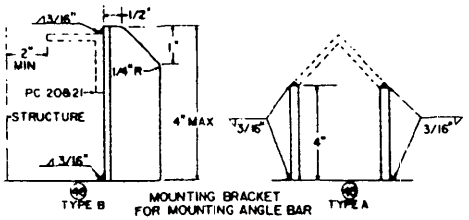


CABLE COVER

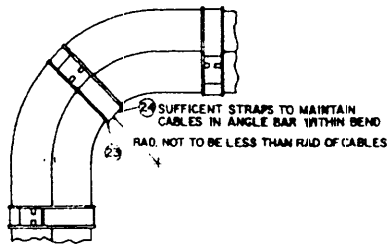


CABLE PROTECTION SHEAR VALVE & SINGLE HULL FTG.

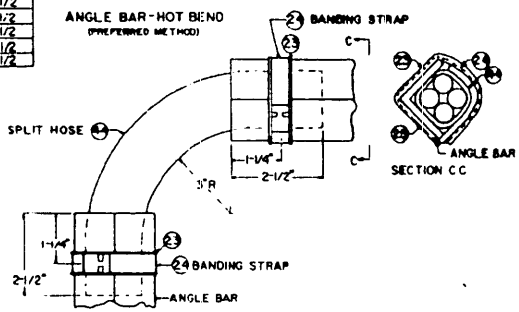
SH 132317214



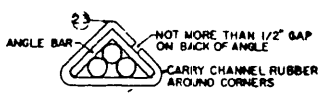
MOUNTING BRACKET FOR MOUNTING ANGLE BAR



ANGLE BAR-HOT BEND
(PREFERRED METHOD)



SPLIT HOSE IN ANGLE BAR RUN
(ALTERNATE METHOD)



THREE OR MORE CABLES

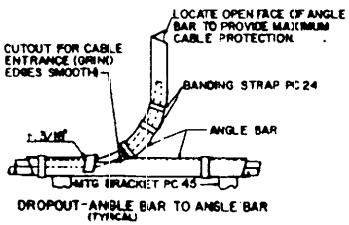


TWO CABLES



ONE CABLE

TYPICAL METHOD OF BANDING CABLES IN ANGLE BAR



DROPOUT-ANGLE BAR TO ANGLE BAR (TYPICAL)

NOTES:

1. CABLES ARE TO BE COMPLETELY PROTECTED FOR THEIR ENTIRE LENGTH BY THE METHODS SHOWN.
2. CABLE PANS PC 51-3, ANGLE BAR PCS 20-21 ARE TUBING PC 4, ARE TO BE MOUNTED FROM THE HULL OR FRAME AT EACH FRAME SPACE OR 3' MAXIMUM SPACING AND NO MORE THAN 6" FROM THE ENDS. CABLES IN PANS ARE TO BE BANDED THRU EACH SLOT PROVIDED AND BANDING IS TO BE SPACED NOT MORE THAN 16" APART. CABLES ANGLE BARS ARE TO BE BANDED EVERY 12" CABLES SHALL NOT SAG. COVERS ARE TO BE DOUBLE BANDED EVERY 16. ALL WELDING IS TO BE ON FRAME LINES ONLY. BANDING OF CABLES ON ANGLE BAR WIREWAYS AND CABLE PANS SHALL BE ONE TURN MINIMUM.
3. CHANNEL RUBBER PC 23, SHALL BE USED IN ALL CASES BETWEEN CABLES AND BANDING STRAPS IN A MANNER TO PREVENT ANY POSSIBLE CONTACT BETWEEN THE BANDING STRAPS AND THE CABLES.
4. ALL WELDING SHALL COMPLY WITH NAVSHIPS 0900-LP-006-9010.
5. ENOUGH CABLE SLACK IS TO BE ALLOWED FOR ONE COMPLETE TURN AROUND THE HULL FITTING CABLE RACK ASSEMBLY. WHERE CABLE IS SHORTENED DUE TO INITIAL MOLDING ERROR. THE CABLE SHALL BE INSTALLED IN RACK AS SHOWN IN FIG. "A-A", SH 4-A-14.
6. CABLE PAN COVERS ARE TO BE INSTALLED AT ALL TIMES EXCEPT DURING INSTALLATION OF CABLES IN THE PANS. CABLE PANS ARE TO BE CLEAN OF ALL FOREIGN MATTER PRIOR TO INSTALLATION OF CABLES
7. RADIUS OF TUBING BENDS ARE TO BE A MINIMUM OF 10 TIMES THE DIAMETER OF THE CABLE INSTALLED WHERE PRACTICABLE.
8. TUBING, PC4, IS NOT TO BE USED EXCEPT IN SPECIAL INSTALLATIONS WHERE SPECIFIED
9. ALL SHARP EDGES SHALL BE ROUNDED AND ALL BURRS SHALL BE REMOVED.
10. TUBING SUPPORT CLAMPS ARE TO BE ASSEMBLED WITHOUT RUBBER LINING AND WELD AS SHOWN IN DETAIL 1A FIGURE 4-A-11
11. FOR OUTBOARD IDENTIFICATION TAGS USE "TY-RAP IDENTIFICATION PLATE PC 56 AND "TY-RAP TIES" PC 57 OR EQUIVALENT. PLATE PC 56 IS TO BE TYPED USING AN ADDRESSOGRAPH SERIES 6300 GRAPHOTYPE 3/16" LETTERING OR EQUIVALENT.
12. ALL CABLE PAN AND ANGLE BAR RUNS ARE TO BE CONTINUOUS AND SHALL BE BROKEN ONLY AS INDICATED ON SHIPS INSTALLATION PLAN.
13. CABLE PAN, ANGLE BARS, MOUNTING BRACKETS AND TUBING SHALL BE PAINTED IN ACCORDANCE WITH THE MATERIAL REQUIREMENTS OF THE SUBMARINE PAINTING SCHEDULE
14. ALL CABLE SLACK IS TO BE TAKEN UP ON THE CABLE RACK AROUND THE HULL FITTING. THERE IS TO BE NO SLACK IN THE CABLE PANS. SEE SECTION "AA" FIGURE 4-A-14 RADIUS OF THE CABLE BENDS ARE TO BE TEN (10) TIMES THE DIAMETER OF THE CABLE WHERE POSSIBLE.
15. BANDING STRAPS SHALL BE INSTALLED SO AS NOT TO DEFORM THE CABLE SHEATH, BUT SHALL BE TIGHTENED SUFFICIENTLY TO PRECLUDE LOOSENING OF CABLE UNDER ENVIRONMENTAL CONDITIONS
16. MOUNTING BRACKET PC 45 TYPE A SHALL BE STAGGERED FROM EACH OTHER.
17. THIS FIGURE SUPERSEDES SHEET 4A11 OF DWG 803-5001027 NAVSHIPS DRAWING NO. 815-1197347 & SECTION 6, SHEET 103 OF 9000-98202-73880

NOTE (A): THE PURPOSE OF FIGURE 4A11-4A14 IS TO PROVIDE GUIDELINES FOR THE INSTALLATION OF ALL CABLING EXTERNAL TO THE SUBMARINE HULL. THEY ILLUSTRATE THE METHODS FOR PROTECTING AND SECURING THE CABLING FROM AN OUTBOARD APPLIANCE TO AN ELECTRICAL HULL FITTING NOT APPLICABLE TO SUBMARINE ANTENNA AND MAST ASSEMBLIES.

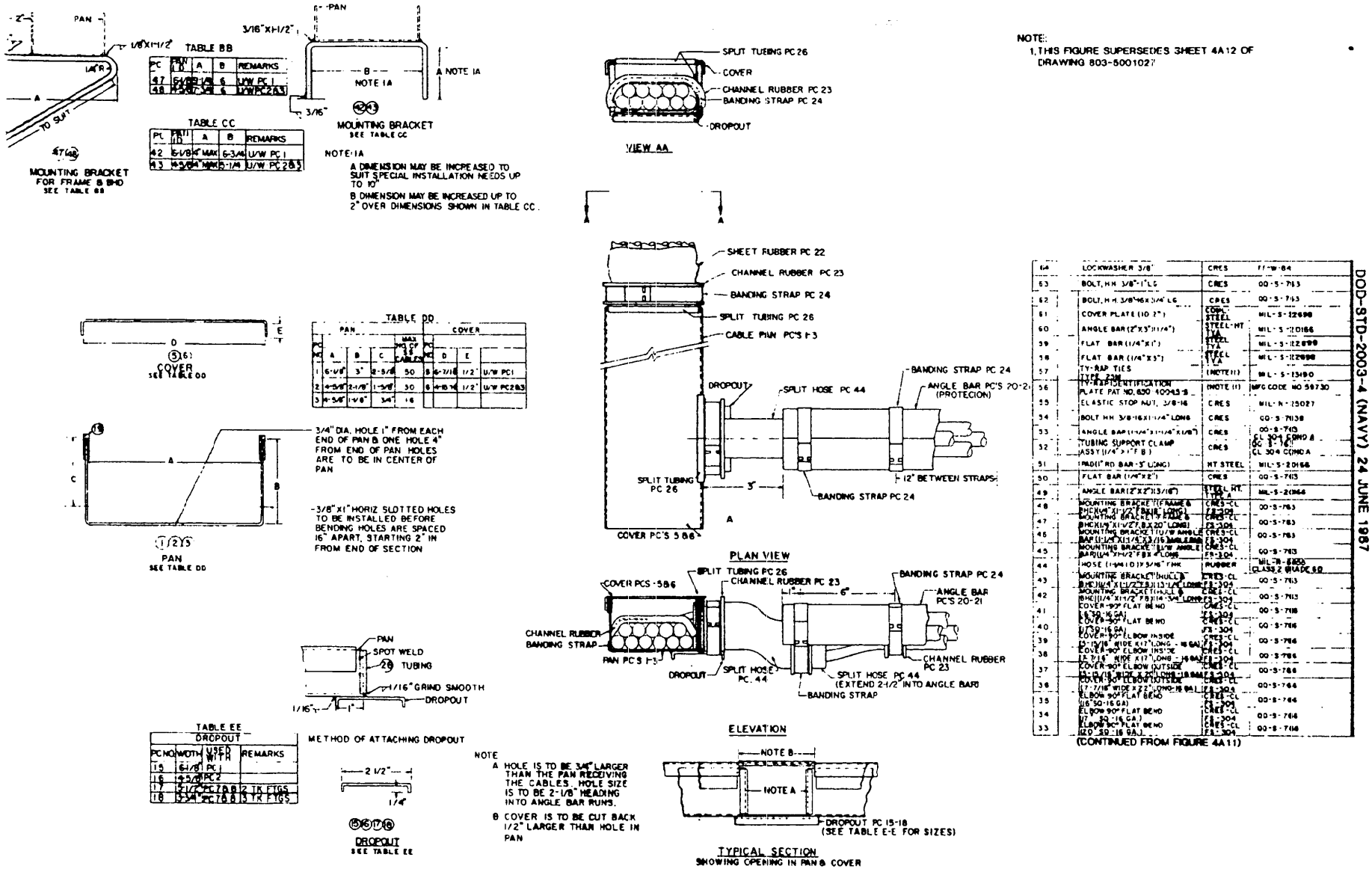
CONTINUED ON FIGURE 4A12

NO.	QTY	NOMENCLATURE	MATERIAL	SPECIFICATION
32		ELBOW-90° INSIDE (6-7/8" WIDE X 18" LONG 16 GA)	CRES-CL FS-304	00-S-766
31		ELBOW-90° INSIDE (12-1/8 WIDE X 20" LONG 16 GA)	CRES-CL FS-304	00-S-766
30		ELBOW-90° INSIDE (12-1/8" WIDE X 22" LONG 16 GA)	CRES-CL FS-304	00-S-766
29		ELBOW-90° OUTSIDE (6" - 7/8" WIDE X 18" LONG 16 GA)	CRES-CL FS-304	00-S-766
28		ELBOW-90° OUTSIDE (6-7/8" WIDE X 16" LONG 16 GA)	CRES-CL FS-304	00-S-766
27		ELBOW-90° OUTSIDE (12 1/8" WIDE X 18" LONG 16 GA)	CRES-CL FS-304	00-S-766
26		TUBING (.250"X.049" WALL)	CRES	MIL-P-1144
25		BUCKLE BANDING U/W PC 24	CRES	00-S-766
24		BANDING STRAP (U/W PC 24 OPT NO.)	CRES-CL FS-304	
23		CHANNEL RUBBER (3/4" W X 1/2" CH X 3/16" H)	RUBBER	MIL-P-6855 CLASS 2 GRADE 60
22		RUBBER-SYNTHETIC 1/8" THK	RUBBER	MIL-R-6855 CLASS 2 GRADE 60
21		ANGLE BAR (2"X2"X3/16")	CRES-CL FS-304	00-S-763
20		ANGLE BAR (1-1/2"X1-1/2"X3/16")	CRES-CL FS-304	00-S-763
19		CHANNEL RUBBER	RUBBER	MIL-R-6855 CLASS 2 GRADE 60
18		DROPOUT (3"X3-3/4" 16 GA)	CRES-CL FS-304	00-S-766
17		DROPOUT (3"X5-1/2" 16 GA)	CRES-CL FS-304	00-S-766
16		DROPOUT (3"X4-5/8" 16 GA)	CRES-CL FS-304	00-S-766
15		DROPOUT (3"X6-1/8" 16 GA)	CRES-CL FS-304	00-S-766
14		COVER OFFSET-RIGHT HAND (13-7/16 X 12" 16 GA) U/W PC 11	CRES-CL FS-304	00-S-766
13		COVER OFFSET-LEFT HAND (13-7/16 X 12" 16 GA) U/W PC 11	CRES-CL FS-304	00-S-766
12		COVER OFFSET-RIGHT HAND (13-5/16 X 12" 16 GA) U/W PC 10	CRES-CL FS-304	00-S-766
11		COVER OFFSET-LEFT HAND (13-5/16 X 12" 16 GA) U/W PC 9	CRES-CL FS-304	00-S-766
10		PAN OFFSET-RIGHT HAND (16-1/4" X 12" 16 GA)	CRES-CL FS-304	00-S-766
9		PAN OFFSET-LEFT HAND (16-1/4" X 12" 16 GA)	CRES-CL FS-304	00-S-766
8		PAN OFFSET-RIGHT HAND (14" X 12" 16 GA)	CRES-CL FS-304	00-S-766
7		PAN OFFSET-LEFT HAND (14" X 12" 16 GA)	CRES-CL FS-304	00-S-766
6		COVER (5-15/16" WIDE-16 GA) U/W PC 283	CRES-CL FS-304	00-S-766
5		COVER (7-7/8" WIDE-16 GA) U/W PC1	CRES-CL FS-304	00-S-766
4		TUBING	CUNI	MIL-T-16420 TABLE AA
3		PAN (8-7/8" WIDE-16 GA-8 SECT	CRES-CL FS-304	00-S-766
2		PAN (8-3/8" WIDE-16 GA-8 SECT	CRES-CL FS-304	00-S-766
1		PAN (12-1/8" WIDE-16 GA-8 SECT	CRES-CL FS-304	00-S-766

LIST OF MATERIAL - QUANTITIES FOR ONE UNIT

FIGURE 4A11 External wireways installation.

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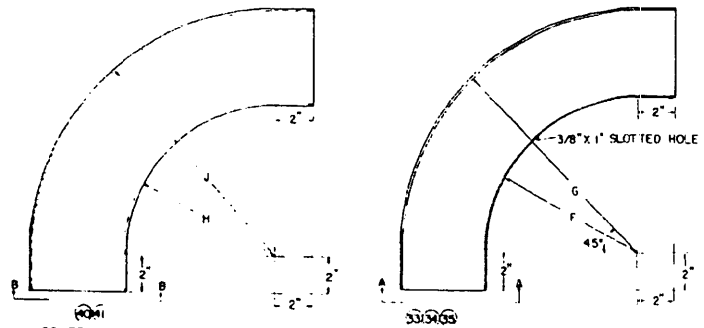
ITEM	NOMENCLATURE	MATERIAL	SPECIFICATION
64	LOCKWASHER 3/8"	CRES	FF-W-84
63	BOLT, H.H. 3/8"-11"LG	CRES	QQ-S-763
62	BOLT, H.H. 3/8"-16X3/4" LG	CRES	QQ-S-763
61	COVER PLATE (10 2")	CORR-RESIST STEEL	MIL-S-12698
60	ANGLE BAR (2"X3"X1/4")	STEEL-HT TYA	MIL-S-21086
59	FLAT BAR (1/4"X1")	STEEL TYA	MIL-S-12698
58	FLAT BAR (1/4"X3")	STEEL TYA	MIL-S-12698
57	TY-RAP TIES TYPE 2 CLS	(NOTE 11)	MIL-S-23190
56	TY-RAP IDENTIFICATION PLATE PAT NO. 630 40043-8	(NOTE 11)	MFG CODE NO 59730
55	ELASTIC STOP NUT, 3/8-16	CRES	MIL-N-25027
54	BOLT HH 3/8-16X1-1/4" LONG	CRES	QQ-S-763
53	ANGLE BAR (1-1/4"X1-1/4"X1/8")	CRES	QQ-S-763
52	TUBING SUPPORT CLAMP ASSY (1/4" I.F.B.)	CRES	CL 304 COND B & E CL 304 COND A
51	PAD (1" RD BAR-3" LONG)	HT STEEL	MIL-S-20166
50	FLAT BAR (1/4"X2")	CRES	QQ-S-763
49	ANGLE BAR (2"X2"X3/16")	STEEL HT TYPE A	MIL-S-20166
48	MOUNTING BRACKET (FRAME & BHD) 1/4"X1-1/2"X8" LONG	CRES-CL FA-304	QQ-S-763
47	MOUNTING BRACKET (FRAME & BHD) 1/4"X1-1/2"X 8"X20" LONG	CRES-CL FA-304	QQ-S-763
46	MOUNTING BRACKET (1/8" ANGLE BAR) (1/4"X1-1/4"X3/16 BARS LONG)	CRES-CL FA-304	QQ-S-763
45	MOUNTING BRACKET (1/8" ANGLE BAR) (1/4"X1-1/2"X8"X4" LONG)	CRES-CL FA-304	QQ-S-763
44	HOSE (1-3/4" I.D. X 3/16" THK)	RUBBER	MIL-H-6850 CLASS 2 GRADE 60
43	MOUNTING BRACKET (HULL & BHD) 1/4"X1-1/2"X8" (1-1/4" LONG)	CRES-CL FA-304	QQ-S-763
42	MOUNTING BRACKET (HULL & BHD) 1/4"X1-1/2"X8" (1-1/4" LONG)	CRES-CL FA-304	QQ-S-763
41	COVER-90° FLAT BEND (10 GA)	CRES-CL FA-304	QQ-S-766
40	COVER-90° FLAT BEND (10 GA)	CRES-CL FA-304	QQ-S-766
39	COVER-90° ELBOW INSIDE (5"X5"X8" WIDE X 12" LONG - 16 GA)	CRES-CL FA-304	QQ-S-766
38	COVER-90° ELBOW INSIDE (5"X5"X8" WIDE X 12" LONG - 16 GA)	CRES-CL FA-304	QQ-S-766
37	COVER-90° ELBOW OUTSIDE (7-7/16" WIDE X 22" LONG - 16 GA)	CRES-CL FA-304	QQ-S-766
36	COVER-90° ELBOW OUTSIDE (7-7/16" WIDE X 22" LONG - 16 GA)	CRES-CL FA-304	QQ-S-766
35	ELBOW 90° FLAT BEND (16 GA)	CRES-CL FA-304	QQ-S-766
34	ELBOW 90° FLAT BEND (7 GA)	CRES-CL FA-304	QQ-S-766
33	ELBOW 90° FLAT BEND (20 GA)	CRES-CL FA-304	QQ-S-766

(CONTINUED FROM FIGURE 4A11)

DOD-STD-2003-4 (NAVY) 24 JUNE 1987

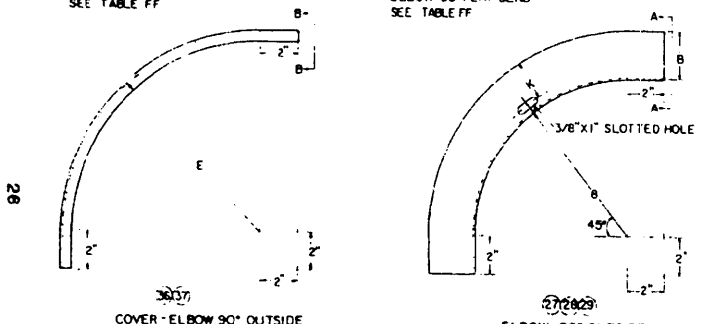
FIGURE 4A12. External wireways installation standard methods (submarines).

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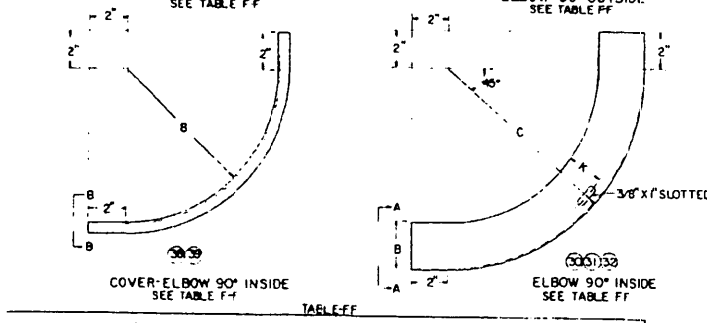
COVER- 90° FLAT BEND
SEE TABLE FF

ELBOW 90° FLAT BEND
SEE TABLE FF



COVER - ELBOW 90° OUTSIDE
SEE TABLE FF

ELBOW 90° OUTSIDE
SEE TABLE FF



COVER-ELBOW 90° INSIDE
SEE TABLE FF

ELBOW 90° INSIDE
SEE TABLE FF

TABLE FF

	ELBOW						COVER					
	A	B	C	F	G	K	PC NO.	D	E	H	J	TYPE
7	9-1/8"	3"				2-5/8"	38	6-7/16"	11-1/16"			90° OUTSIDE
8	4-3/8"	2-1/8"				1-5/8"	37	4-15/16"	10-3/16"			90° OUTSIDE
9	4-3/8"	1-1/8"				3/4"						
0	6-1/8"	3"	11-1/16"			2-5/8"	38	6-7/16"				90° INSIDE
1	4-3/8"	2-1/8"	10-3/16"			1-3/8"	39	4-15/16"				90° INSIDE
2	4-3/8"	1-1/8"	9-3/16"			3/4"						
3	6-1/8"	3"		6"	14-3/16"	2-5/8"	40	6-7/16"		7-29/32"	14-11/32"	90° FLAT
4	4-3/8"	2-1/8"		6"	12-11/16"	1-5/8"	41	4-15/16"		7-29/32"	7-27/32"	90° FLAT
5	4-3/8"	1-1/8"		6"	12-11/16"	3/4"						

THESE 90° ELBOWS ARE TO BE CUT TO OBTAIN ELBOWS OF LESSER DEGREES

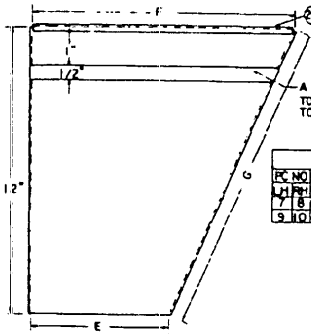
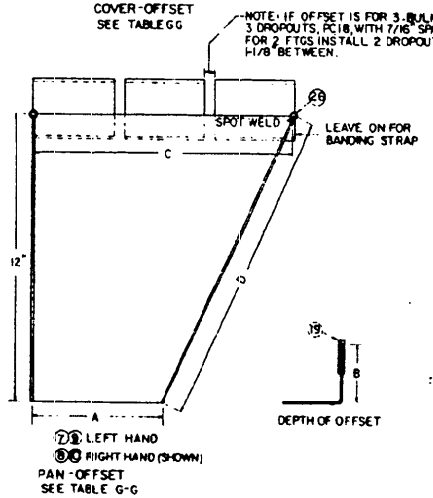


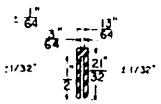
TABLE GG

PC NO.	OFFSET A	OFFSET B	OFFSET C	OFFSET D	PC NO.	OFFSET-COVER E	OFFSET-COVER F	OFFSET-COVER G	REMARKS
7/8	6-1/8"	3"	12"	13-3/8"	11/12	6-7/16"	9-3-5/16"	13-3/8"	
9/10	4-3/8"	2-1/8"	12"	14-1/8"	13/14	4-15/16"	10-3/16"	14-1/8"	

① LEFT HAND
② RIGHT HAND (SHOWN)
COVER-OFFSET
SEE TABLE GG



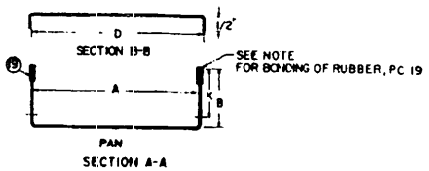
⑦ LEFT HAND
⑧ RIGHT HAND (SHOWN)
PAN-OFFSET
SEE TABLE G-G



⑨ CHANNEL RUBBER
(SEE BONDING NOTES)

BONDING NOTES FOR PC 19

1. SOLVENT WIPE THE EDGES OF THE CABLE PAN IN THE AREA TO BE BONDED WITH INHIBISOL OR TRICHLOROETHYLENE.
2. APPLY A COAT OF CARBOLINE NEOPRENE 100 PRIMER BY BRUSHING AND ALLOW BETWEEN TWO HOURS AND 36 HOURS DRYING TIME AT ROOM TEMPERATURE.
3. IN THE MEANTIME SOLVENT WIPE THE AREA OF THE NEOPRENE EXTRUSION TO BE BONDED WITH INHIBISOL OR TRICHLOROETHYLENE.
4. AFTER THE NEOPRENE 100 PRIMER IS DRY, BRUSH APPLY A COAT OF CARBOLINE F-1 ADHESIVE TO BOTH THE METAL AND NEOPRENE RUBBER SURFACES. ALLOW A 20 TO 30 MINUTE DRYING TIME AT ROOM TEMPERATURE BEFORE JOINING.
5. FOR PIECE NUMBERS SHOWN HEREON, SEE FIGURES 4A11&4A12.



SH 132317216

FIGURE 4A13. External wireways installation standard methods (submarines).

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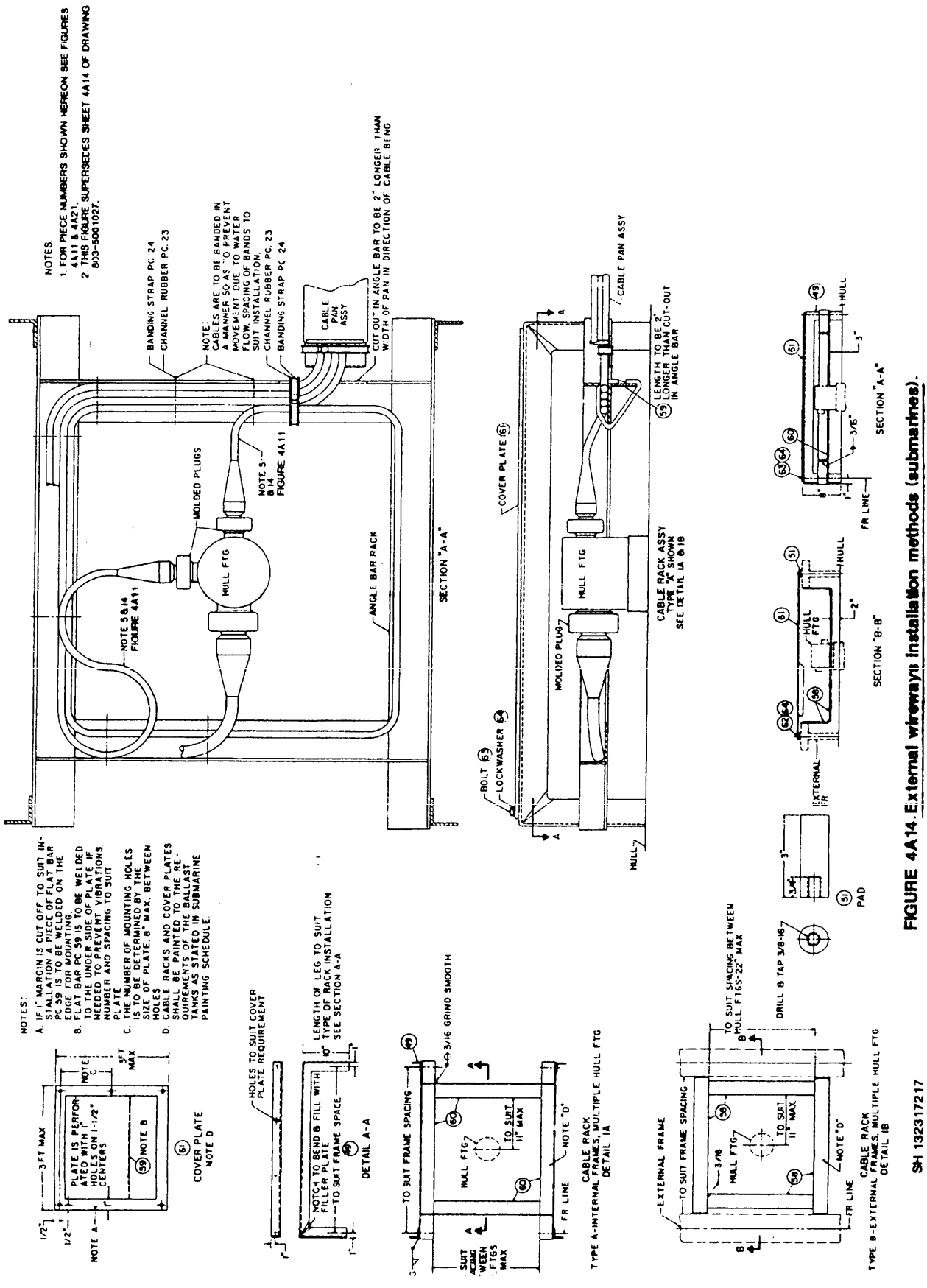


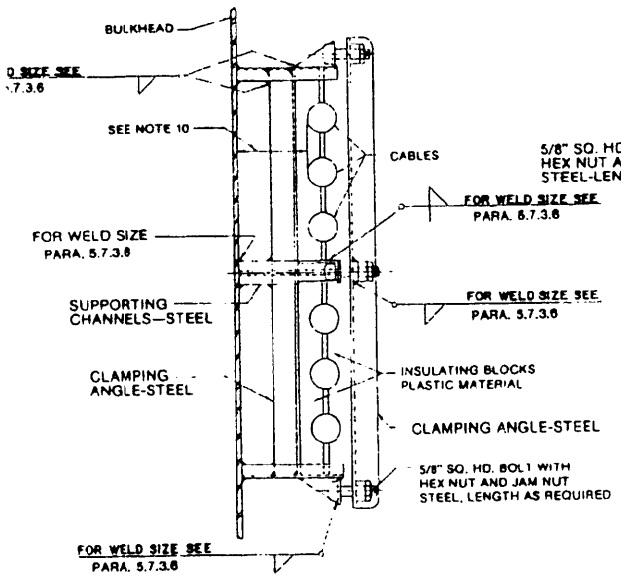
FIGURE 4A14. External wireways installation methods (submarines).

SH 132317217

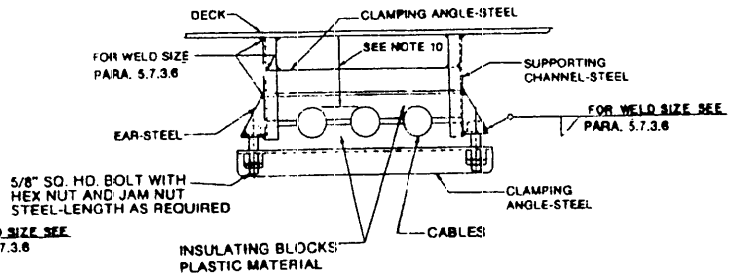
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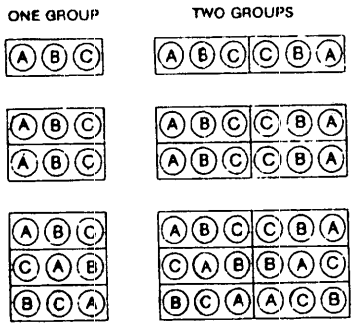
4B11
TWO GROUPS AND ONE TIER OF CABLES
FROM STEEL DECKS OR BULKHEADS



4B12
ONE GROUP AND ONE TIER OF CABLES
FROM STEEL DECKS OR BULKHEADS



4B13
CABLE ARRANGEMENTS
AND PHASE RELATIONSHIP



NOTES.
1. WHERE CABLES ENTER PROPULSION ELECTRICAL EQUIPMENT, CARE SHALL BE EXERCISED TO AVOID BUNCHING OR CLOSELY GROUPING THE CABLES IN ANY MANNER THAT MAY RESTRICT DISSIPATION OF HEAT FROM THE CABLES.
2. PHASE ARRANGEMENTS AS SHOWN IN CHART ON THIS FIGURE SHALL APPLY TO HORIZONTAL, VERTICAL OR INCLINED WIREWAY RUNS. ANY ARRANGEMENTS OF CABLES OTHER THAN SHOWN MUST BE SPECIFICALLY APPROVED BY NAVSEA
3. PROPULSION CABLES IN WIREWAYS SHALL BE RUN SEPARATE FROM OTHER CABLES.
4. CABLE ENDS SHALL BE WATER-SEALED BY TERMINATING IN WATER-SEAL SOLDERLESS LUGS IN ACCORDANCE WITH MIL-STD- 98-1 GROUP A
5. WHERE CABLES TO PROPULSION EQUIPMENT ARE REQUIRED TO ENTER THROUGH STUFFING TUBES, THEY SHALL BE INSTALLED IN ACCORDANCE WITH MIL-STD- 98-3
6. IN ORDER TO INSURE THAT THE PROPER ARRANGEMENT OF SINGLE CONDUCTOR CABLES IS MAINTAINED, THE PHASE OF SINGLE CONDUCTOR CABLES SHALL BE INDICATED (IN ACCORDANCE WITH SECTION 304 OF THE GENERAL SPECIFICATIONS FOR SHIPS) AT SUITABLE INTERVALS ALONG THE LENGTH OF EACH CABLE AND AT THE EXTREMETIES
7. UNLESS OTHERWISE APPROVED NO CABLE SUPPORT SHALL BE MOUNTED IN SUCH A MANNER AS TO BE DIRECTLY CONNECTED TO STRENGTH FRAMES OR BEAMS, PARTICULARLY WHERE NEAR THE SKIN OF THE SHIP AND AT OR BELOW THE WATERLINE.
8. AVOID RUNNING PROPULSION CABLE WIREWAYS IN LOCATIONS WHERE AMBIENT TEMPERATURES EXCEED 50°C. DURING OPERATION.
9. HANGERS SHALL BE SPACED AT INTERVALS OF NOT MORE THAN 16" O.C.
10. MINIMUM CLEARANCE BETWEEN CABLES AND BULKHEADS SHALL BE 4 INCHES.
11. THIS FIGURE SUPERSEDES SHEET 4B1 OF DRAWING 803-5001027 AND SECTION 2, SHEET 131, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

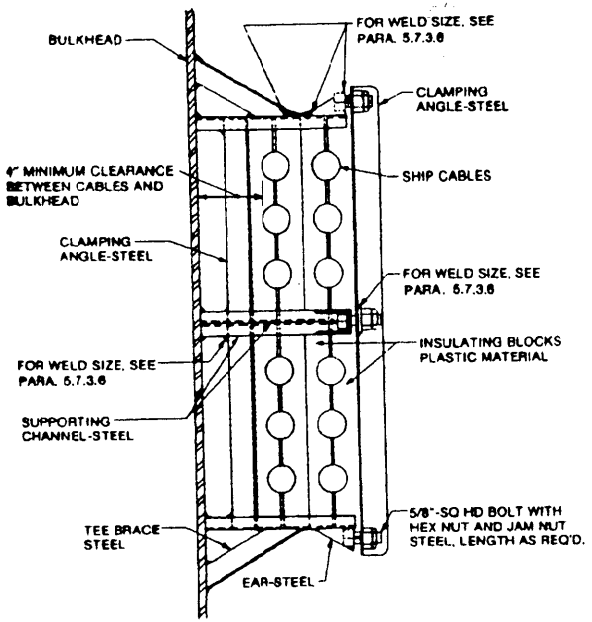
SH 132317218

FIGURE 4B1. Cableways for A.C. propulsion cables (surface ship).

NOTE:

1. THIS FIGURE SUPERCEDES SHEET 482 OF DRAWING 803-5001027 AND SECTION 2 SHEET 132 ,OF DRAWING , NAVSEC NO. 9002-S6202-73980.

4B21
TWO GROUPS AND TWO TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS



4B22
ONE GROUP AND TWO TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS

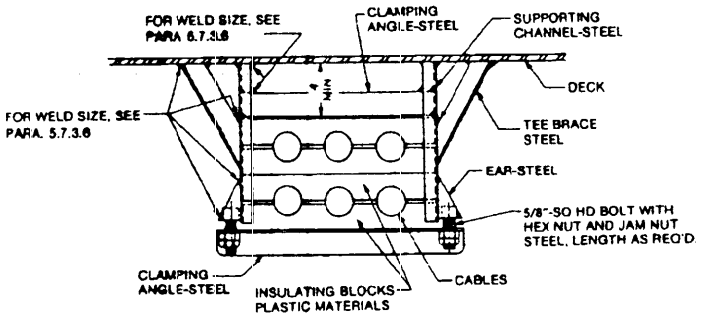
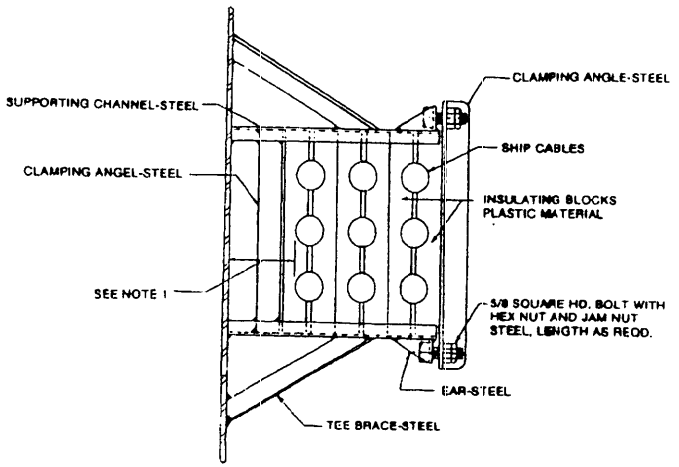


FIGURE 4B2. Cableways for A.C. propulsion cables (surface ship).

SH 132317219

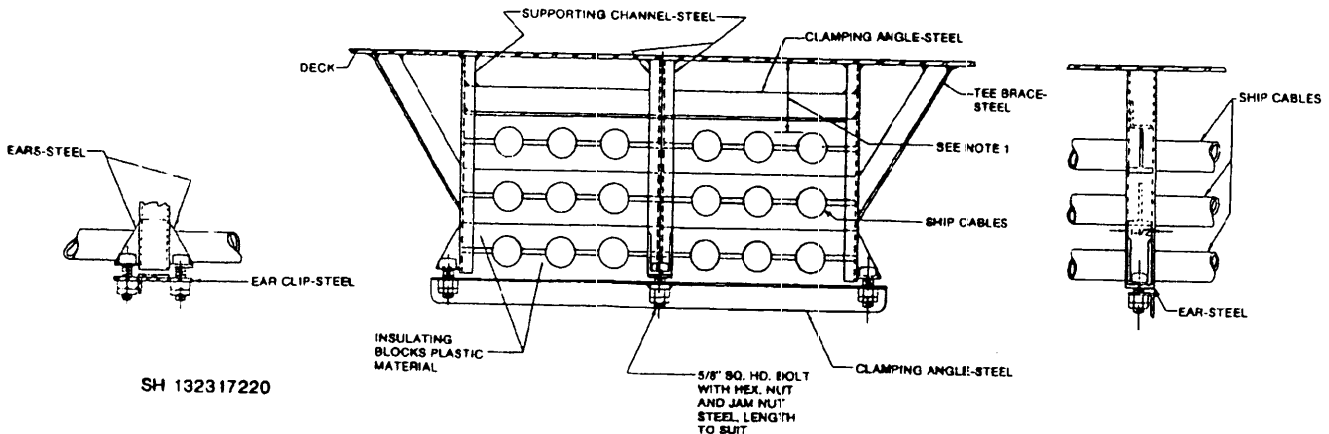
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4B31
ONE GROUP AND THREE TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS



NOTE:
1. MINIMUM CLEARANCE BETWEEN CABLES AND BULKHEADS/DECKS SHALL BE 4 INCHES.
2. ALL WELDING SHALL BE AS PER PARA. 5.7.3.6
3. THIS FIGURE SUPERSEDES SHT. 483 OF DRAWING 803-5001027 AND SECTION 2, SHEET 133 OF DRAWING NAVSEC NO. 9000-S6202-73980

4B32
TWO GROUPS AND THREE TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS



SH 132317220

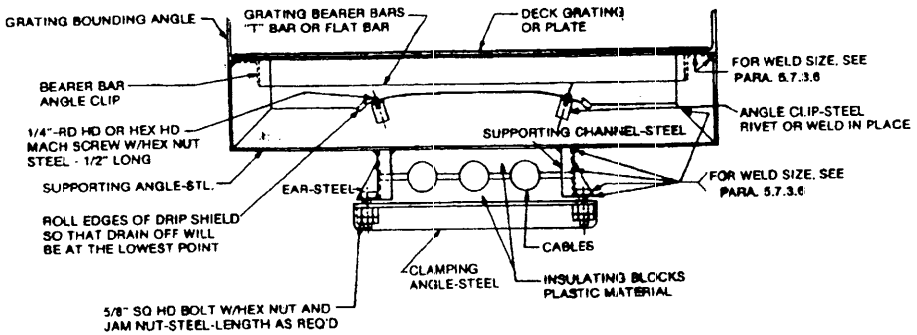
FIGURE 4B3. Cableways for A.C. propulsion cables (surface ships).

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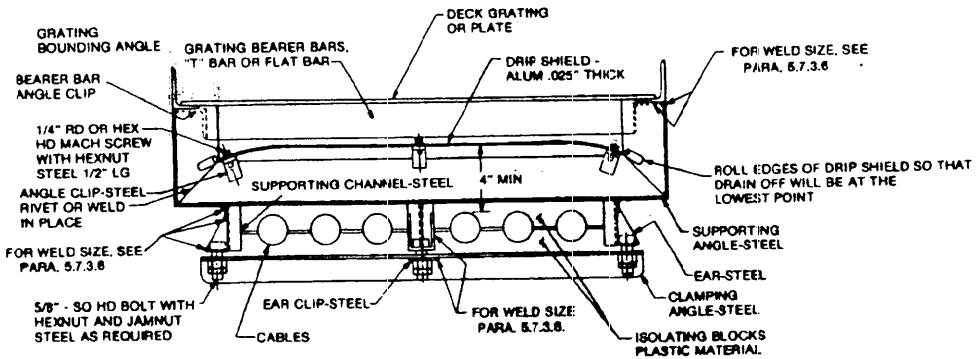
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NOTE:
1. THIS FIGURE SUPERSEDES SHT 4B4 OF DRAWING 803-5001027 AND SECTION 2, SHEET 134, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B41
ONE GROUP AND ONE TIER OF CABLES
UNDER DECK GRATINGS



4B42
TWO GROUPS AND ONE TIER OF CABLE
UNDER DECK GRATINGS



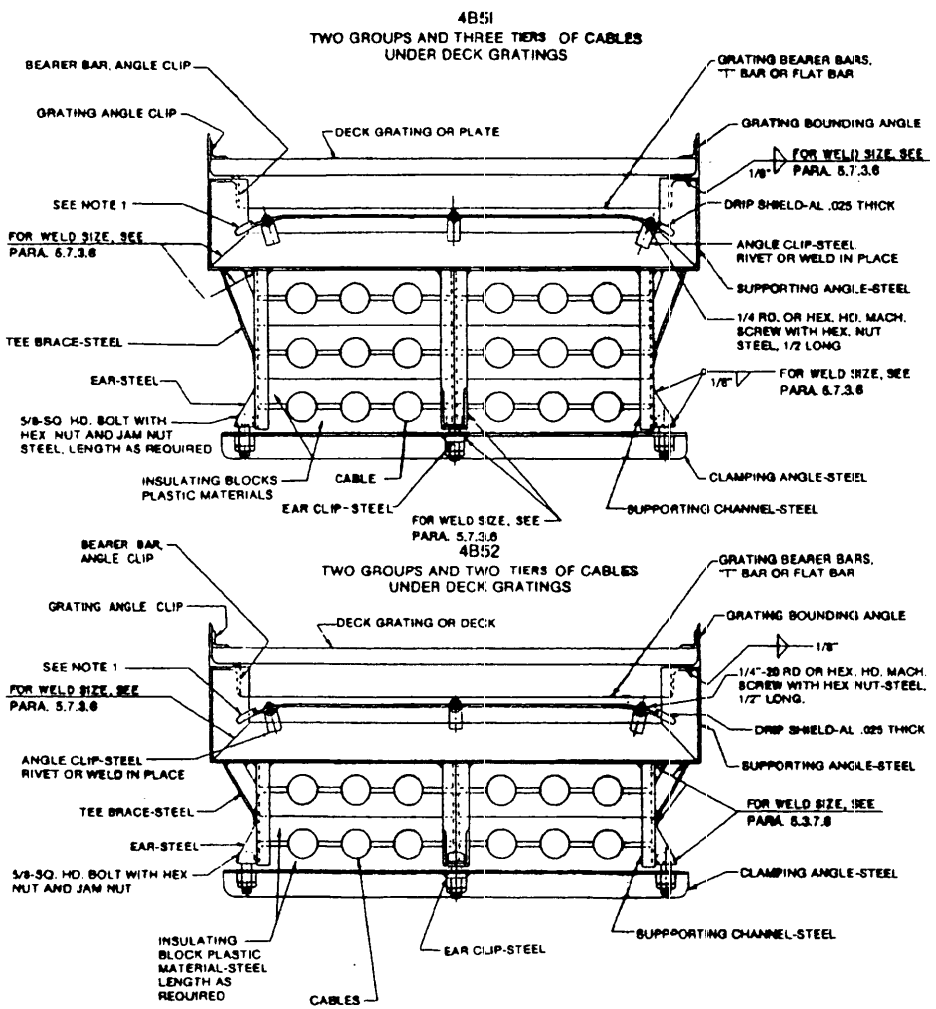
H 132317221

FIGURE 4B4. Support for A.C. propulsion cables (surface ships).

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NOTES:

1. ROLL EDGES OF DRIP SHIELD SO THAT DRAIN OFF WILL BE AT LOWEST POINT.
2. THIS FIGURE SUPERSEDES SHEET 485 OF DRAWING 803-5001027 AND SECTION 2, SHEET 135 OF DR DRAWING, NAVSEC NO. 9000-98202-73980.



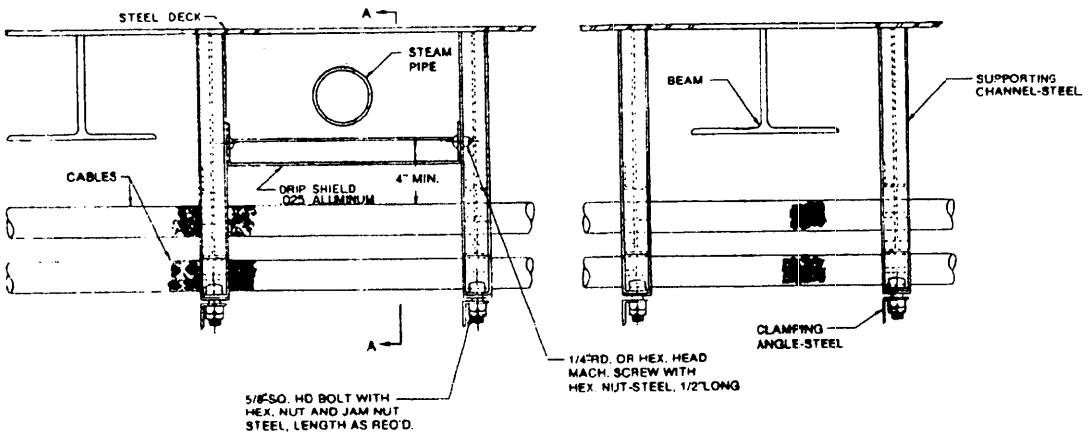
SH 132317222

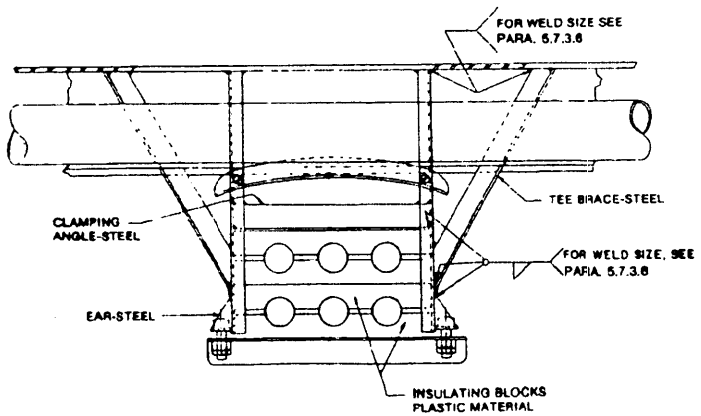
FIGURE 4B5. Cableways for A.C. propulsion cables (surface ships).

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4B61

CABLE RUNS ACROSS STEEL BEAMS
SHOWING DRIP SHIELD UNDER STEAM PIPE





SECTION A-A

SH 132317223

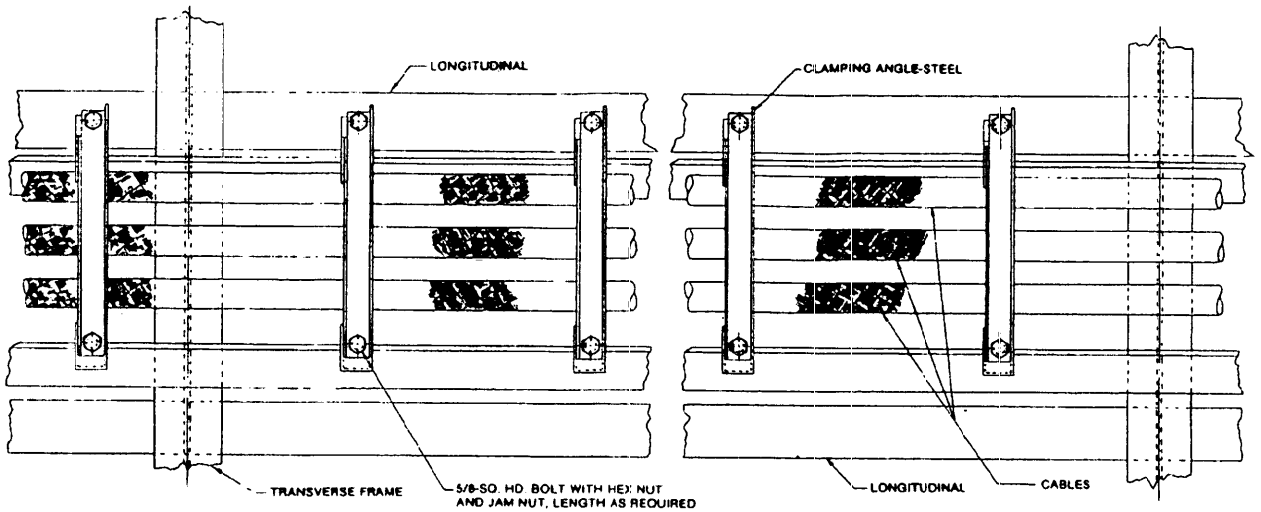
NOTE:

1 THIS FIGURE SUPERSEDES SHEET 4B6 OF DRAWING 803-5001027 AND SECTION2, SHEET 134, OF DRAWING, NAVSEC NO. 800-56202-73980

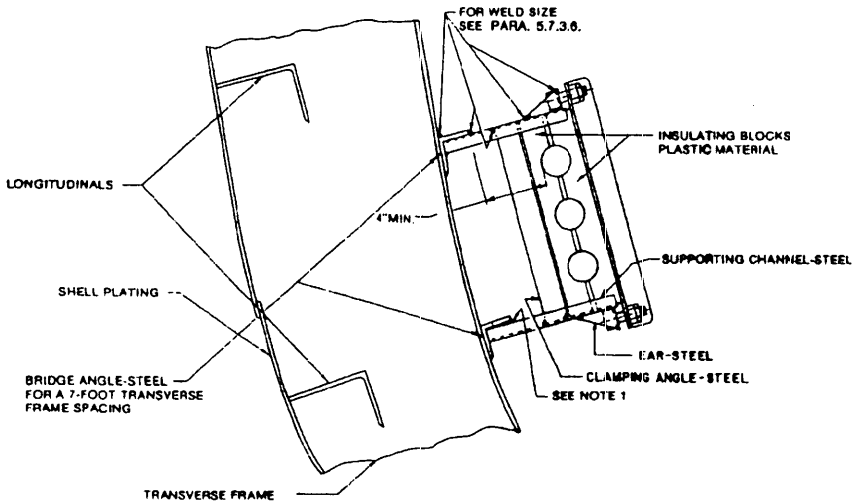
FIGURE 4B6. Cableways for A.C. propulsion cables (surface ships).

NOTE:

1. FILL SPACE BETWEEN CHANNEL AND BRIDGE ANGLE WITH PLASTIC SEALER , MIL-H-3084, TYPE HF
2. THIS FIGURE SUPERSEDES SHEET 4B7 OF DWG. 803-5001027 AND SECTION 2, SHEET 137 OF DRAWING, NAVSEC NO 9000-86202-73980



END VIEW



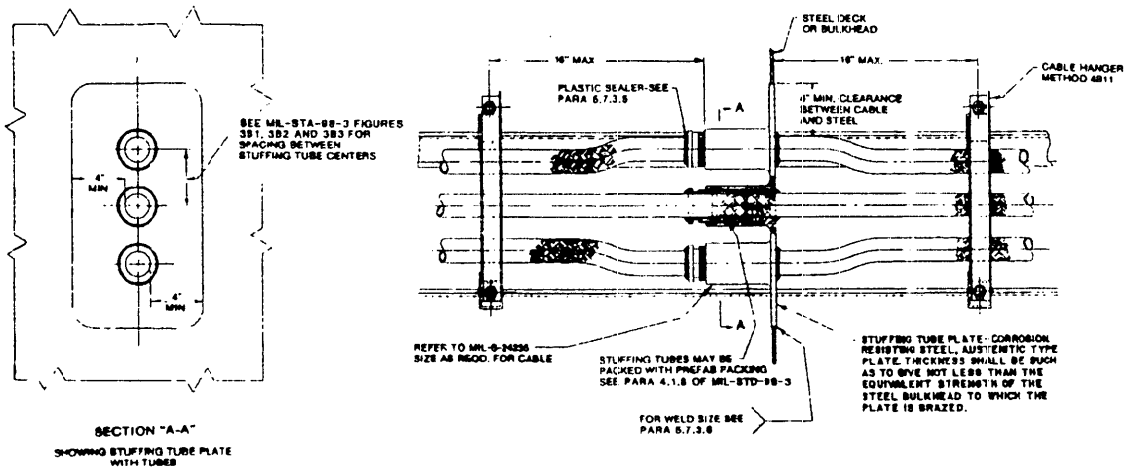
:H 132317224

FIGURE 4B7. Cableways for A.C. propulsion cables (surface ships).

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4B81

PASSING CABLES
THROUGH STEEL BULKHEADS OR DECKS
USING PLATE WITH STUFFING TUBES - INTEGRALLY CAST



NOTES:
1 THIS FIGURE SUPERSEDES SHEET 4B8 OF DRAWING 803-5001027 AND SECTION 4, SHEET 51, OF DRAWING NAVSEC NO. 9000-S6202-73980.

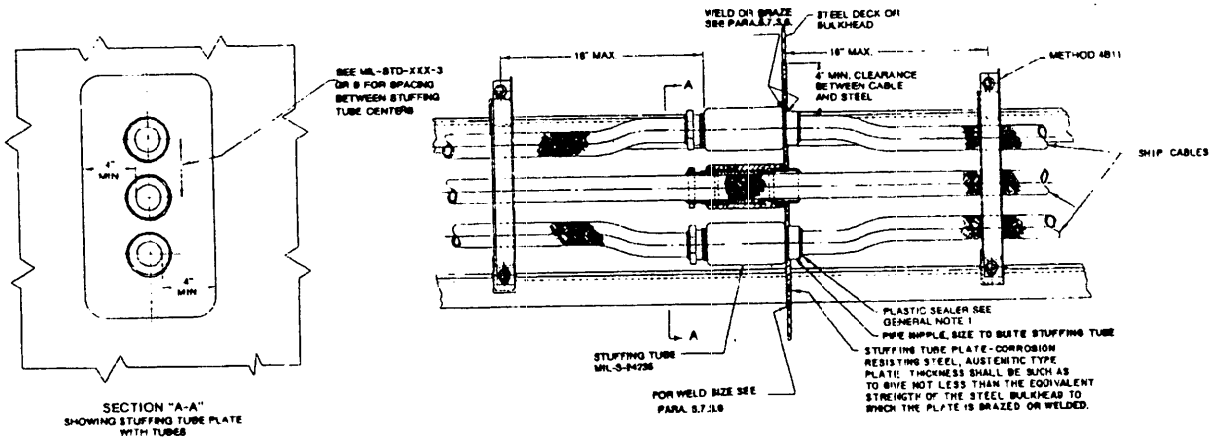
SH 132317225

FIGURE 4B8. Cableways for A.C. propulsion cables (surface ships).

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4B9.1

PASSING CABLES
THROUGH STEEL BULKHEADS OR DECKS
USING PLATE WITH STUFFING TUBES-FABRICATED



NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B9 OF DRAWING 803-5001027 AND SECTION 4, SHEET 52, OF DRAWING NAVSEC NO. 9000-S6202-73980.

SH 132317226

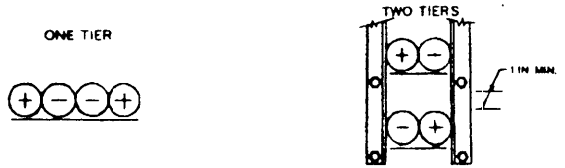
FIGURE 4B9. Cableways for A.C. propulsion cables (surface ships).

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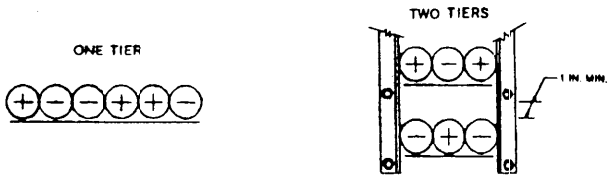
4B101

ARRANGEMENTS FOR TWO SINGLE CONDUCTOR CABLES PER LEG



4B102

ARRANGEMENTS FOR THREE SINGLE CONDUCTOR CABLES PER LEG



NOTES:

1. THE METHODS SHOWN INDICATE THE RECOMMENDED ARRANGEMENTS OF DIRECT CURRENT SINGLE CONDUCTOR PROPULSION CABLES IN WIREWAYS TO REDUCE TO A MINIMUM EXTERNAL MAGNETIC FIELDS DUE TO THE PROPULSION CURRENT. EACH METHOD ILLUSTRATES THE ARRANGEMENT OF CABLES FOR ONE CIRCUIT, CONSISTING OF A POSITIVE AND A NEGATIVE LEG, EACH LEG CONTAINING TWO OR MORE SINGLE CONDUCTOR CABLES IN PARALLEL WHENEVER PRACTICABLE. EACH CIRCUIT SHOULD CONTAIN AN EVEN NUMBER OF SINGLE CONDUCTOR CABLES PER LEG IN PREFERENCE TO AN ODD NUMBER PER LEG.

2. THE ARRANGEMENTS APPLY TO HORIZONTAL, VERTICAL OR INCLINED WIREWAYS. WHEN SEVERAL PROPULSION CIRCUITS ARE INSTALLED IN ONE WIREWAY, EACH CIRCUIT SHOULD BE ARRANGED AS INDICATED. THE CIRCUITS MAY BE LOCATED ADJACENT ON THE SAME HANGER OR HANGERS OR SEPARATELY ON INDIVIDUAL HANGERS.

3. THE EXTERNAL MAGNETIC FIELDS DUE TO DIRECT CURRENT IN DOUBLE CONDUCTOR CABLES ARE RELATIVELY SMALL IF THE TWO CONDUCTORS IN EACH CABLE ARE CONNECTED TO CARRY CURRENT OF OPPOSITE POLARITY. ACCORDINGLY, WHEN SEVERAL DOUBLE CONDUCTOR CABLES ARE USED IN PARALLEL, THE TWO CONDUCTORS IN EACH CABLE SHOULD BE OF OPPOSITE POLARITY.

4. CABLE CONNECTIONS TO THE TERMINALS OF ELECTRICAL EQUIPMENT SHOULD BE ACCOMPLISHED WITH A MINIMUM AMOUNT OF FANNING OF THE INDIVIDUAL CONDUCTORS. LOOPS FORMED BY CONDUCTORS OF OPPOSITE POLARITY SHOULD BE REDUCED IN AREA TO A MINIMUM, AND AVOIDED WHERE VER POSSIBLE. WHEN UNAVOIDABLE, THE PLANE OF THE LOOP FORMED BY THE CONDUCTORS OF OPPOSITE POLARITY IN EACH CIRCUIT SHOULD LIE VERTICAL IN PREFERENCE TO HORIZONTAL.

5. IN ORDER TO INSURE THAT THE PROPER ARRANGEMENT OF SINGLE CONDUCTOR CABLES IS MAINTAINED, THE POLARITY OF SINGLE CONDUCTOR CABLES SHOULD BE INDICATED AT SUITABLE INTERVALS ALONG THE LENGTH OF EACH CABLE AND AT THE EXTREMITIES.

6. PROPULSION CABLES IN WIREWAYS SHALL RUN SEPARATE FROM OTHER CABLES WHERE PRACTICABLE.

7. PROPULSION CABLES SHALL BE ADEQUATELY CLAMPED TO AVOID LOOSENING OR DISLODGEMENT DUE TO VIBRATION, SHOCK, OR MAGNETIC STRESSES.

8. AVOID RUNNING PROPULSION CABLE WIREWAYS IN LOCATION WHERE AMBIENT TEMPERATURES EXCEED 50°C. DURING OPERATION, AS IN VICINITY OF BOILERS, EVAPORATORS, STEAMPIPES, ETC.

9. WHERE CABLES ENTER PROPULSION ELECTRICAL EQUIPMENT, CARE SHALL BE EXERCISED TO AVOID BUNCHING OR CLOSELY GROUPING THE CABLES IN ANY MANNER THAT MAY RESTRICT DISSIPATION OF HEAT FROM THE CABLES.

10. THE ENDS OF PROPULSION CABLES SHALL BE ADEQUATELY SEALED BY APPROVED METHODS TO AVOID THE ENTRANCE OF WATER OR MOISTURE.

11. DETAILS NOT SPECIFICALLY COVERED IN D.C. PROPULSION METHODS SHALL BE IN ACCORDANCE WITH THE APPLICABLE STANDARD METHODS.

12. THIS FIGURE SUPERSEDES SHEET 4B10 OF DRAWING 803-800 1027 AND SECTION 2, SHEET 129, OF DRAWING NAVSEC NO. 9000-86202-7 5980.

*MAXIMUM ALLOWABLE CURRENT CAPACITIES FOR D.C. PROPULSION CABLES IN WIREWAYS AND RACKS

PERCENTAGE OF CURRENT RATINGS SPECIFIED IN DESIGN DATA SHEET DDS314-2 AT 50°C AMBIENT

NO. OF HANGERS HORIZONTAL OR VERTICAL	INDIVIDUAL CABLES ON EACH HANGER LOCATED ADJACENT WITHOUT SPACING				INDIVIDUAL CABLES ON EACH HANGER SPACED 1/2" APART			
	CABLES IN HORIZONTAL HANGERS NEXT TO BULKHEADS WITHOUT INSULATION	CABLES IN HORIZONTAL HANGERS NEXT TO BULKHEADS WITH INSULATION	CABLES IN HORIZONTAL HANGERS UNDER DECKS WITHOUT INSULATION OR IN VERTICAL HANGERS NEXT TO BULKHEADS WITHOUT INSULATION	CABLES IN HORIZONTAL HANGERS UNDER DECKS WITH INSULATION OR IN VERTICAL HANGERS NEXT TO BULKHEADS WITH INSULATION	CABLES IN HORIZONTAL HANGERS NEXT TO BULKHEADS WITHOUT INSULATION	CABLES IN HORIZONTAL HANGERS NEXT TO BULKHEADS WITH INSULATION	CABLES IN HORIZONTAL HANGERS UNDER DECKS WITHOUT INSULATION OR IN VERTICAL HANGERS NEXT TO BULKHEADS WITHOUT INSULATION	CABLES IN HORIZONTAL HANGERS UNDER DECKS WITH INSULATION OR IN VERTICAL HANGERS NEXT TO BULKHEADS WITH INSULATION
1**	95	90	90	80	100	95	100	95
2**	84	80	80	70	100	95	100	95
3**	80	75	70	60	95	90	95	90

*FOR PROPULSION CABLES IN ENCLOSED DUCTS, CABLES SHALL BE RATED AT 85% OF VALUES OBTAINED BY USING THE ABOVE TABULATION.

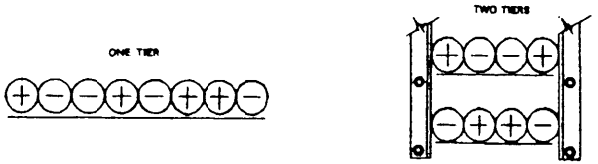
**THE ABOVE RATINGS ARE BASED ON FOUR OR MORE CABLES ON EACH HANGER. FOR LESS THAN FOUR CABLES PER HANGER AN ADDITIONAL 5% INCREASE IN THE PERCENTAGES SPECIFIED IN THE ABOVE TABLE IS PERMISSIBLE.

SH 132317227

FIGURE 4B10. Cableways for D.C. propulsion (surface ships).

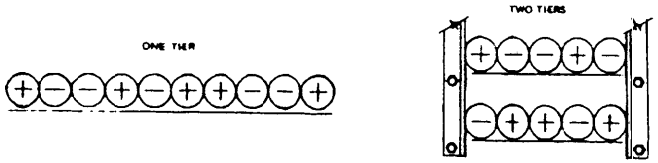
4B111

ARRANGEMENTS FOR FOUR SINGLE CONDUCTOR CABLES PER LEG



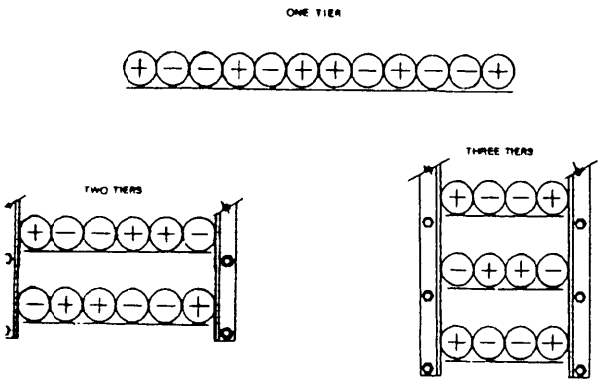
4B112

ARRANGEMENTS FOR FIVE SINGLE CONDUCTOR CABLES PER LEG



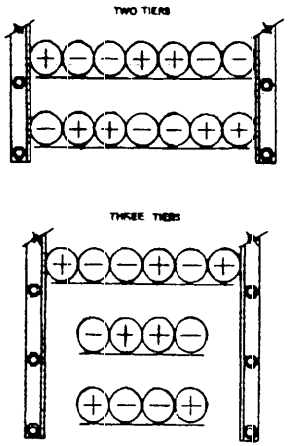
4B113

ARRANGEMENTS FOR SIX SINGLE CONDUCTOR CABLES PER LEG



4B114

ARRANGEMENTS FOR SEVEN SINGLE CONDUCTOR CABLES PER LEG



NOTES:
1. MAINTAIN A MINIMUM OF 1 IN. BETWEEN THE ROW OF CABLES AND THE ABOVE OR ADJACENT STRUCTURE.
2. THIS FIGURE SUPERSEDES SHEET 4B11 OF DRAWING 803-5001027 AND SECTION 2, SHEET 130, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

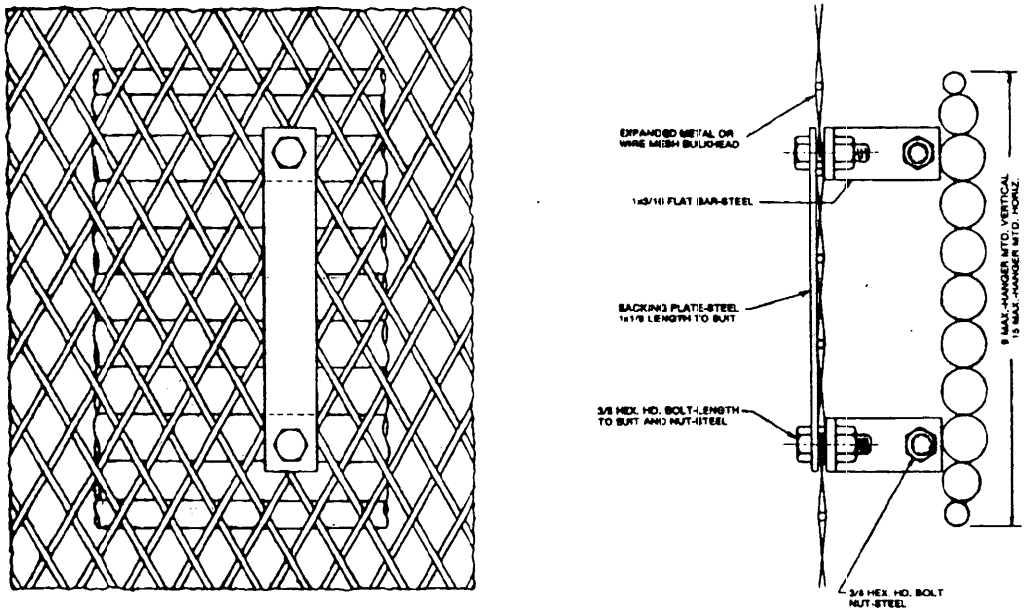
:H 132317228

FIGURE 4B11. Cableways for D.C. propulsion cables (surface ships).

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NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B12 OF DRAWING 803-5001027 AND SECTION 2, SHEET 177, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B121
EXPANDED METAL OR WIRE MESH BULKHEADS



SH 132317229

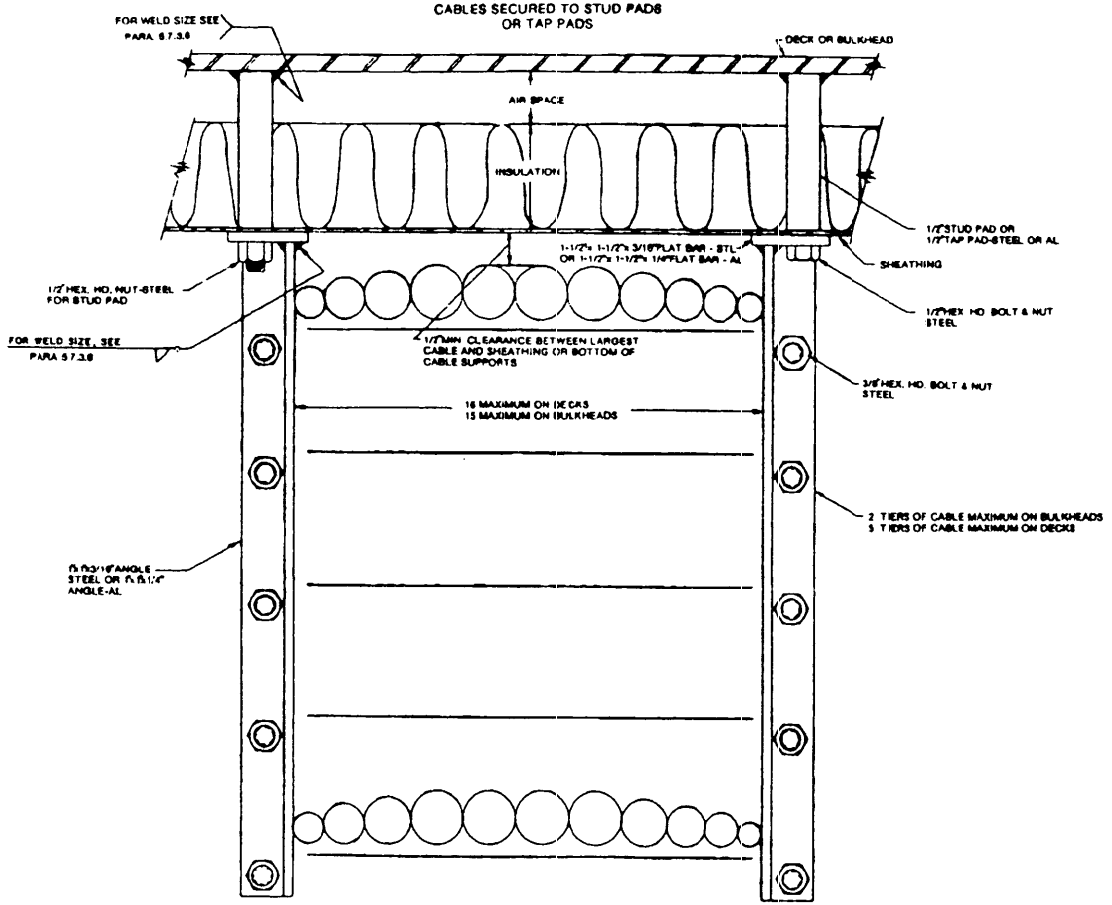
FIGURE 4B12. Single cableway on non structural bulkhead (surface ships).

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NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B13 OF DRAWING 803-5001027 AND SECTION 2, SHEET 113, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B131

WATERTIGHT DECK OR BULKHEAD WITH INSULATION AND SHEATHING CABLES SECURED TO STUD PADS OR TAP PADS



SH 132317230

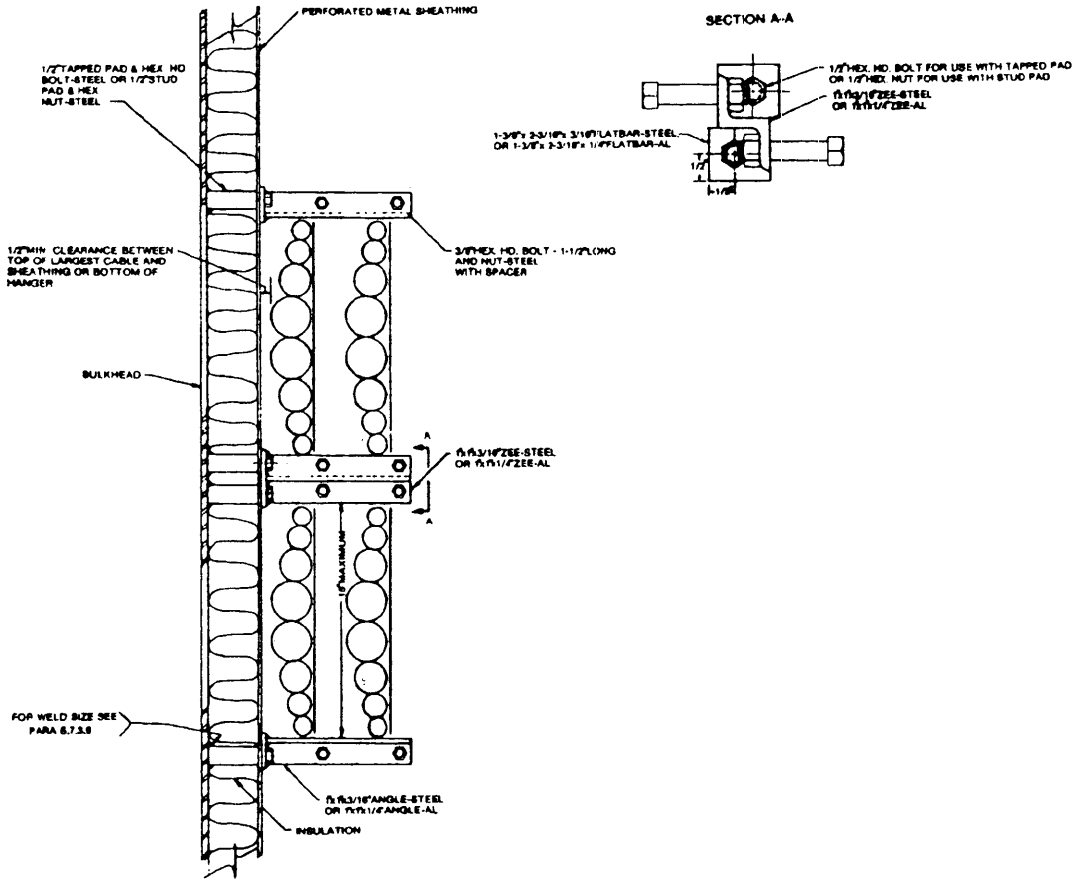
FIGURE 4B13. Cableways in acoustical locations (surface ships).

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NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B14 OF DRAWING 803-5001027 AND SECTION 2, SHEET 111, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B141

TWO BANKS OF CABLES ON WATERTIGHT BULKHEADS
WITH INSULATION AND SHEATHING
CABLE RACKS SECURED WITH BOLTS IN TAPPED PADS
OR WITH NUTS ON STUD PADS

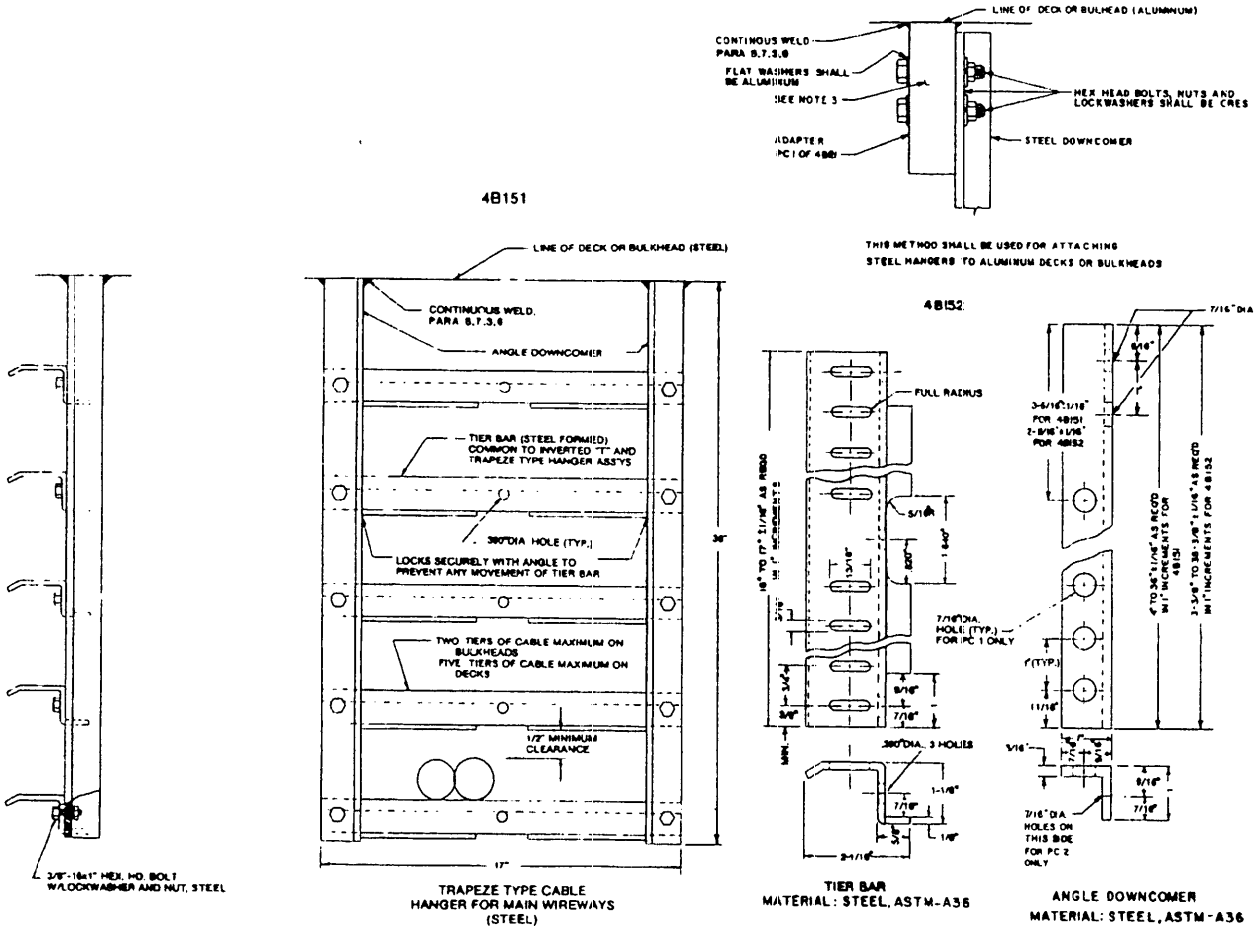


SH 132317231

FIGURE 4B14. Cableways in acoustical locations (surface ships).

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NOTES.

1. SPACING, NUMBER OF TIERS AND CABLE WEIGHT SHALL NOT EXCEED THE VALUES SPECIFIED BELOW. DOWNWARD EXTENSION TO THE LESSER CONDITION OF FEWER TIERS FOR HANGER ASSEMBLIES (WITH CABLE LOAD NOT EXCEEDING THE MAXIMUM PER TIER AS SPECIFIED BELOW) IS ACCEPTABLE

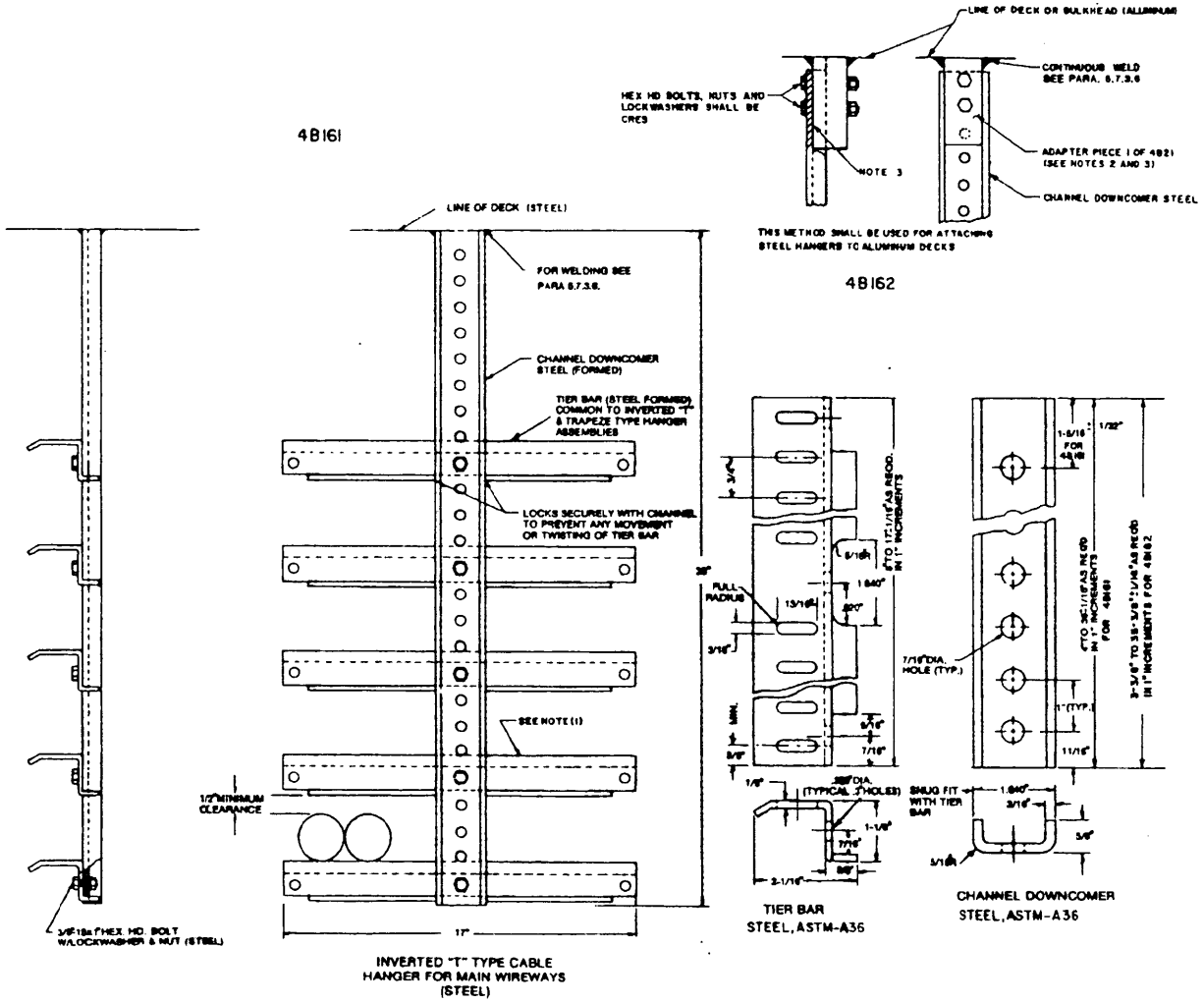
CABLE RUN	TYPE ASS'Y	CENTER TO CENTER SPACING (INCHES)	MAX NO OF TIERS	MAX CABLE WT (LBS.) SUPPORTED BY EA TIER
HORIZONTAL	DECK MTD 36"x17"	32	5	91
VERTICAL	BLK HD MTD 20"x17"	24	7	66

2. SHARP CORNERS SHALL BE SLIGHTLY ROUNDED. ALL PARTS SHALL HAVE SMOOTH SURFACES.
3. CONTACT AREAS BETWEEN ADAPTOR, DOWNCOMER AND ATTACHING BOLTS SHALL BE PAINTED WITH TWO COATS OF ZINC CHROMATE PRIMER.
4. SEE 4C22 & 4C23 FOR INSTALLATION OF BANDING STRAPS.
5. THIS FIGURE SUPERSEDES SHEET 4B16 of DRAWING 803-8001027 AND SECTION 2, SHEET 226, OF DRAWING NAVSEC NO. 8000-S82202-73980

FIGURE 4B15. Multiple cableways on steel decks and bulkheads (surface ships).

SH 132317232

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NOTES:

1. SPACING, NUMBER OF TIERS AND CABLE WEIGHT SHALL NOT EXCEED THE VALUES SPECIFIED BELOW. DOWNWARD EXTENSION TO THE LESSER CONDITION OF FEWER TIERS FOR HANGER ASSEMBLIES (WITH CABLE LOAD NOT EXCEEDING THE MAXIMUM PER TIER AS SPECIFIED BELOW) IS ACCEPTABLE.

CABLE RUN	TYPE ASSY	CENTER TO CENTER SPACING (INCHES)	MAX NO OF TIERS	MAX CABLE WT. (LBS.) SUPPORTED BY EACH TIER
HORIZONTAL	DECK MTD 36"x17"	39	5	91

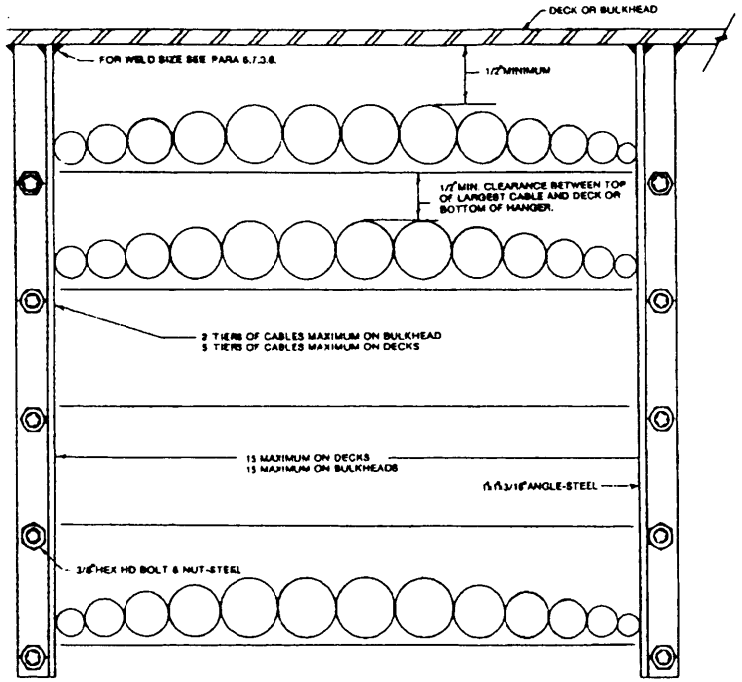
2. SHARP CORNERS SHALL BE SLIGHTLY ROUNDED. ALL PARTS SHALL HAVE SMOOTH SURFACES.
3. CONTACT AREAS BETWEEN ADAPTER DOWNCOMER AND ATTACHING BOLTS SHALL BE PAINTED WITH TWO COATS OF ZINC CHROMATE PRIMER.
4. SEE 4C22 & 4C23 FOR INSTALLATION OF BANDING STRAPS.
5. THIS FIGURE SUPERSEDES SHEET 4B16 OF DRAWING 803-6001097 AND SECTION 2, SHEET 23A, OF DRAWING NO. 8080-88202-73880.

FIGURE 4B16. Multiple cables on steel decks and bulkheads (surface ships).

NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B17 OF DWG 803-5001027 AND SECTION 2, SHEET 213 OF DRAWING NAVSEC NO. 9000-S6202-73960

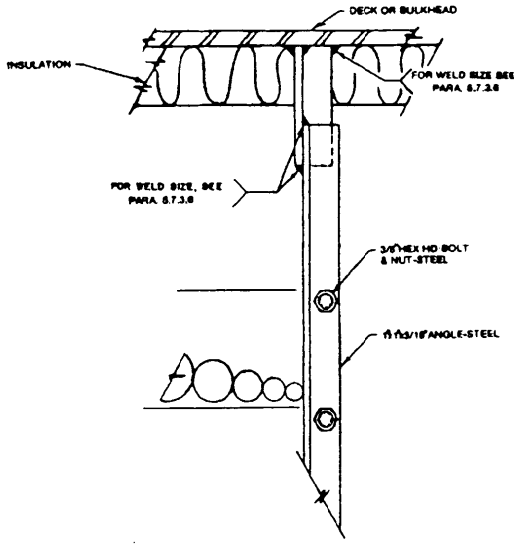
4B171

MULTIPLE TIERS OF CABLE
SUPPORTED FROM STEEL DECK OR BULKHEADS
WITHOUT INSULATION



4B172

MULTIPLE TIERS OF CABLES
SUPPORTED FROM STEEL DECKS OR BULKHEADS
WITH INSULATION



SH 132317234

FIGURE 4B17. Multiple cables on steel decks or bulkheads.

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NOTES:
1. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.8.
2. THIS FIGURE SUPERSEDES SHEET 4B18 OF DRAWING 803-5001027 AND SECTION 2, SHEET 217 OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B181
CABLES SUPPORTED
ON INTERMEDIATE SUPPORTS WITH INSULATION
BETWEEN BULKHEAD STIFFENERS

4B182
CABLES SUPPORTED
ON INTERMEDIATE SUPPORTS
BETWEEN BULKHEAD STIFFENERS

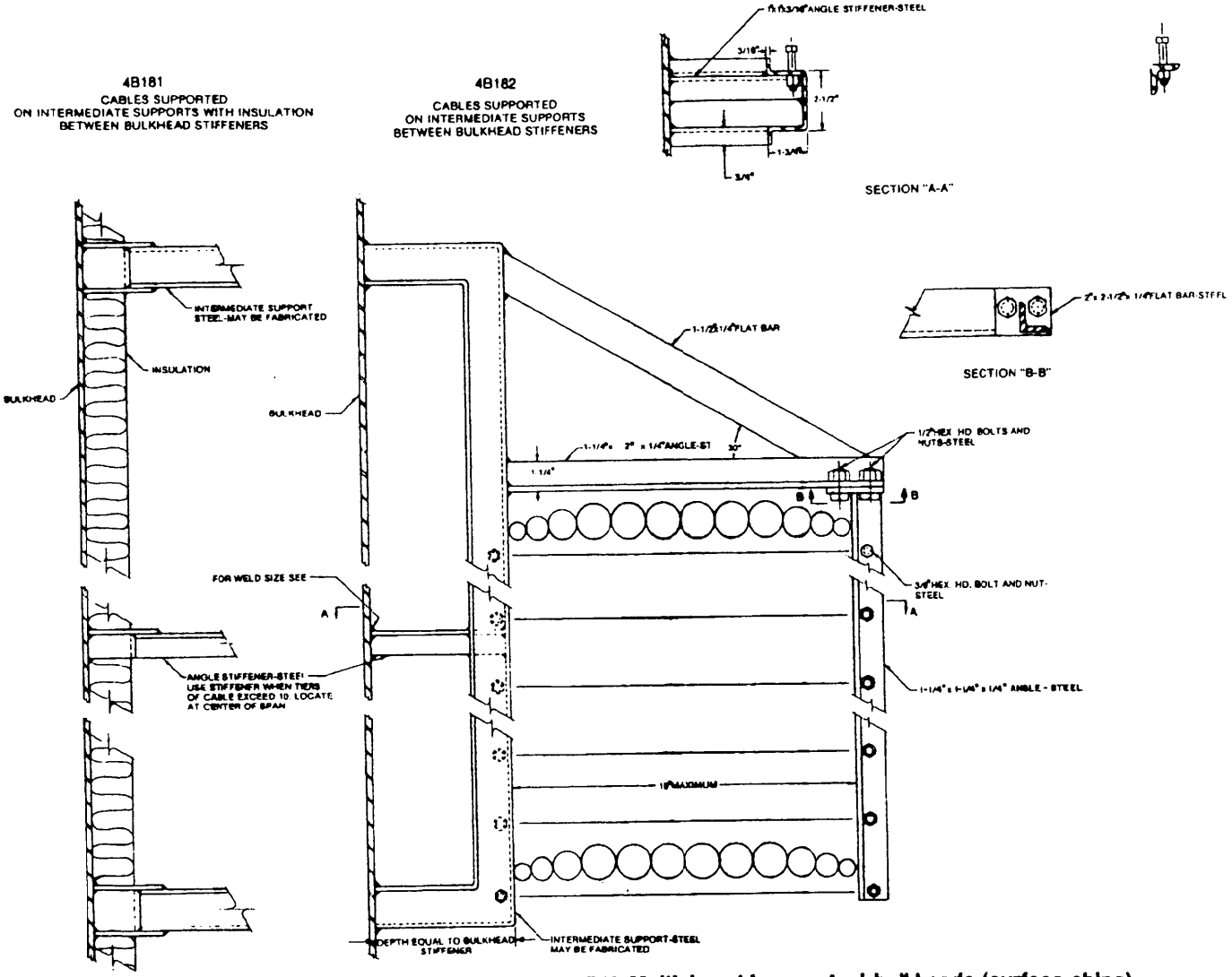
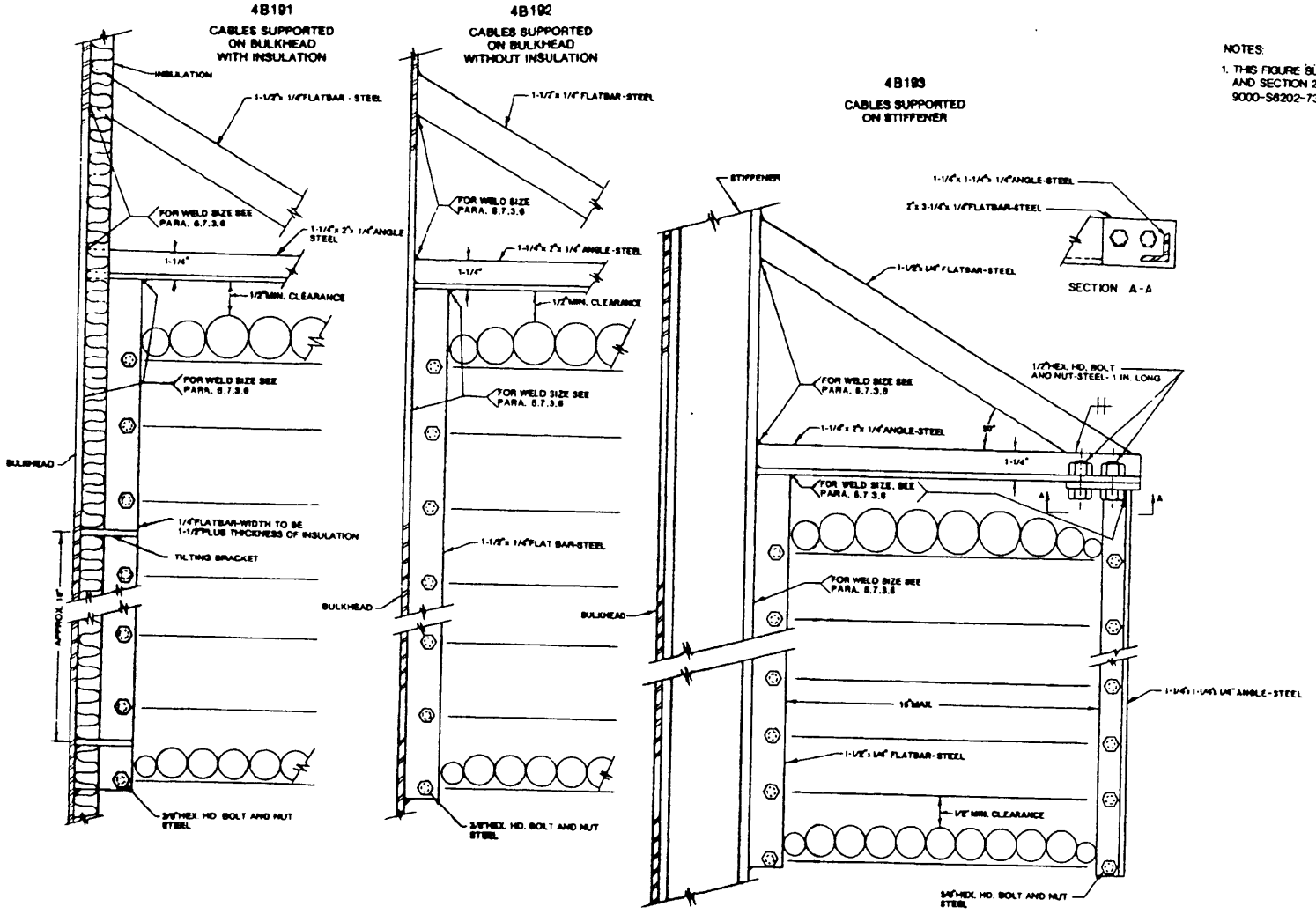


FIGURE 4B18. Multiple cables on steel bulkheads (surface ships).

SH 132317235

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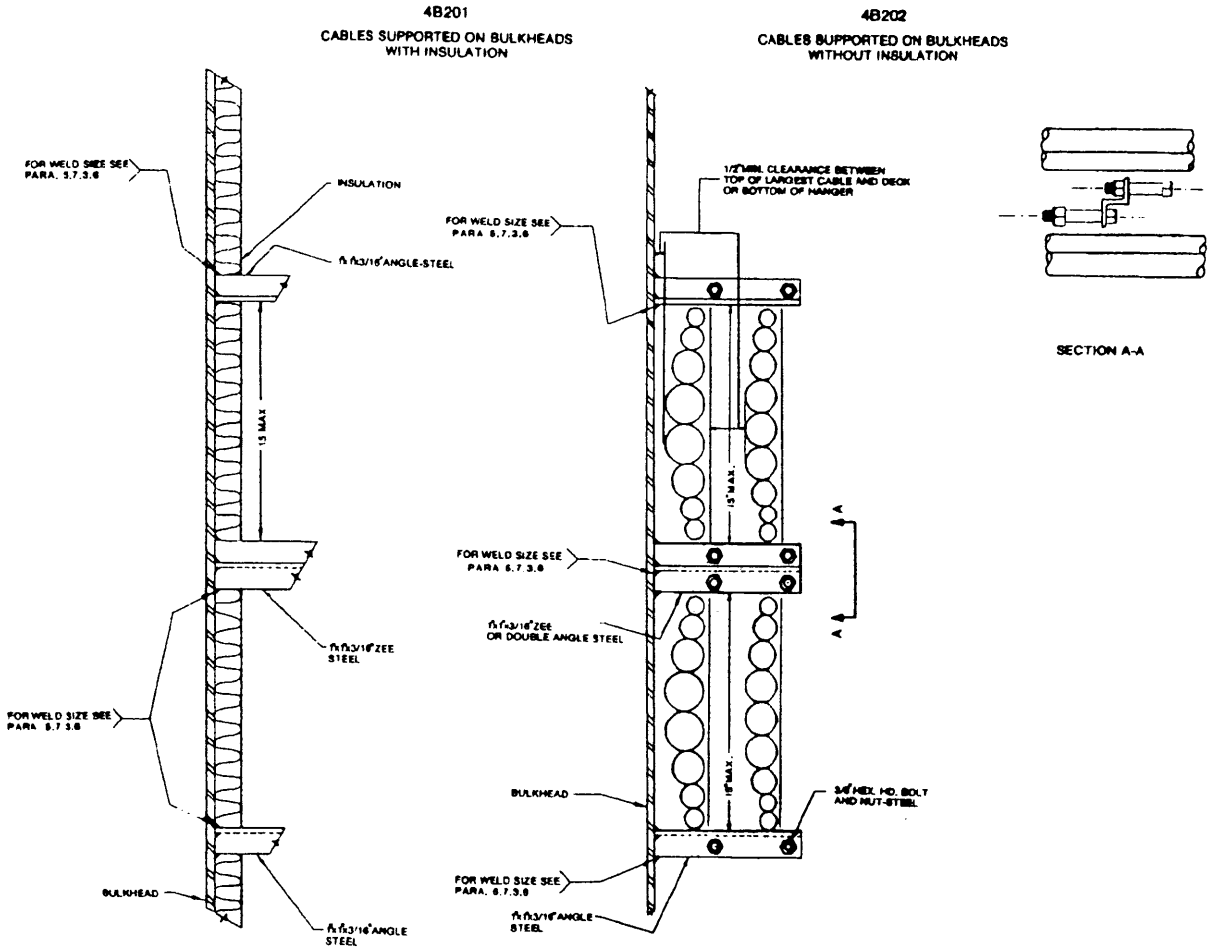
SH 132317236

NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B19 OF DRAWING 803-5001027 AND SECTION 2, SHEET 216, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

FIGURE 4B19. Multiple cables on steel bulkheads (surface ships).

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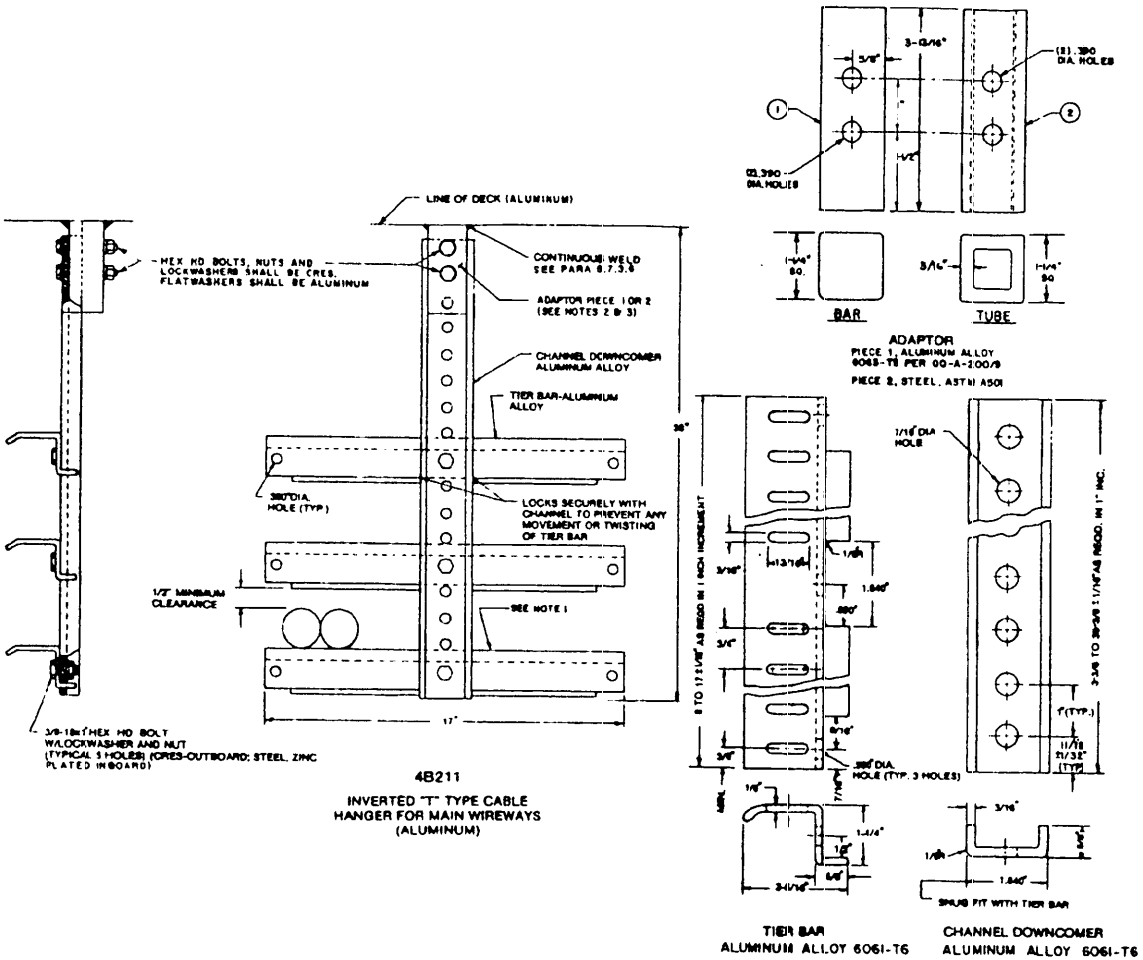
SH 132317237

NOTES:

1. THREE ROWS MAY BE USED IN CONGESTED LOCATIONS ON REQUEST AND SUBJECT TO SPECIFIC APPROVAL.
2. THIS FIGURE SUPERSEDES SHEET 4B20 OF DRAWING 803-5001027 AND SECTION 2, SHEET 214, OF DRAWING NAVSEC NO. 9000-S6202-73980.

FIGURE 4B20. Multiple cables on steel bulkheads (surface ships).

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4B211

INVERTED "T" TYPE CABLE HANGER FOR MAIN WIREWAYS (ALUMINUM)

NOTES:

1. SPACING, NUMBER OF TIERS, AND CABLE WEIGHT SHALL NOT EXCEED THE VALUES SPECIFIED BELOW. DOWNWARD EXTENSION TO THE LESSER CONDITION OF FEWER TIERS FOR HANGER ASSY'S (WITH CABLE LOAD NOT EXCEEDING THE MAXIMUM PER TIER AS SPECIFIED BELOW) IS ACCEPTABLE.

CABLE RUN	TYPE ASSY.	CENTER TO CENTER SPACING (INCHES)	MAX. NO. OF TIERS	MAX. CABLE WT. (LBS.) SUPPORTED BY EACH TIER
HORIZONTAL	DECK MTD. 36"x17"	16	3	45.5

2. ADAPTOR, PC 1 USED TO MAINTAIN SIMILAR METAL COMPATABILITY FOR WELDING TO SHIPS ALUM. STRUCTURE
3. FOR MOUNTING ALUMINUM HANGERS ON STEEL DECKS AND BULKHEADS, USE ADAPTOR PC 2 STEEL ASTM A501. CONTACT AREAS BETWEEN ADAPTOR, DOWNCOMER AND ATTACHING BOLTS SHALL BE PAINTED WITH TWO COATS OF ZINC CHROMATE PRIMER
4. SHARP CORNERS SHALL BE SLIGHTLY ROUNDED. ALL PARTS SHALL HAVE SMOOTH SURFACES.
5. DELETED
6. FOR SPACING OF HANGERS SEE FIGURE 4B55.
7. SEE 4C22 & 4C23 FOR INSTALLATION OF BANDING STRAPS.
8. THIS FIGURE SUPERSEDES SHEET 4B21 OF DRAWING 803-5001027 AND SECTION 2, SHEET 223, OF DRAWING NAVSEC NO. 9000-S6202-73980.

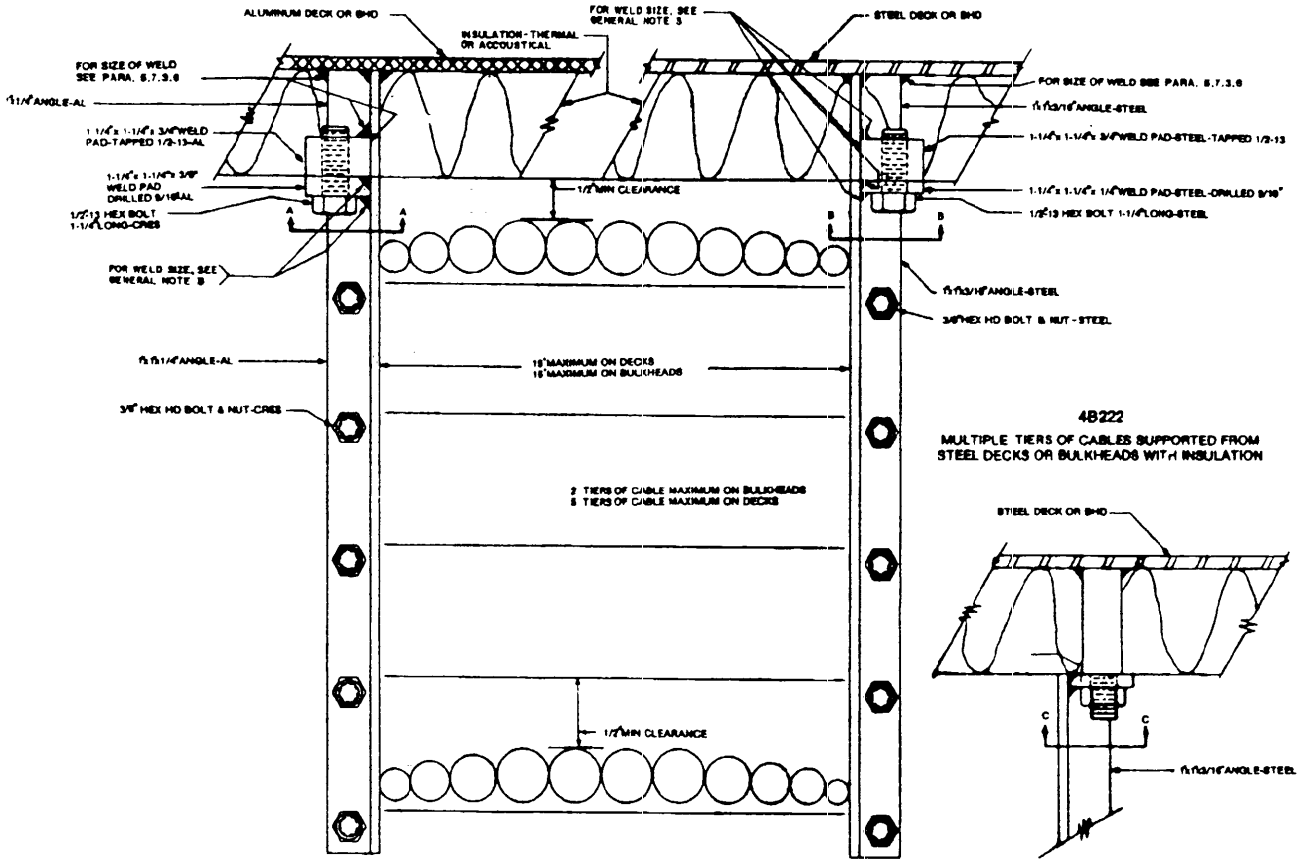
SH 132317238

FIGURE 4B21. Multiple cableways support from aluminum or steel decks & bulkheads (surface ships).

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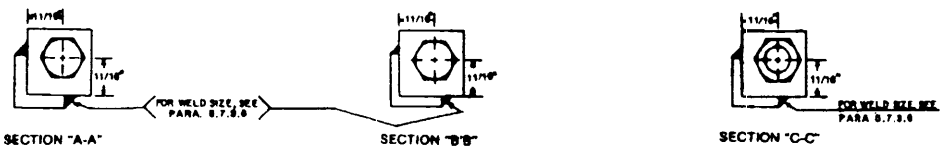
4B221

MULTIPLE TIERS OF CABLES SUPPORTED FROM STEEL AND ALUMINUM DECKS OR BULKHEADS WITH INSULATION



4B222

MULTIPLE TIERS OF CABLES SUPPORTED FROM STEEL DECKS OR BULKHEADS WITH INSULATION



NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B22 OF DRAWING 803-6001027 AND SECTION 2, SHEET 215, OF DRAWING, NAVSEC NO. 9000-56202-7398D.

SH 132317239

FIGURE 4B22. Multiple cableways on insulated steel or aluminum decks or bulkheads (surface ships).

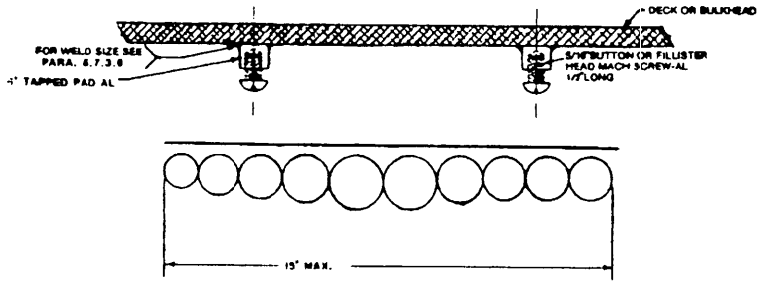
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Check the source to verify that this is the current version before use.

NOTES:

1. METHODS SHOWN ON THIS FIGURE MAY BE USED IN LIEU OF THOSE SHOWN ON FIGURE 4B24.
2. THIS FIGURE SUPERSEDES SHEET 4B23 OF DRAWING 803-5001027 AND SECTION 2, SHEET 201, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

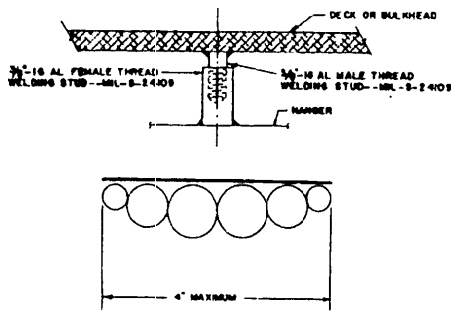
4B231

SINGLE TIER SECURED WITH TAPPED PADS ON ALUMINUM DECKS AND BULKHEADS



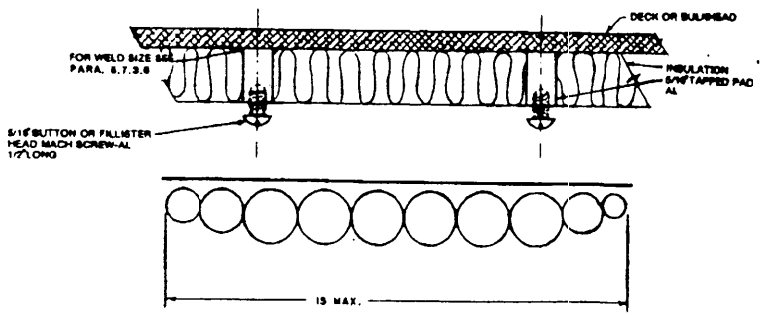
4B233

SINGLE TIER SECURED WITH A SINGLE WELD STUD ON ALUMINUM DECKS AND BULKHEADS MAXIMUM CABLE WEIGHT 3 POUNDS PER SUPPORT



4B232

SINGLE TIER SECURED WITH TAPPED PADS ON INSULATED ALUMINUM DECKS AND BULKHEADS



SH 132317240

FIGURE 4B23. Single cableway or aluminum structures (surface ships).

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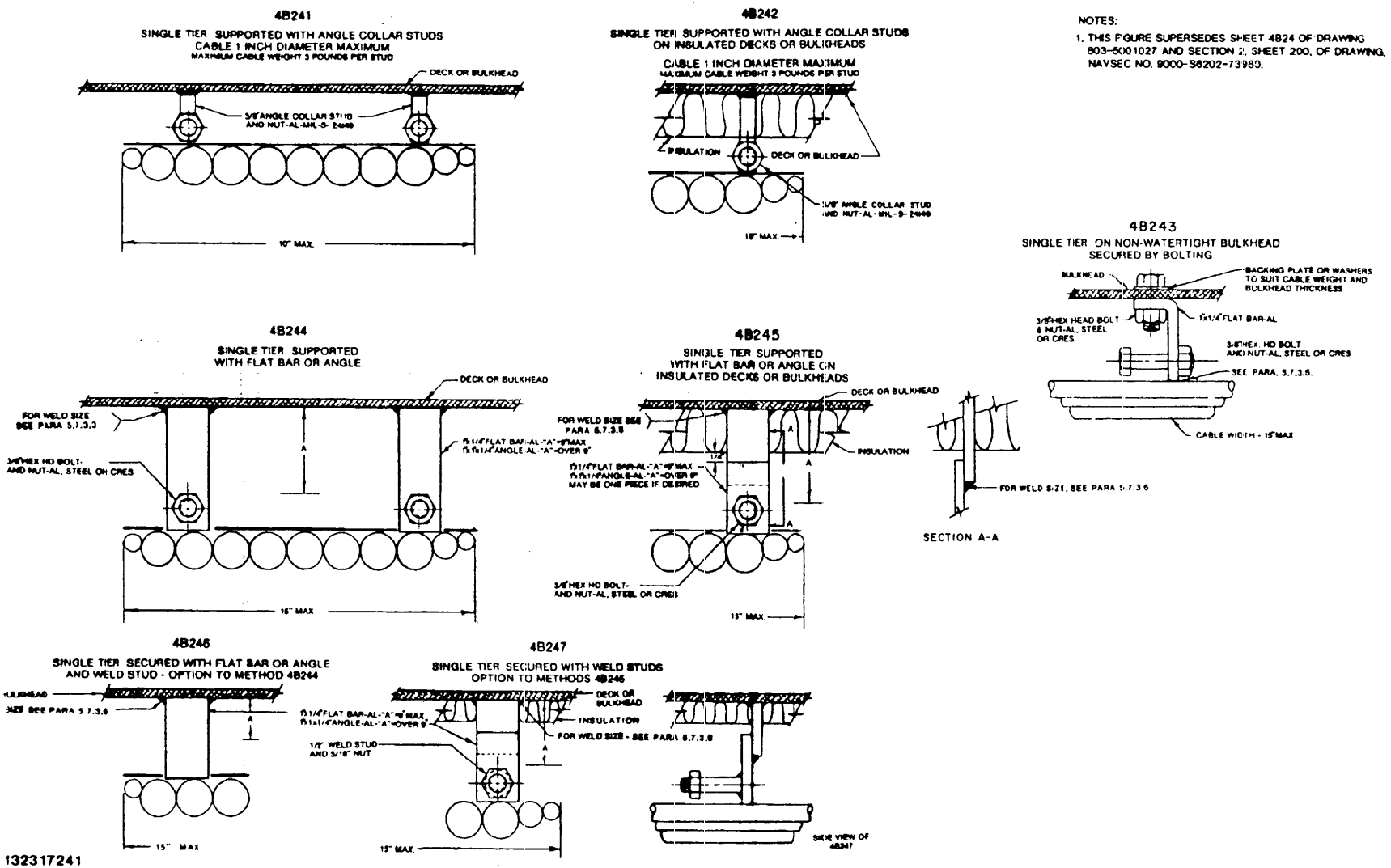


FIGURE 4B24. Single cableway on aluminum structure (surface ships).

NOTES:

1. THE NUMBER OF HANGERS IN GROUPS BETWEEN LONGITUDINALS SHALL BE DETERMINED ON A BASIS OF THE NUMBER OF CABLES WHICHEHICH CAN BE SUITABLY CARRIED THROUGH LIGHTENING HOLES IN THE TRANSVERSE WEBS.
2. THIS FIGURE SUPERSEDES SHEET 4B25 OF DRAWING 803-5001027 AND SECTION 2, SHEET 183, OF DRAWING NAVSEC NO. 9000-S6202-73980

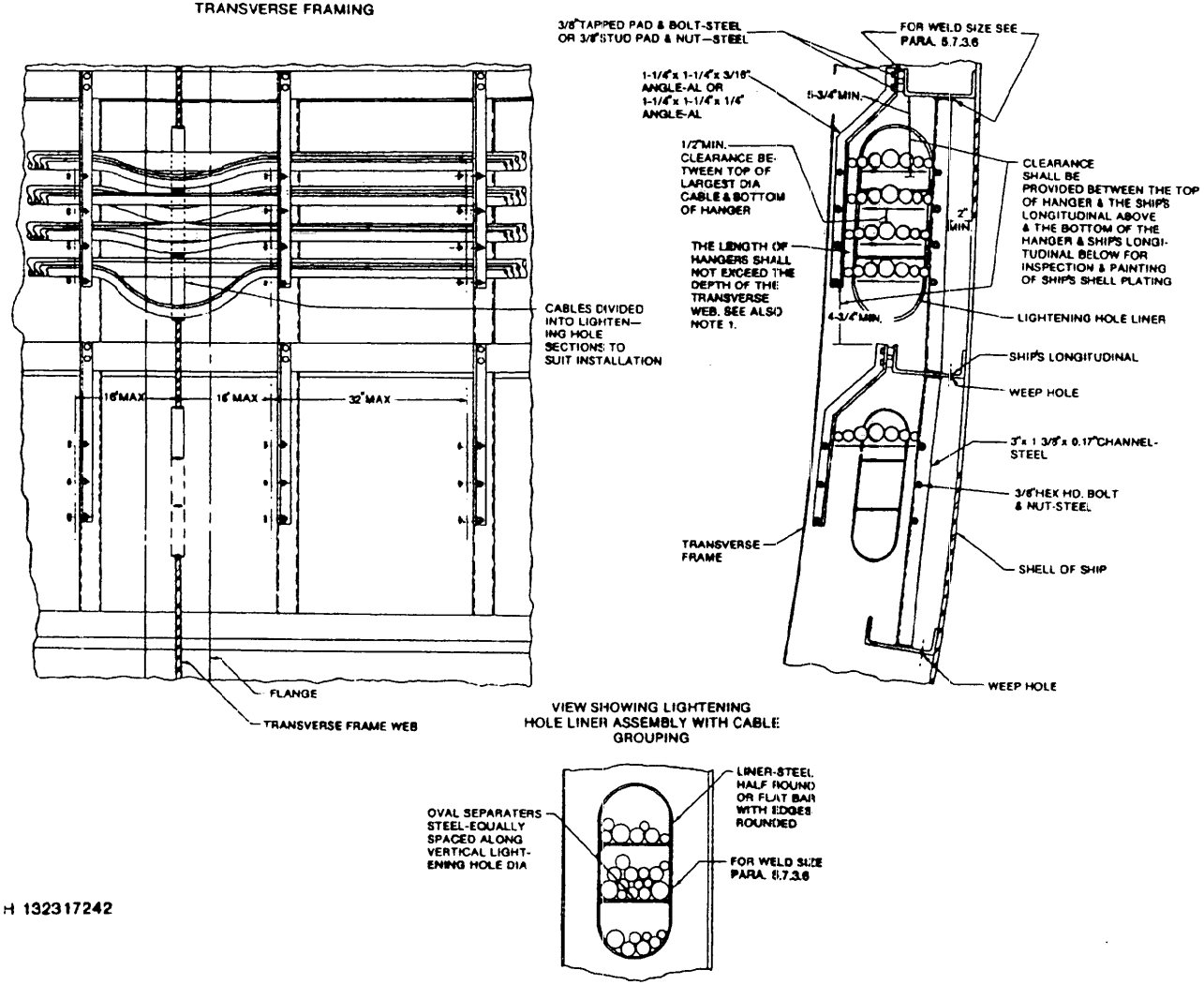


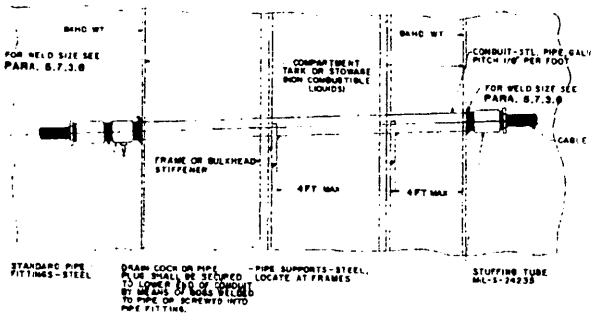
FIGURE 4B25. Multiple cableways in machinery spaces (surface ships).

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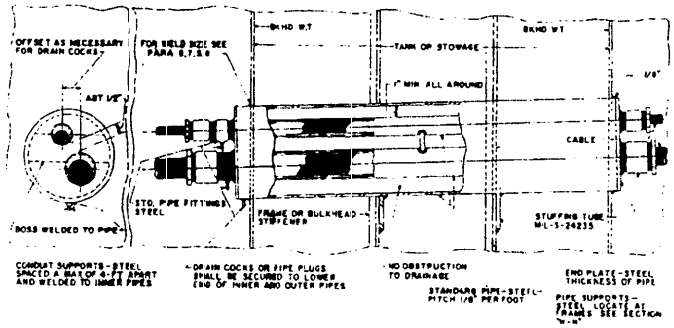
4B261

CABLES THROUGH CONDUIT
WATERTIGHT INSTALLATION
NON-HAZARDOUS LOCATION



4B263

CABLES THROUGH HAZARDOUS LOCATIONS
CONTAINING COMBUSTIBLE LIQUIDS
WATERTIGHT INSTALLATIONS
TWO OR MORE CABLES

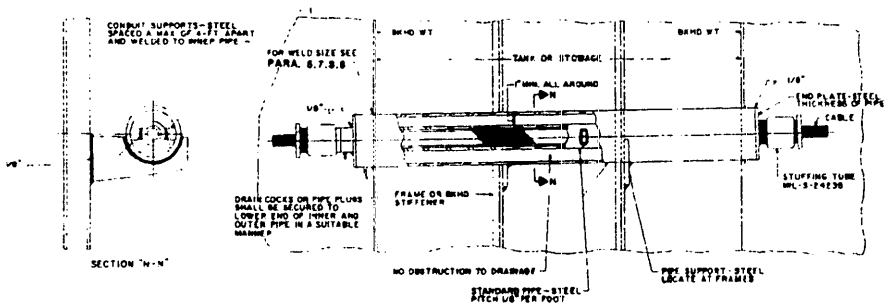


NOTES:

1. UP TO 3 CABLES MAY BE USED IN THE INNER PIPE, PROVIDING NO MORE THAN TWO SIMULTANEOUSLY LOADED CABLES SHALL BE CONTIGUOUS TO EACH OTHER. THIS IS AN ALTERNATIVE METHOD AND WILL REQUIRE AN END PLATE ON THE PROTRUDING INNER PIPE WITH THE RISER TOPS TO BE WELDED DIRECTLY TO THE END PLATE.
2. THE STANDARD PIPE FOR METHODS 4B262 AND 4B263 SHALL BE CLEAN AND FREE OF ALL FOREIGN CONTAMINATION.
3. THIS FIGURE SUPERSEDES SHEET 4B26 OF DWG. 803-5001027 AND SECTION 2, SHEET 155, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B262

CABLES THROUGH HAZARDOUS LOCATIONS
CONTAINING COMBUSTIBLE LIQUIDS
WATERTIGHT INSTALLATIONS
A SINGLE CABLE



SH 132317243

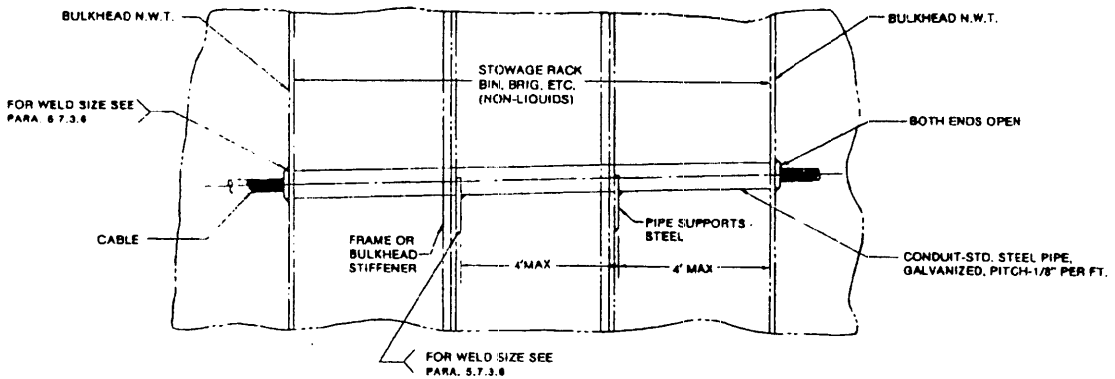
FIGURE 4B26. Cableways through storage spaces & tanks.

4B271

CABLES THROUGH CONDUIT
NON-WATERTIGHT INSTALLATIONS
NON-HAZARDOUS LOCATIONS

NOTE:

1. THIS FIGURE SUPERSEDES SHEET 4B27 OF DWG. 803-5001027 AND SECTION 2, SHEET 154, OF DRAWING, NAVSEC NO. 9000-S6202-73980.



SH 132317244

FIGURE 4B27. Cableways through storage spaces & tanks (surface ships).

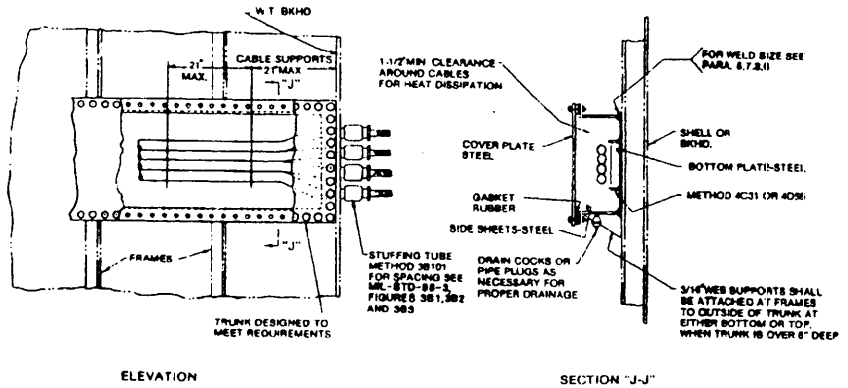
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. METHOD 4B28.1 MAY BE USED FOR FIRE PROTECTION.
2. THIS FIGURE SUPERSEDES SHEET 4B26 OF DRAWING 803-5001027 AND SECTION 2, SHEET 153, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

4B28.1

CABLES IN TRUNKS W.T. OR O.T

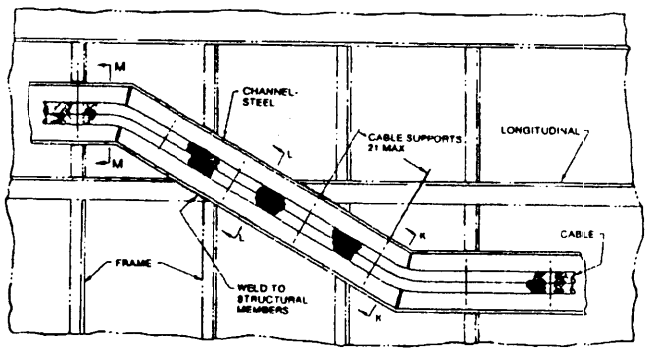


ELEVATION

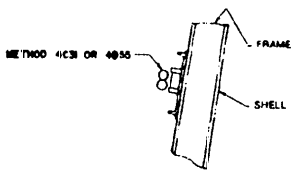
SECTION "J-J"

4B28.2

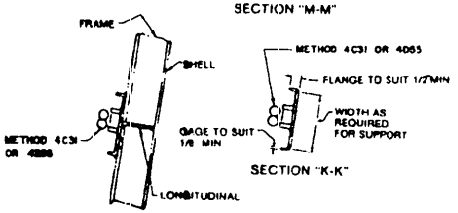
CABLES IN CHANNEL



ELEVATION - LOOKING OUTBOARD



SECTION "M-M"



SECTION "K-K"

SECTION "L-L"

SH 132317245

FIGURE 4B28. Cableways through trunks and channels (surface ships).

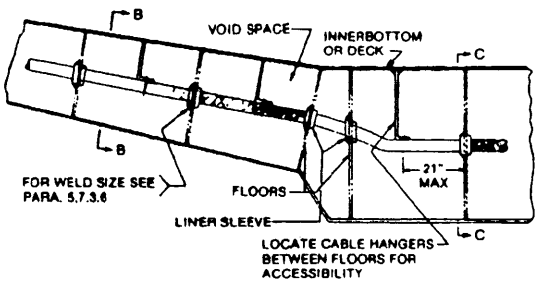
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Check the source to verify that this is the current version before use.

NOTES:

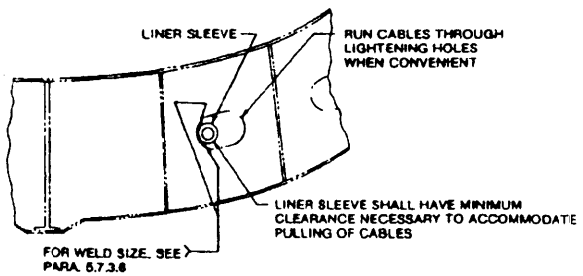
1. A MAX. OF 3 CABLES MAY BE USED IN THE INNER PIPE, PROVIDING NO MORE THAN TWO SIMULTANEOUSLY LOADED CABLES SHALL BE CONTIGUOUS TO EACH OTHER. THIS IS AN ALTERNATIVE METHOD AND WILL REQUIRE AN END PLATE ON THE PROTRUDING INNER PIPE WITH THE RISER TOPS TO BE WELDED DIRECTLY TO THE END PLATE.
2. THIS FIGURE SUPERSEDES SHEET 4B29 OF DRAWING 803-5001027 AND SECTION 2, SHEET 152, OF DRAWING NAVSEC NO. 9000-86202-73960.

4B291
CABLES THROUGH VOID SPACES
NOT SUBJECT TO FLOODING



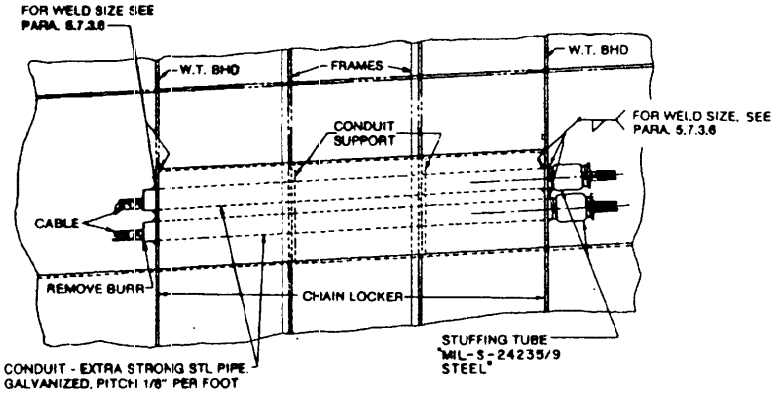


SECTION B-B

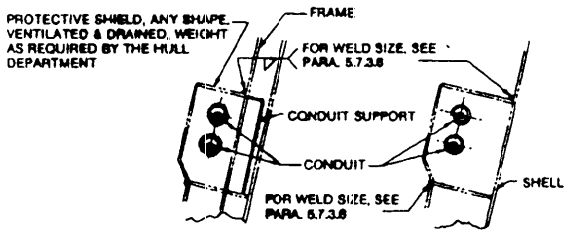


SECTION C-C

4B292
CABLES THROUGH CHAIN LOCKERS



ELEVATION LOOKING OUTBOARD



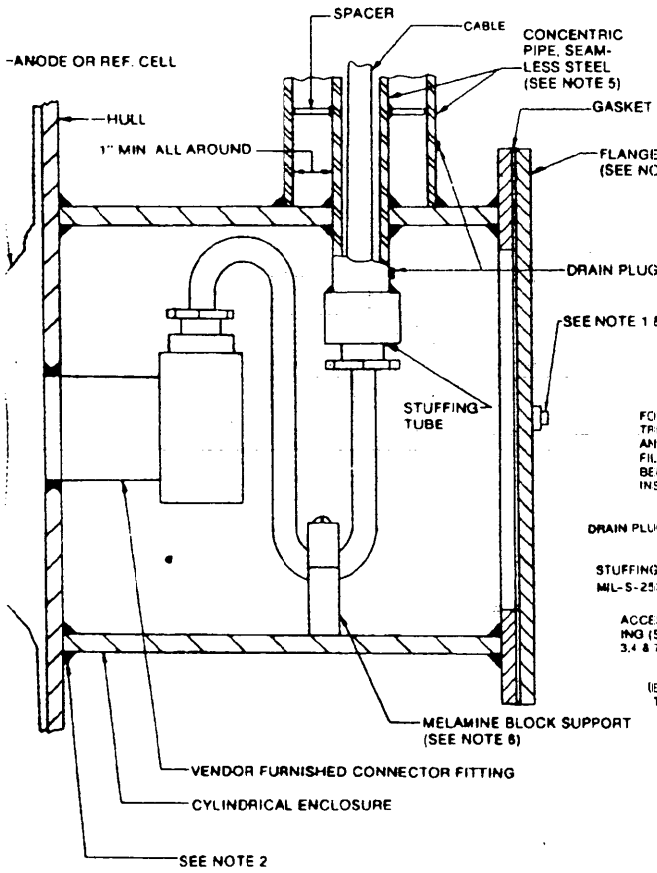
SECTION AT FRAME

SECTION BETWEEN FRAME

FIGURE 4B29. Cables through void spaces and chain lockers (surface ships).

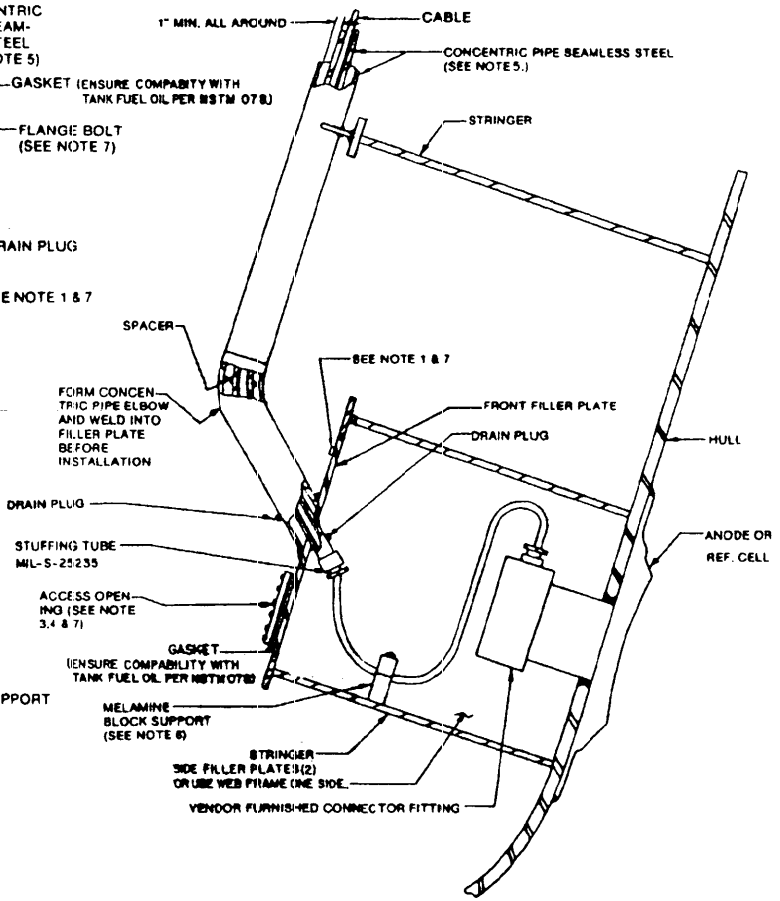
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Check the source to verify that this is the current version before use.

4B301
CABLE THROUGH CONDUIT IN FUEL OIL TANKS
WATERTIGHT INSTALLATION-SINGLE CABLE



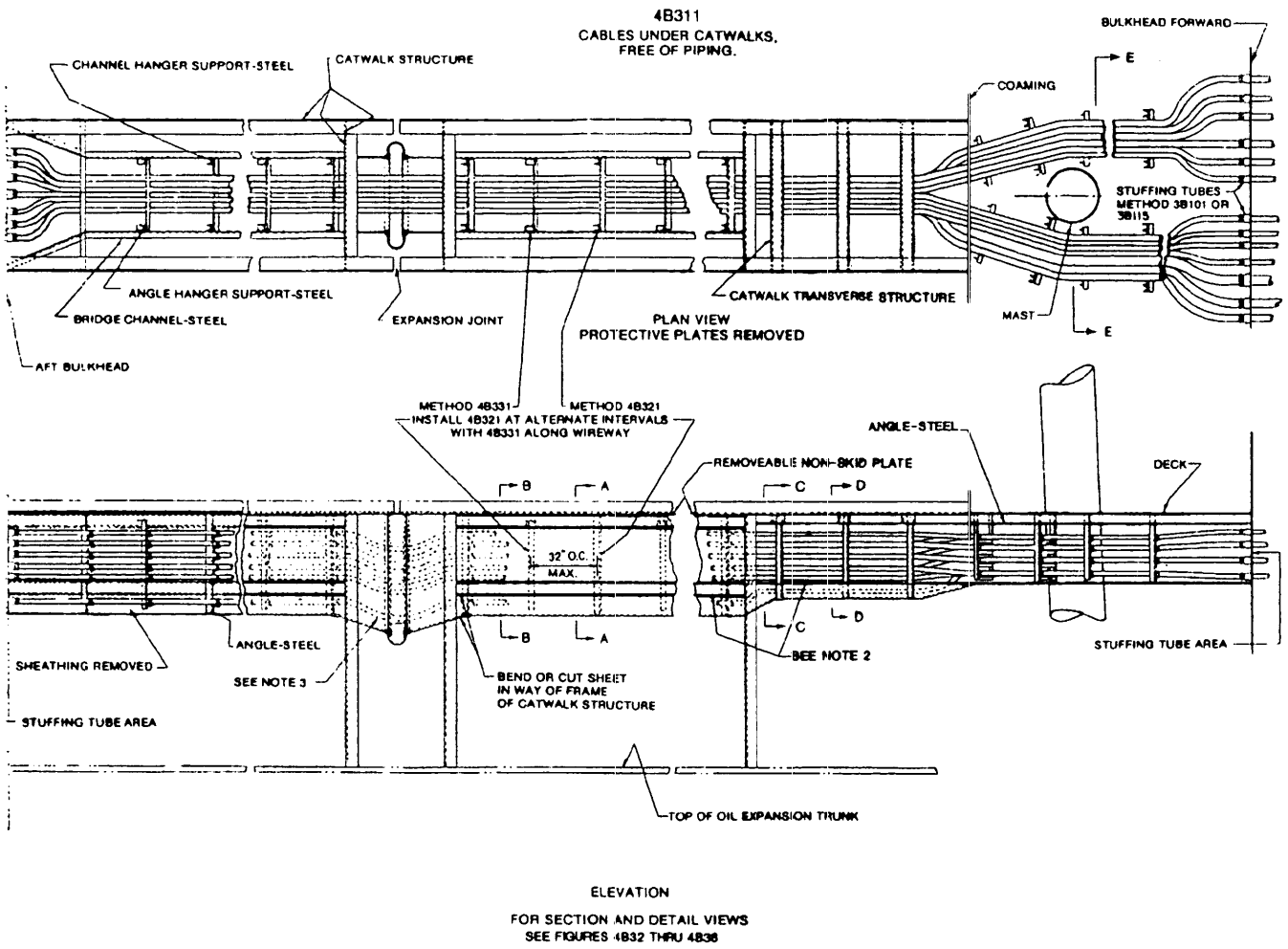
SH 132317247

4B302
ALTERNATE ENCLOSURE METHOD
(USING STRINGERS OF LONGITUDINALS)



NOTES:
1. INSTALL 3/4 INCH PIPE PLUG AND BOSS IN ENCLOSURE FOR PRESSURE TEST.
2. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.8 AND SHALL BE TESTED OIL TIGHT.
3. ALTERNATE LOCATION FOR ACCESS OPENING WILL BE ON SIDE FILLER PLATE REQUIRING A HEAVIER PLATE.
4. ACCESS COVER 2" GREATER THAN ACCESS OPENING ALL AROUND.
5. THE INNER CONCENTRIC PIPE SHALL BE GALVANIZED. THE OUTER CONCENTRIC PIPE SHALL BE PAINTED TO CONFORM TO THE TANK PAINT SYSTEM.
6. 1/8 INCH FIBER SHEET MAY BE CEMENTED IN PLACE TO PREVENT CABLE CHAFING IN LIEU OF MELAMINE SUPPORT.
7. ALL BOLTS, NUTS, WASHERS, AND PLUGS SHALL BE OF A CORROSIVE RESISTANT MATERIAL.
8. THIS FIGURE SUPERSEDES SHEET 4B30 OF DRAWING 803-5001027 AND SECTION 2, SHEET 151, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

FIGURE 4B30. Cableways through fuel oil tanks for cathodic protection system (surface ships).



SH 132317248

NOTES:

1. ALL MILD STEEL MATERIAL USED FOR SUPPORTING AND PROTECTING CABLES UNDER CATWALK, SHALL BE ZINC COATED AFTER DRILLING AND PRIOR TO INSTALLATION THE COMPLETE INSTALLATION SHALL BE PAINTED.
2. BOLTS USED TO SECURE PROTECTIVE SHEATH SHALL BE SPACED 6 INCHES O.C. APPROX. SECTIONS OF PROTECTIVE SHEATHING SHALL BE 5 LB. PLATE AND NOT EXCEED APPROX 64 INCHES IN LENGTH (SIDE AND BOTTOM OF WIREWAY)
3. PROVIDE SUFFICIENT SLACK IN CABLES FOR WORKING OF EXPANSION JOINT. ALLOW 2" MIN. CLEARANCE BETWEEN CABLES AND BOTTOM FRAMING.
4. THIS FIGURE SUPERISEDES SHEET 4B31 OF DWG. 803-5001027 AND SECTION 2, SHEET 158, OF DRAWING NAVSEC NO. 9000-S0202-73980.

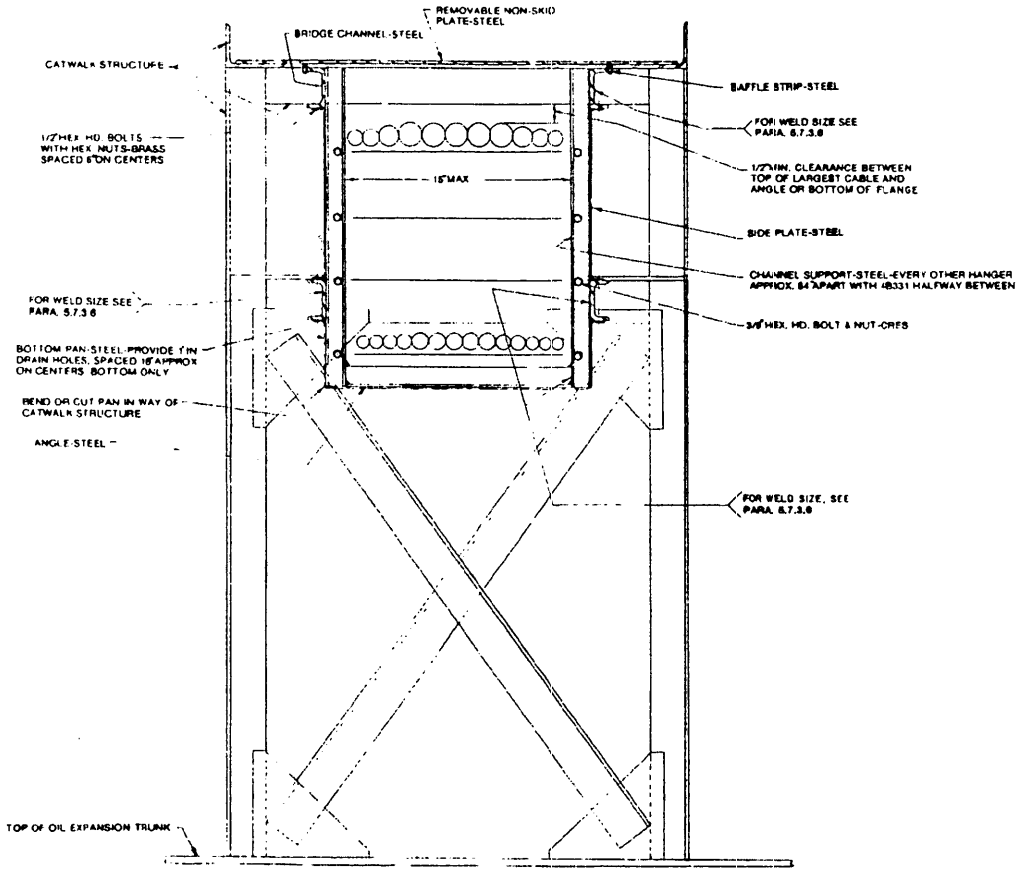
FIGURE 4B31. Topside cableways on tankers.

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Check the source to verify that this is the current version before use.

4B321

VIEW SHOWING SUPPORT OF CABLES AND SHEATHING
USING SQUARE TUBE HANGERS
ATTACHED TO BRIDGE CHANNELS



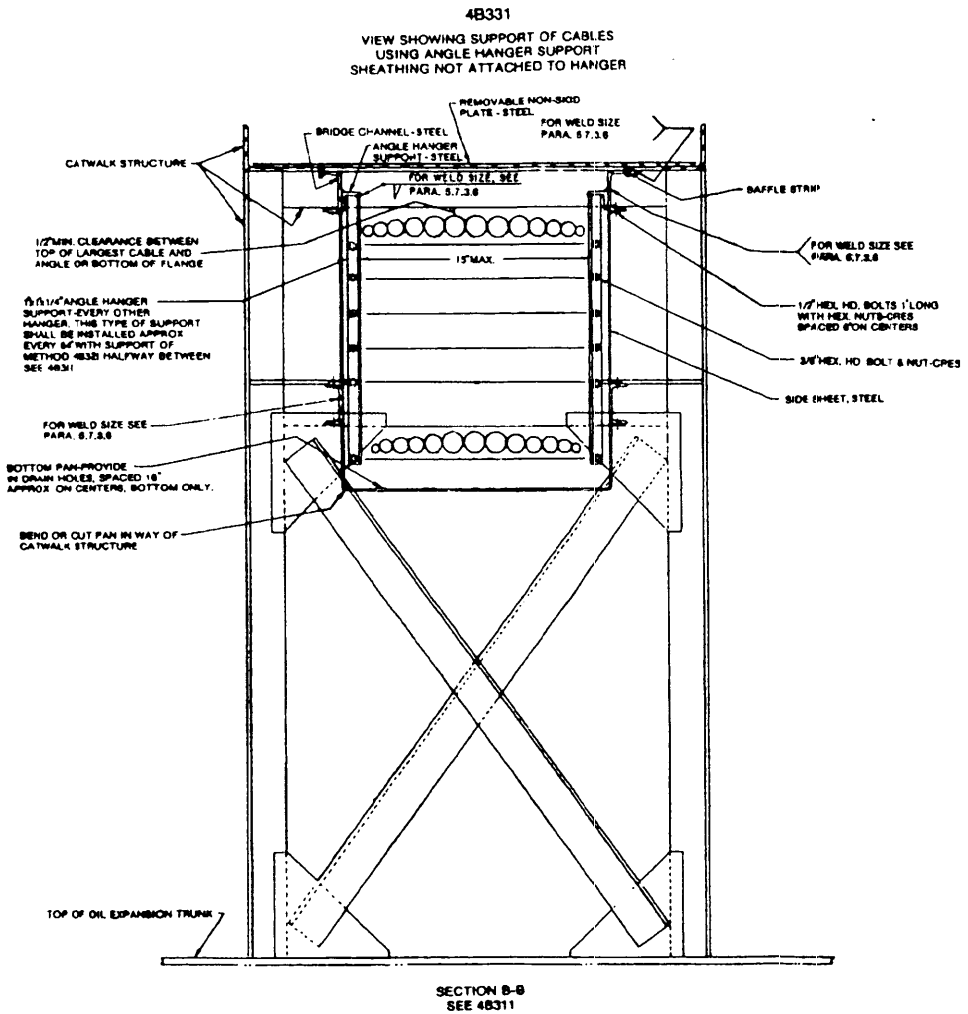
SECTION A-A
SEE 4B311

SH 132317249

NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B32 OF DRAWING 803I-5001027 AND SECTION 2, SHEET 159, OF DRAWING, NAVSEC NO. 9000-S6202-/3980.

FIGURE 4B32. Topside cableways on tankers.



SH 132317250

NOTES:

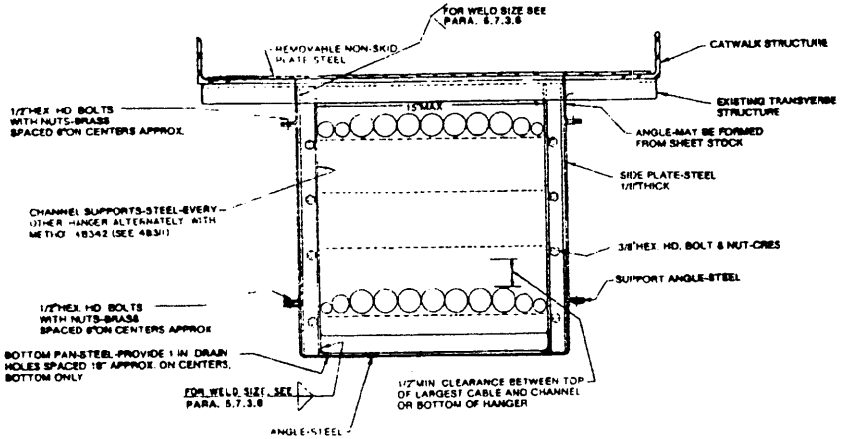
1. THIS FIGURE SUPERSEDES SHEET 4B33 OF DRAWING 803-5001027 AND SECTION 2, SHEET 160, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

FIGURE 4B33. Topside cableways on tankers.

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4B341

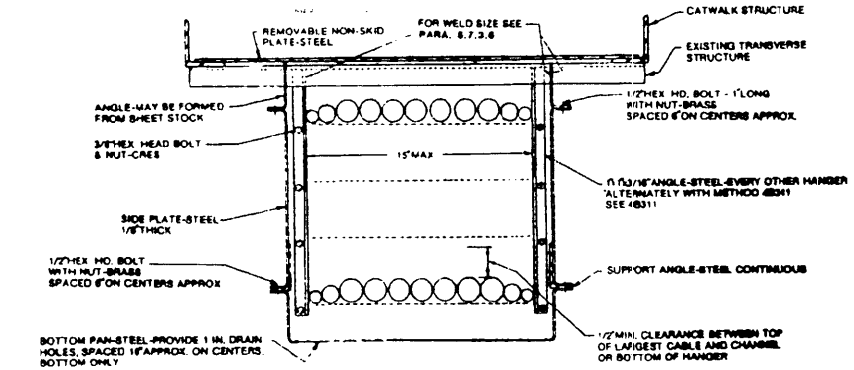
VIEW SHOWING CHANNEL HANGER SUPPORT
SUSPENDED FROM
CATWALK TRANSVERSE STRUCTURE



SECTION C-C
SEE 4B311

4B342

VIEW SHOWING ANGLE HANGER SUPPORT
SUSPENDED FROM
CATWALK TRANSVERSE STRUCTURE



SECTION D-D
SEE 4B311

NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B34 OF DWG. 803-6001027 AND SECTION 7, SHEET 181 OF DRAWING. NAVSEC NO. 6000-S6202-73960

FIGURE 4B34. Topside cableways on tankers.

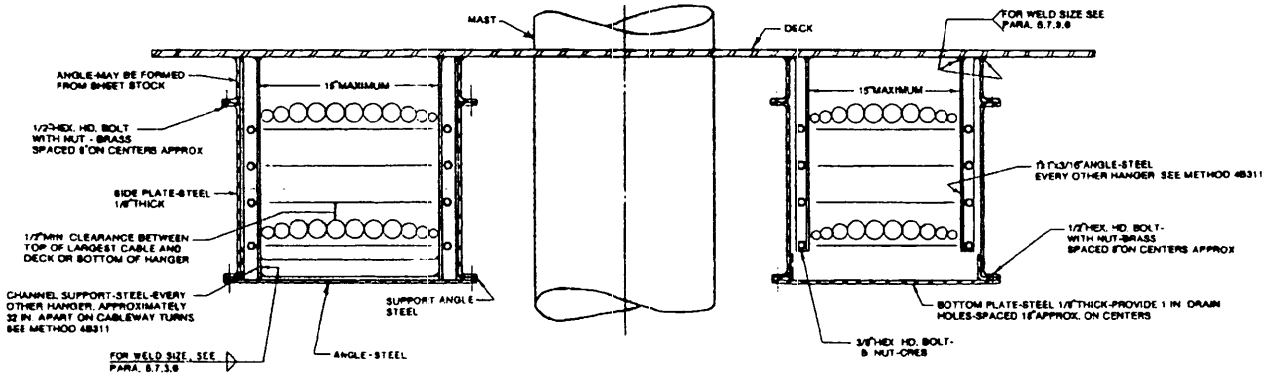
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B35 OF DRAWING 803-5001027 AND SECTION 2, SHEET 162, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4E351

VIEW SHOWING CABLEWAY DIVIDED TO PASS AN OBSTRUCTION



SECTION E-E
SEE 4B311

SH 132317252

FIGURE 4B35 Topside cableways on tankers.

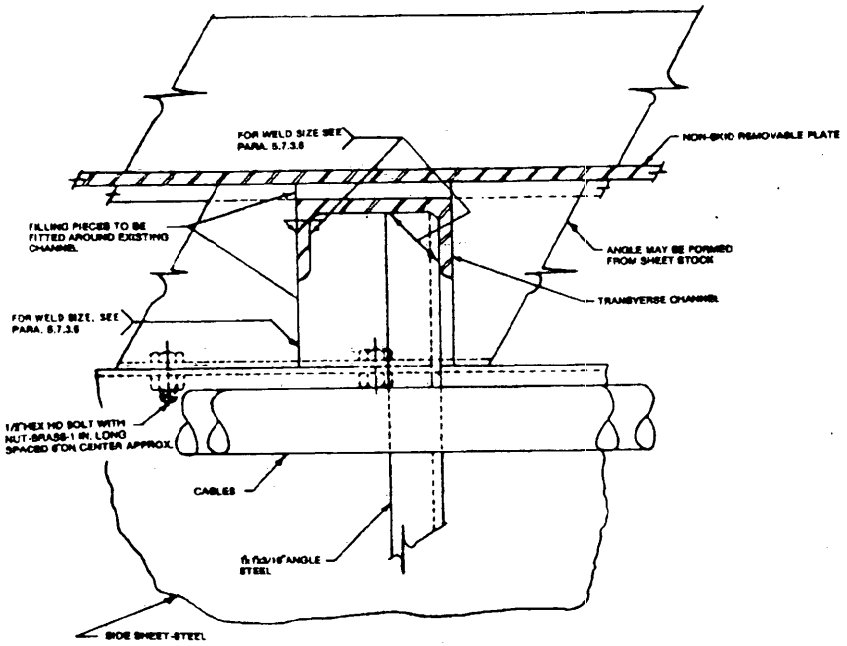
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B36 OF DRAWING 803-5001027 AND SECTION 2, SHEET 163, OF DRAWING NAVSEC NO. 9000-S6202-73980.

4B361

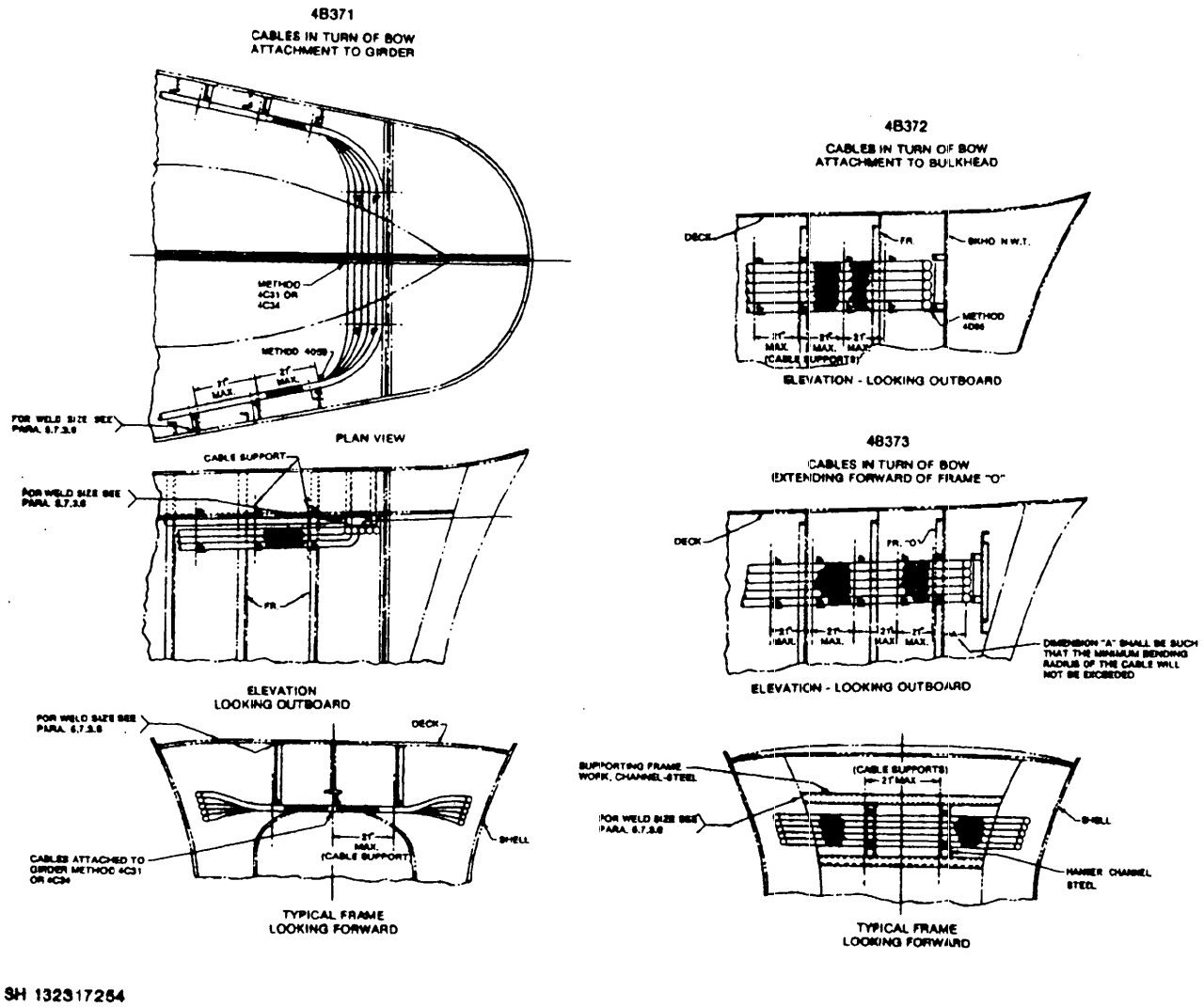
VIEW SHOWING
HANGER SUPPORT
ATTACHED TO TRANSVERSE MEMBER
SEE METHOD 4B311



SH 132317253

FIGURE 4B36. Topside cableways on tankers.

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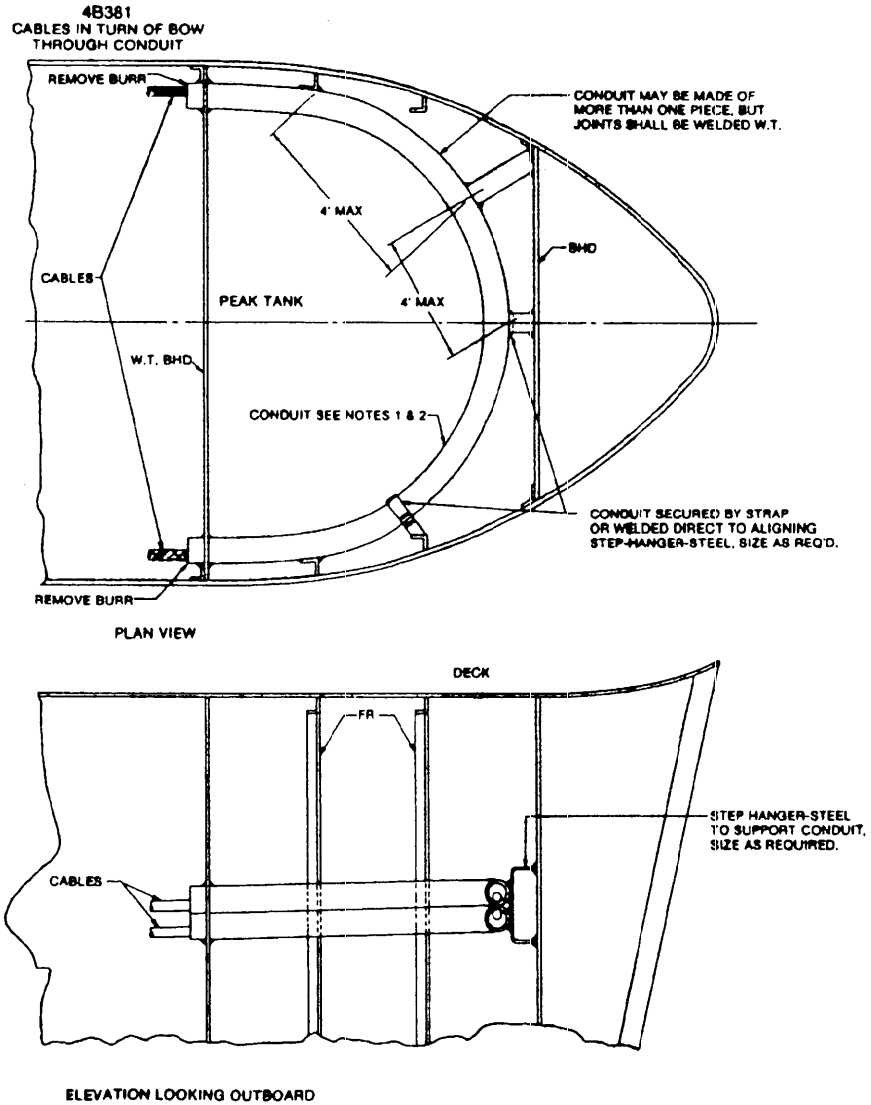
SH 132317254

NOTES:

1. WHEN INSULATION IS USED WITH DEGAUSSING CABLES, USE CORRESPONDING INSULATION METHODS FOR CABLE HANGER METHODS LISTED.
2. USE APPROPRIATE BALLISTIC METHODS FOR DEGAUSSING CABLE INSTALLATIONS AS REQUIRED.
3. METHODS SHOWN ON THIS FIGURE MAY BE USED FOR GENERAL INSTALLATIONS WHERE APPLICABLE.
4. THIS FIGURE SUPERSEDES SHEET 4B37 OF DRAWING 803-5001027 AND SECTION 2, SHEET 144, OF DRAWING NAVSEC NO. 9000-S6202-73980.

FIGURE 4B37. Cable support for degaussing cables (surface ships).

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NOTES:
1. GRADE CONDUIT 1/8" PER FOOT MINIMUM.
2. CABLE CONDUIT SHALL BE OF COPPER NICKEL PIPE.
3. METHODS SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATION WHERE APPLICABLE.
4. THIS FIGURE SUPERSEDES SHEET 4B38 OF DRAWING 803-5001027 AND SECTION 2, SHEET 145, OF DRAWING NAVSEC NO. 9000-S8202-73980.

SH 132317255

FIGURE 4B38. Cable support for degaussing cables (surface ships).

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Check the source to verify that this is the current version before use.

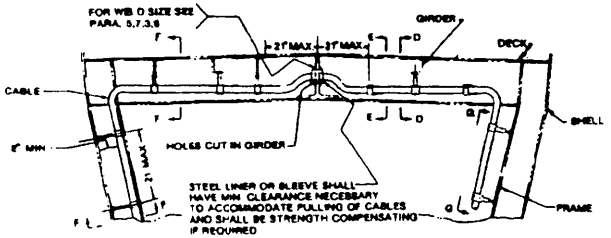
METHODS SHOWN ON THIS SHEET MAY BE USED INSTALLATIONS WHERE APPLICABLE

NOTES:

1. METHODS SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATION WHERE APPLICABLE.
2. THIS FIGURE SUPERSEDES SHEET 4B39 OF DRAWING 803-9001027 AND SECTION 2, SHEET 146, OF DRAWING, NAVSEC NO. 9000-98202-73980.

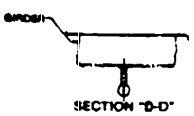
4B391

CABLE RUNS ATHWART SHIP OVERHEAD



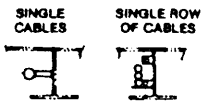
4B393

CABLES RUNNING ACROSS GIRDERS



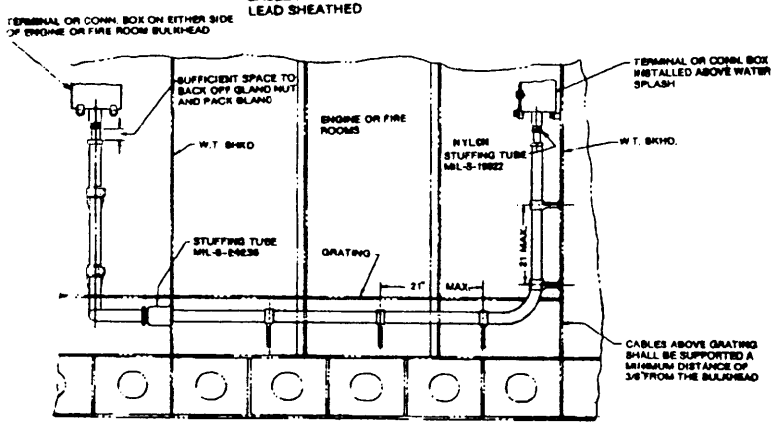
4B394

CABLES MOUNTED ON WEBS

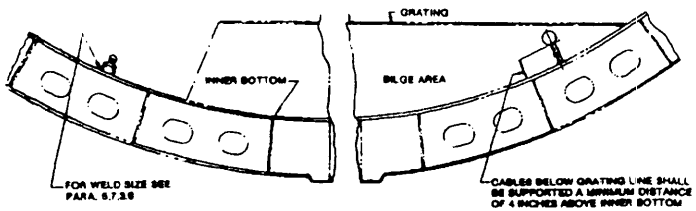


4B392

CABLE IN BILGES LEAD SHEATHED



ELEVATION LOOKING-OUTBOARD

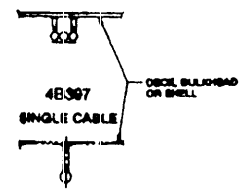


VIEW ATHWART SHIP

4B395

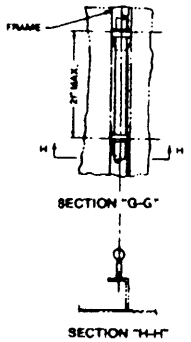
CABLES MOUNTED BETWEEN GIRDERS OR FRAMES

SINGLE ROW



4B396

SINGLE CABLES ON FRAMES



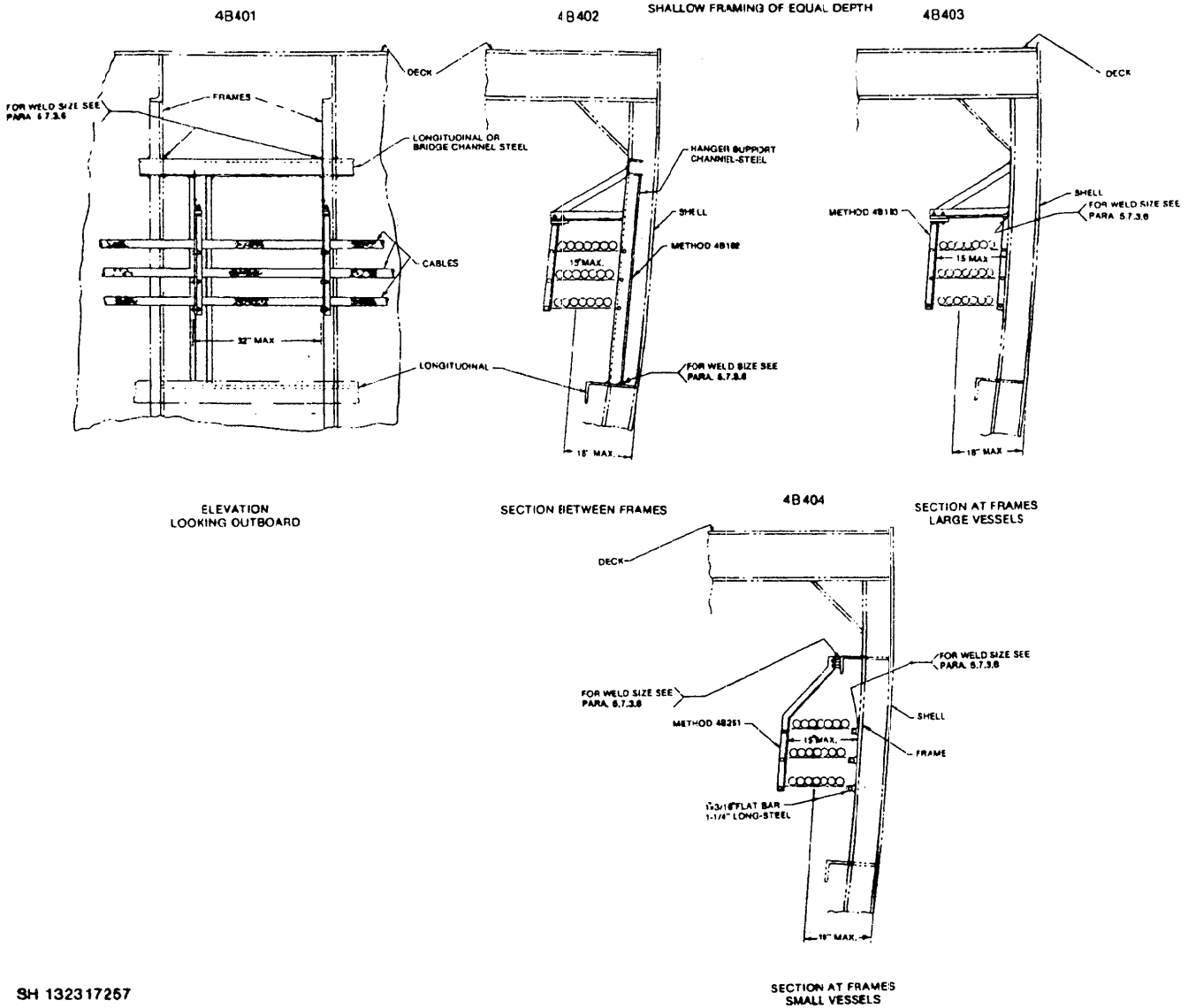
SH 132317256

FIGURE 4B39. Cable support for degaussing cables (surface ships).

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NOTES:

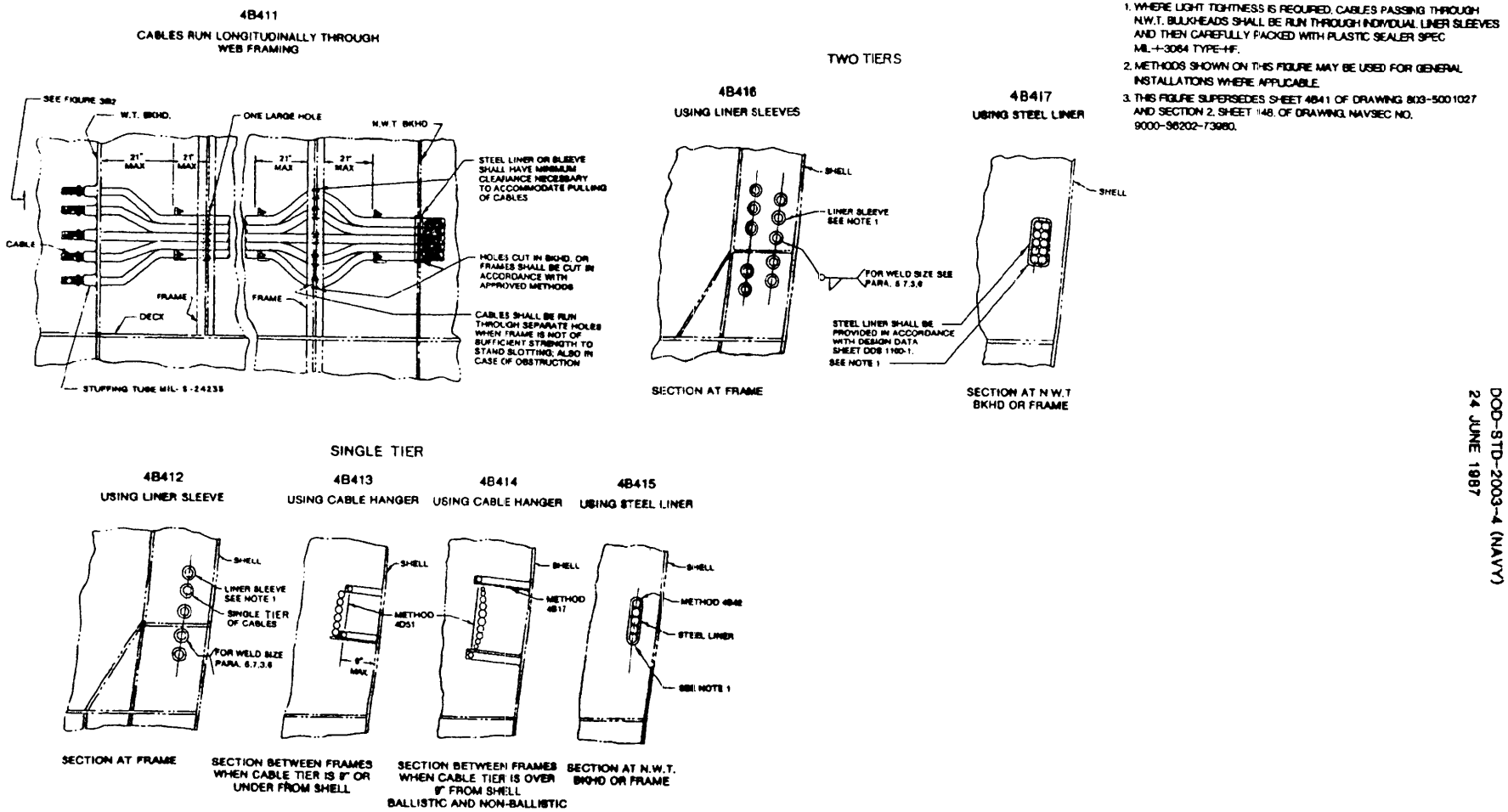
1. METHODS SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATIONS WHERE APPLICABLE.
2. THIS FIGURE SUPERSEDES SHEET 4B40 OF DRAWING 803-5001027 AND SECTION 2, SHEET 147, OF DRAWING NAVSEC NO. 9000-S6202-73980.



SH 132317267

FIGURE 4B40. Cable support for degaussing cables (surface ships).

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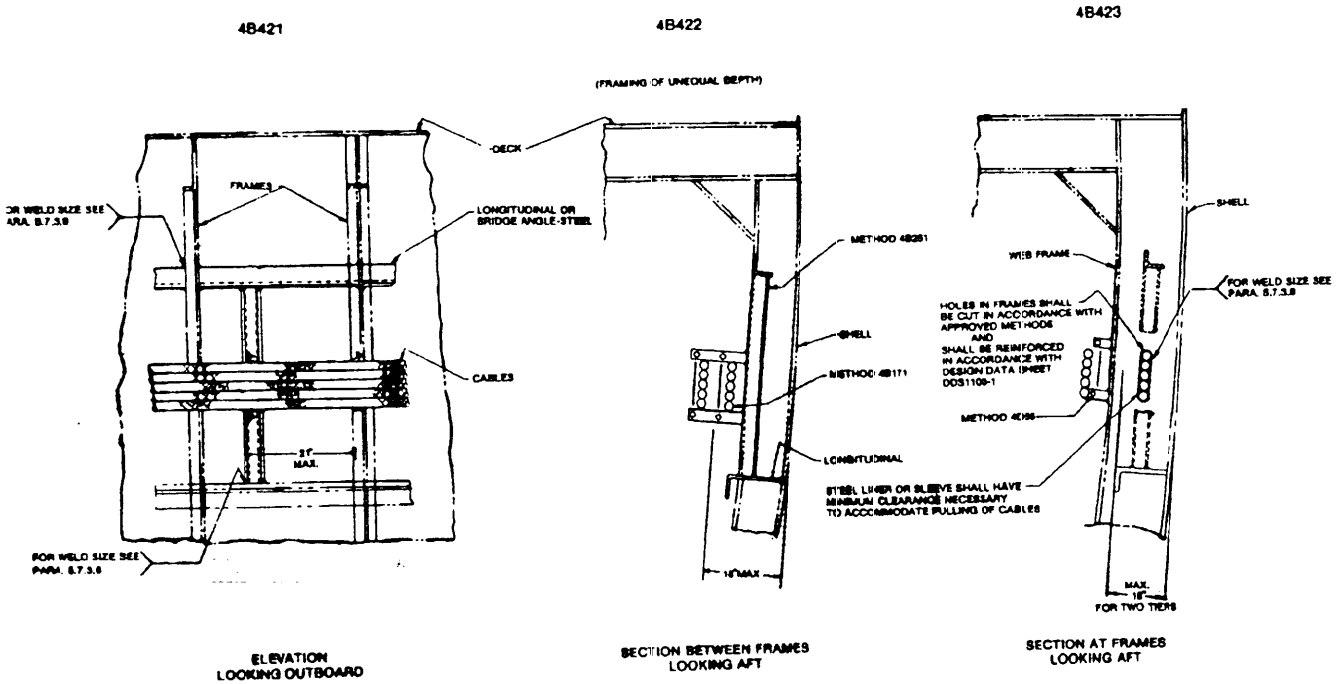
NOTES:

1. WHERE LIGHT TIGHTNESS IS REQUIRED, CABLES PASSING THROUGH N.W.T. BULKHEADS SHALL BE RUN THROUGH INDIVIDUAL LINER SLEEVES AND THEN CAREFULLY PACKED WITH PLASTIC SEALER SPEC MIL-I-3064 TYPE-HF.
2. METHODS SHOWN ON THIS FIGURE MAY BE USED FOR GENERAL INSTALLATIONS WHERE APPLICABLE.
3. THIS FIGURE SUPERSEDES SHEET 4B41 OF DRAWING 803-5001027 AND SECTION 2, SHEET 1148, OF DRAWING NAVSEC NO. 9000-96202-73980.

SH 132317258

FIGURE 4B41. Cable support for degaussing cables (surface ships).

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NOTES:

1. METHODS SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATIONS WHERE APPLICABLE.
2. THIS FIGURE SUPERSEDES SHEET 4B42 OF DRAWING 803-5001027 AND SECTION 2, SHEET 148 OF DRAWING NAVSEC NO. 9000-S6202-73980.

SH 132317259

FIGURE 4B42. Cable support for degaussing cables (surface ships).

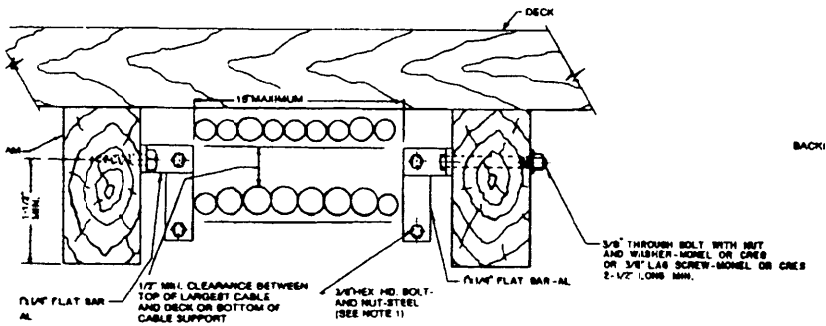
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. FOR NON-MAGNETIC APPLICATION SEE PARA. 5.7.3.2.
2. THIS FIGURE SUPERSEDES SHEET 4B43 OF DRAWING 803-5001027 AND SECTION 2, SHEET 82, OF DRAWING NAVSEC NO. 9010-S6202-73980.

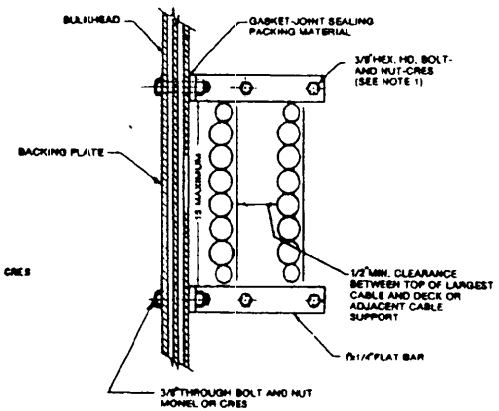
4B431

CABLES RUNS BETWEEN BEAMS
(CABLE BANDS NOT REQUIRED)



4B432

WATERTIGHT BULKHEADS
UNDER 1 INCH THICK

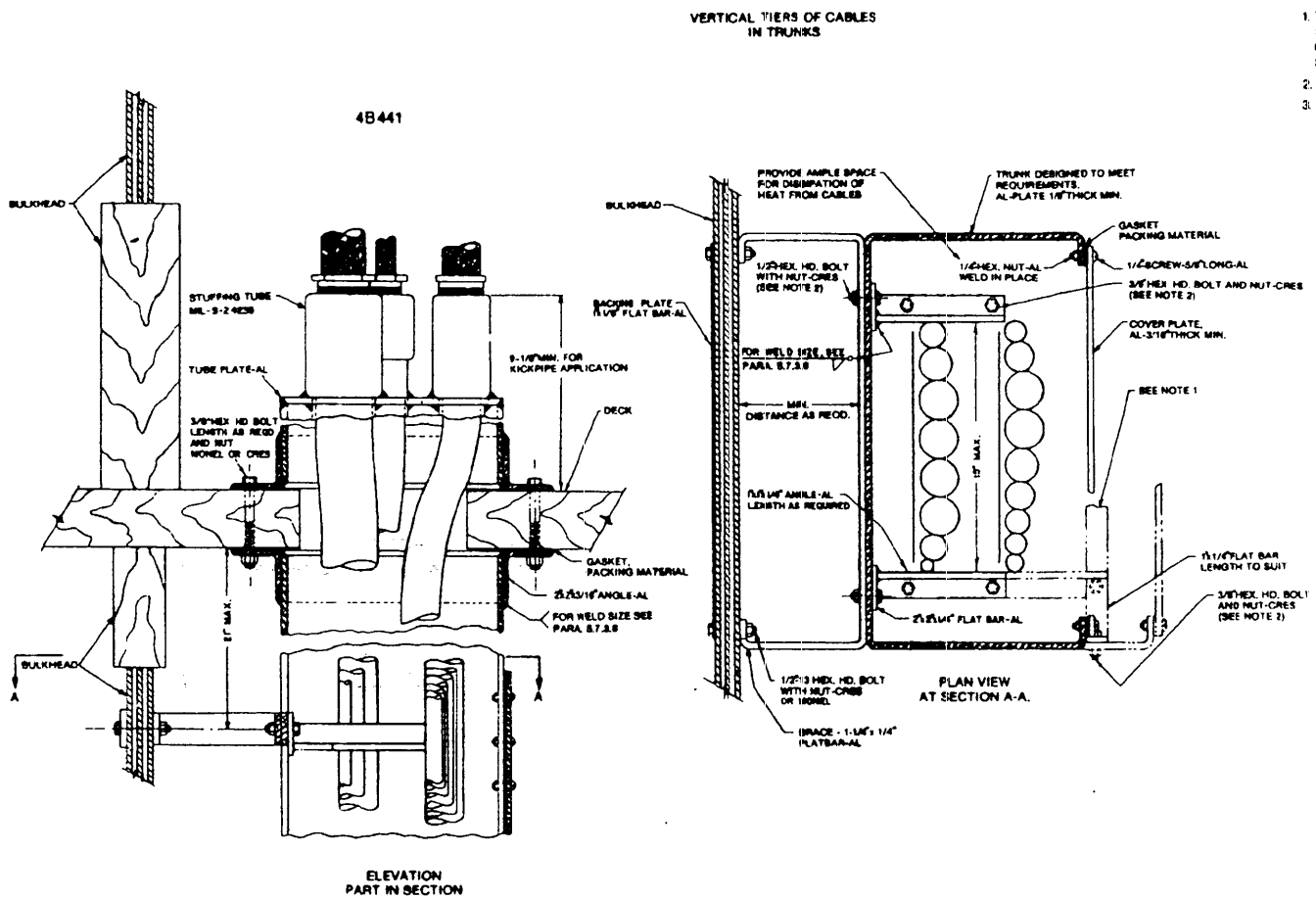


1 132317260

FIGURE 4B43. Cableways on wooden ships (surface ships).

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Check the source to verify that this is the current version before use.

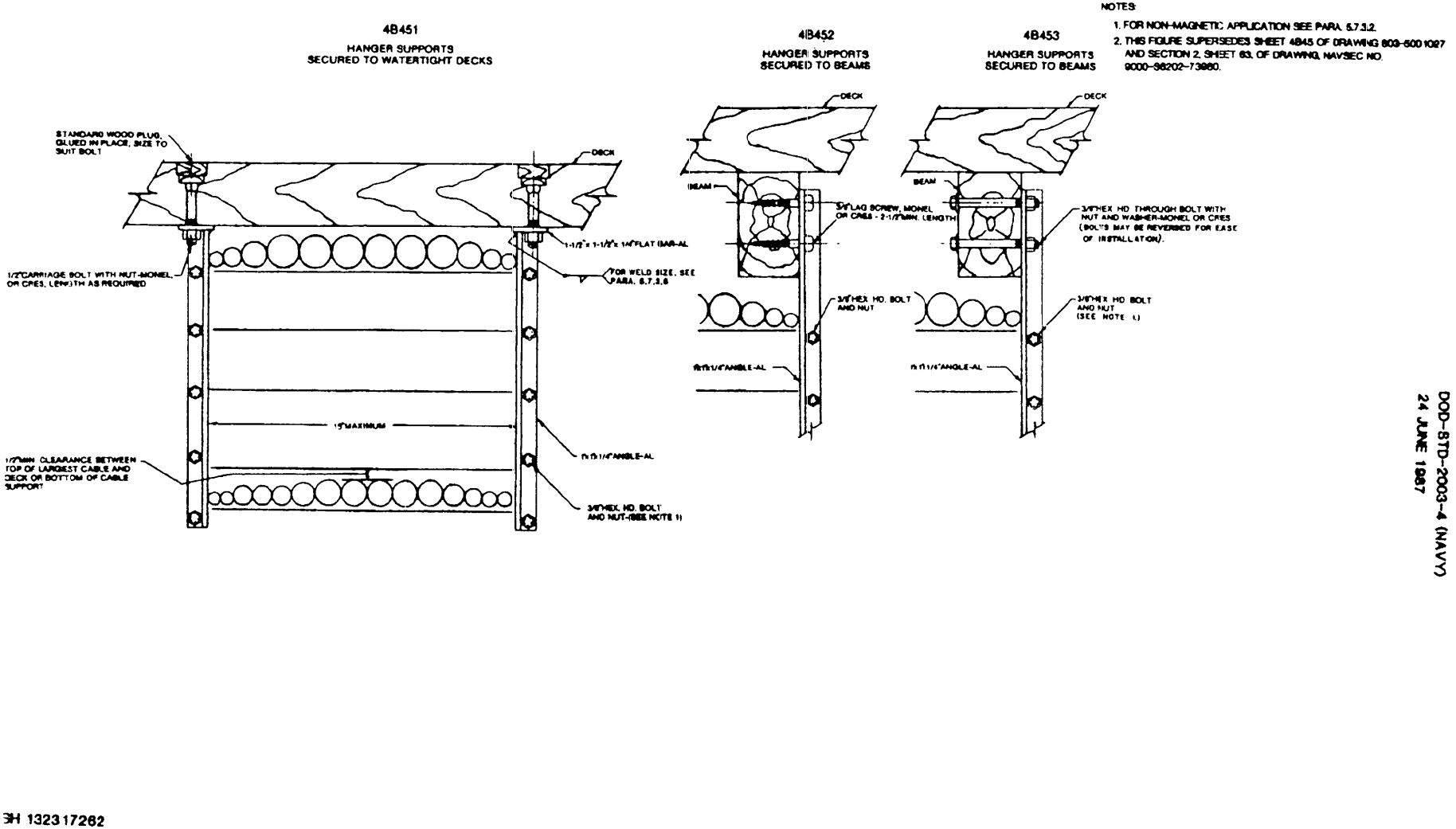


NOTES:

1. TWO ROWS OF CABLES WITH A 9" LENGTH OF CABLE SPAN SHALL BE CONSIDERED NORMAL. IF A LONGER SPAN OR THREE ROWS OF CABLES ARE REQUIRED, SECURE THE OUTER END OF HANGER SUPPORT TO THE WALL OF THE TRUNK.
2. FOR NONMAGNETIC APPLICATION SEE PARA. 5.7.3.2.
3. THIS FIGURE SUPERSEDES SHEET 4B44 OF DRAWING 803-6001027 AND SECTION 2, SHEET 63, OF DRAWING NAVSEC NO.

SH 132317261

FIGURE 4B44. Cableways for wooden vessels (surface ships).



3H 132317262

FIGURE 4B45. Cableways for wooden ships (surface ships).

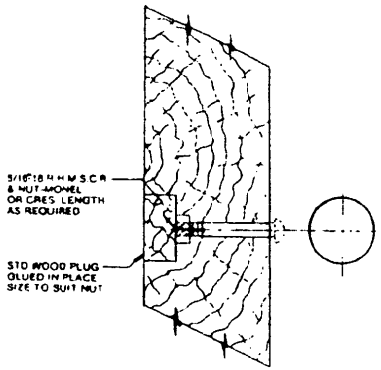
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. METHODS SHOWN ON THIS SHEET ARE FOR USE WITH THE CABLES UP TO 2 IN. IN DIA FOR CABLES GREATER THAN 2 IN.IN DIA. USE METHODS SHOWN ON FIGURES 4B48, 4B49 & 4B50.
2. THIS FIGURE SUPERSEDES SHEET 4B46 OF DRAWING 803-5001027 AND SECTION 2, SHEET 192, OF DRAWING NAVSEC NO. 9000-S6202-73980.

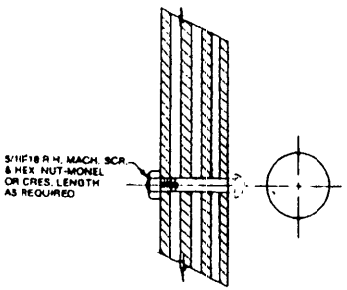
4B461

WATERTIGHT BULKHEADS OVER 1" THICK
WHERE MOISTURE OR DRIP IS PRESENT
CABLE SUPPORTED VERTICALLY



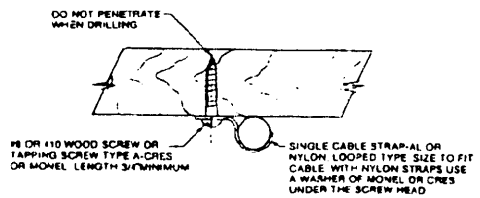
4B462

WATERTIGHT BULKHEADS 1" MAX. THICKNESS
WHERE MOISTURE OR DRIP IS PRESENT
CABLE SUPPORTED HORIZONTALLY



4B463

DECKS OR BULKHEADS OVER 1" THICK
ALSO FOR BEAMS
CABLES WITH A MAXIMUM DIA OF 5/8"
SINGLE SUPPORT ALUM CABLE STRAPS



SH 132317263

4B464

DECKS OR BULKHEADS OVER 1" THICK
ALSO FOR BEAMS

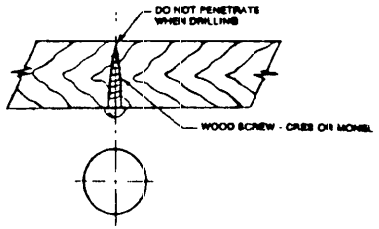
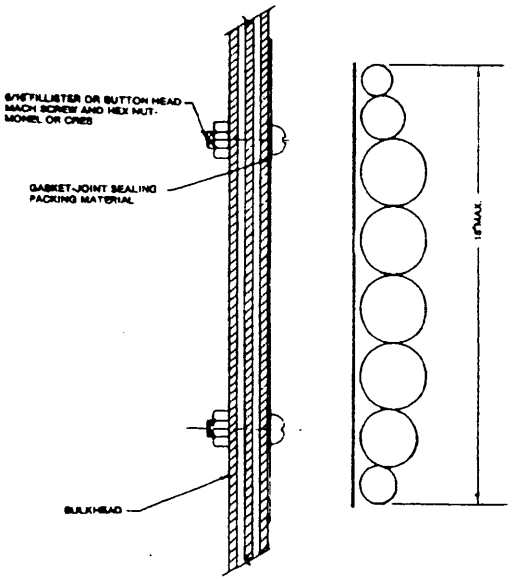


FIGURE 4B46. Cableways for wooden hull ships (surface ships).

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4B471

SINGLE TIER OF CABLES
SUPPORTED FROM WATERTIGHT BULKHEADS
UNDER 1 INCH THICKNESS



SH 132317264

NOTES:

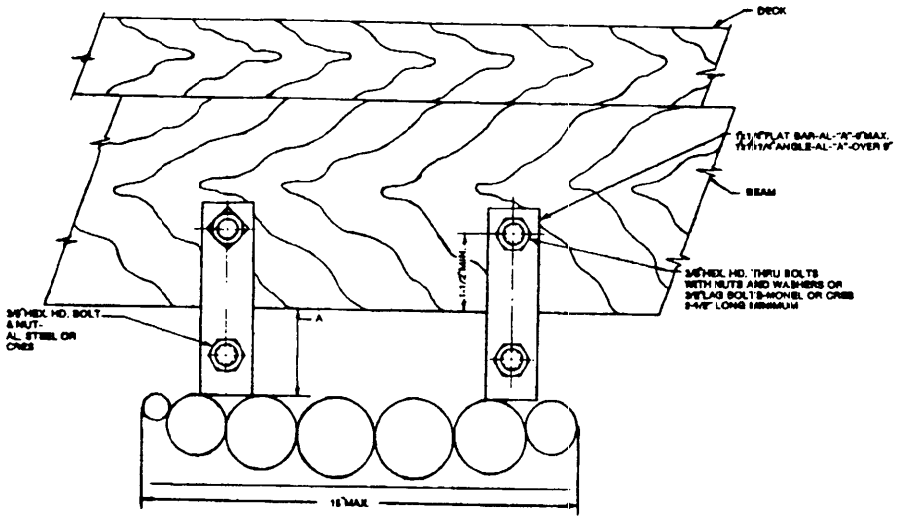
1. METHODS SHOWN ON THIS SHEET MAY BE USED IN LIEU OF THOSE SHOWN ON FIGURES 4B48, 4B49 I 4B50.
2. THIS FIGURE SUPERSEDES SHEET 4B47 OF DRAWING 803-5001027 AND SECTION 2, SHEET 208, OF DRAWING NAVSEC NO. 9000-S6202-73980.

FIGURE 4B47. Cableways for wooden ships (surface ships).

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Check the source to verify that this is the current version before use.

4B481

A SINGLE TIER OF CABLES SUPPORTED FROM BEAMS



NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B48 OF DRAWING 803-6001027 AND SECTION 2, SHEET 207, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

SH 132317265

FIGURE 4B48. Cableways for wooden ships (surface ships).

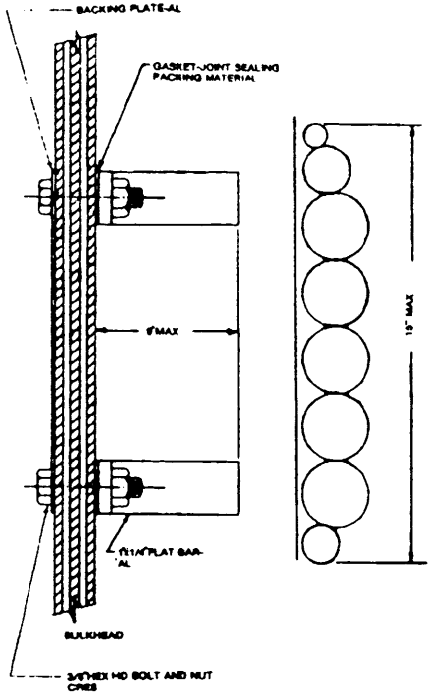
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NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4B49 OF DRAWING 803-5001027 AND SECTION 2, SHEET 208, OF DRAWING, NAVSEC NO. 9000-98202-73980.

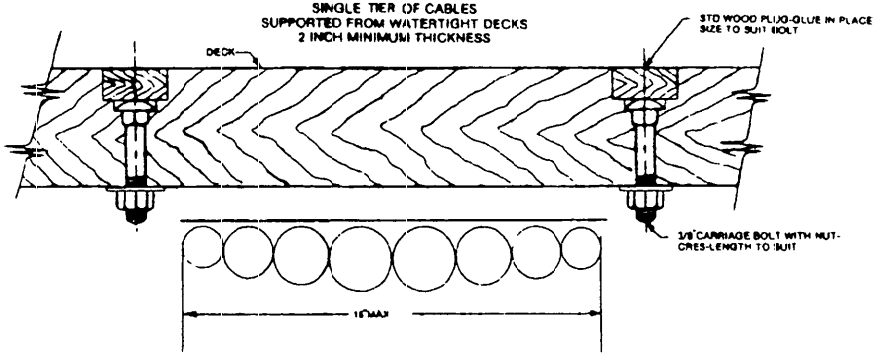
4B491

SINGLE TIER OF CABLE
SUPPORTED FROM WATERTIGHT BULKHEADS
UNDER 1 INCH IN THICKNESS



4B492

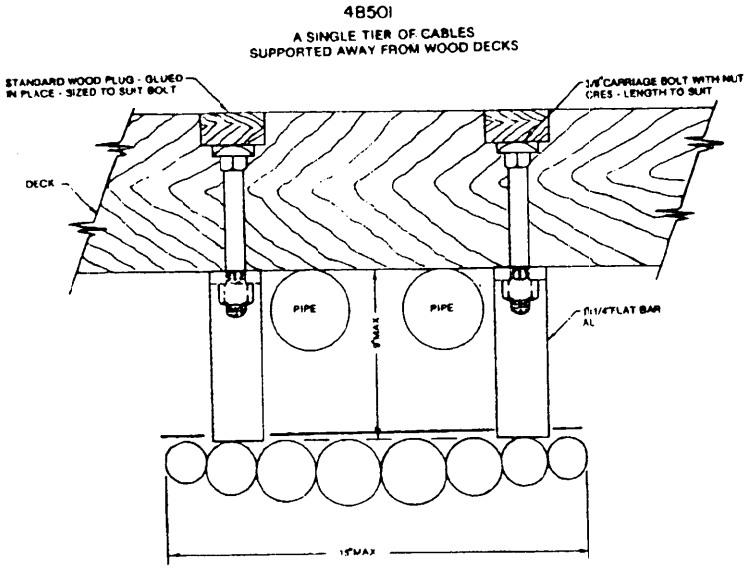
SINGLE TIER OF CABLES
SUPPORTED FROM WATERTIGHT DECKS
2 INCH MINIMUM THICKNESS



SH 132317268

FIGURE 4B49. Cableways for wooden ships (surface ships).

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Check the source to verify that this is the current version before use.

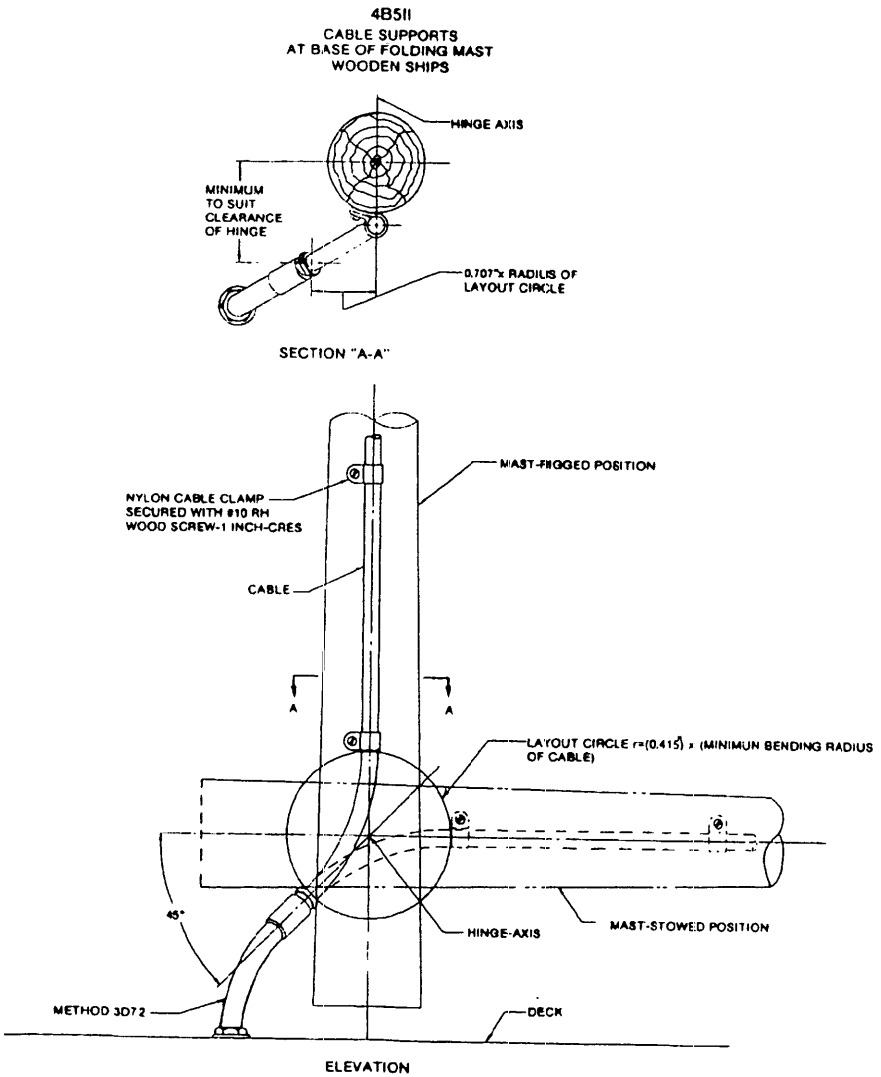


NOTES:
1 THIS FIGURE SUPERSEDES SHEET 4B50 OF DRAWING 803-6001027 AND SECTION 2, SHEET 209, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

SH 132317267

FIGURE 4B50. Cableways for wooden ships.

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Check the source to verify that this is the current version before use.



NOTES:

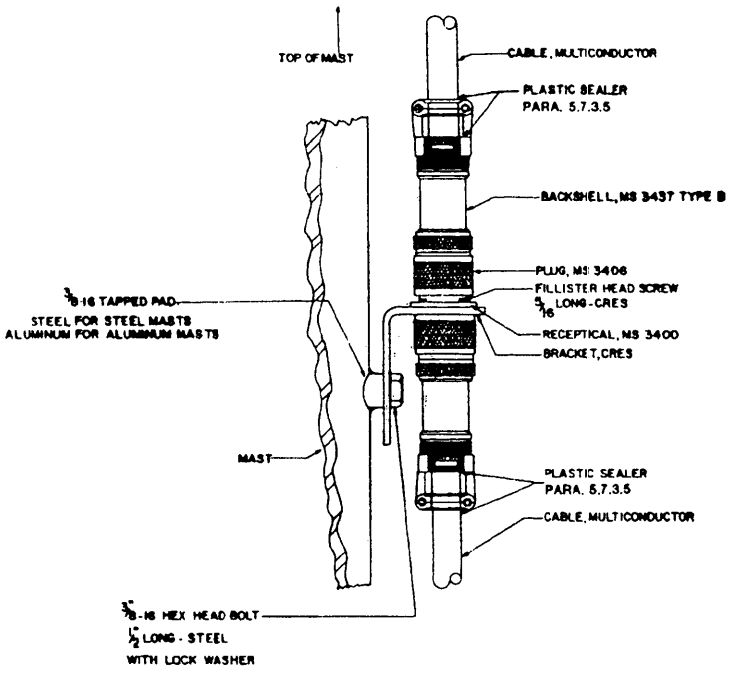
1. THE DIMENSION USED FOR "r" SHOULD NOT BE LESS THAN 4 INCHES.
2. WHERE MORE THAN ONE CABLE IS INVOLVED USE THE BENDING RADIUS OF THE LARGEST CABLE TO FIND "r".
3. CABLE SHOULD BE FIRMLY FASTENED WITH MAST IN 45° POSITION TO INSURE PROPER LENGTH FOR FLEXING.
4. THIS FIGURE SUPERSEDES SHEET 4B51 OF DRAWING 803-5001027 AND SECTION 2, SHEET218, DRAWING, NAVSEC NO. 9000-S6202-73960.

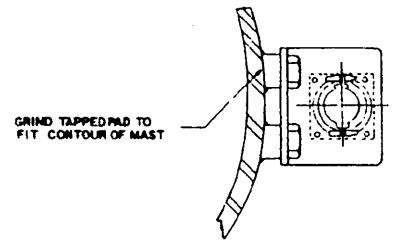
FIGURE 4B51. Cableways on folding masts (surface ships).

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Check the source to verify that this is the current version before use.

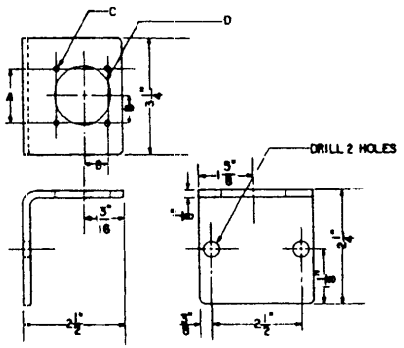
4B521
COUPLING OF MULTIPLE CABLES
CONNECTORS PER MIL-C-5015





SH 132317269

4B522
MOUNTING BRACKET DETAILS



SHELL SIZE	DIMENSIONS A	B	C DIA	D DIA
8	.1010	.000	.031	.000-.005
10	.1162	0.341	.120	1.125
12	.1458	0.378	.120	1.125
16	.1250	0.823	.120	1.375
20	.1378	0.488	.147	1.500
24	.1562	0.941	.147	1.813
32	1.750	0.875	.173	2.063
36	1.938	0.967	.173	2.313

NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4B52 OF DRAWING 803-5001027 AND SECTION 1, SHEET 58 AND SECTION 2, SHEET 219 OF DRAWING, NAVSEC NO. 9000-S8202-73980

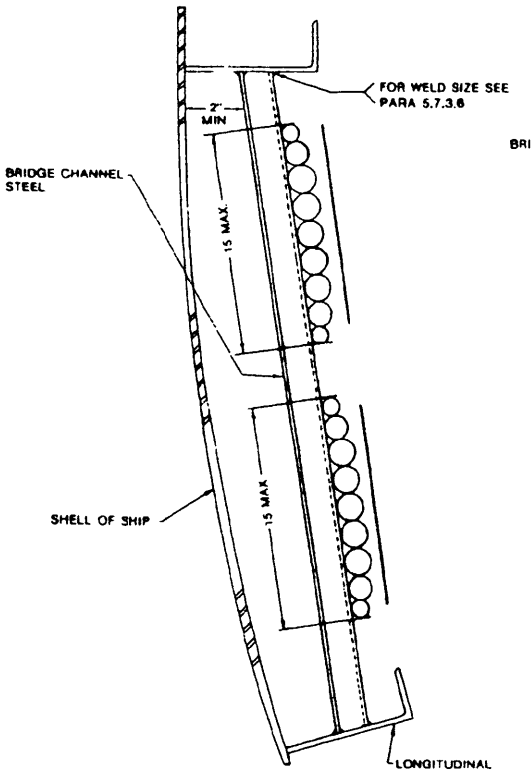
FIGURE 4B52. Coupling of cables on masts with connectors.

MAIN RUN CABLES
SMALL VESSELS

NOTES:

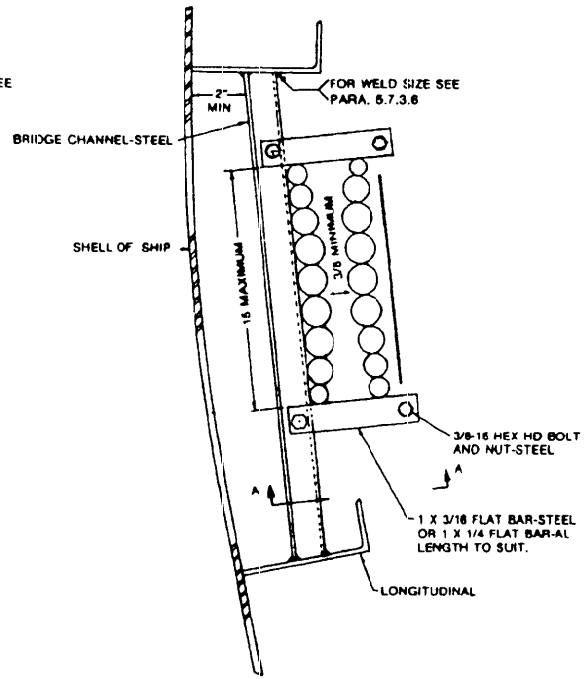
1. THIS FIGURE SUPERSEDES SHEET 4B53 OF DRAWING 803-5001027 AND SECTION 2, SHEETS (9) & (9), OF DRAWING NAVSEC NO. 9000-S6202-73980.

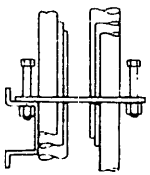
4B531
CABLES THROUGH ENGINE AND BOILER ROOMS
SUPPORTED BETWEEN SHIPS LONGITUDINALS



3H 132317270

4B532
CABLES THROUGH ENGINE AND BOILER ROOMS
SUPPORTED BETWEEN SHIPS LONGITUDINALS.





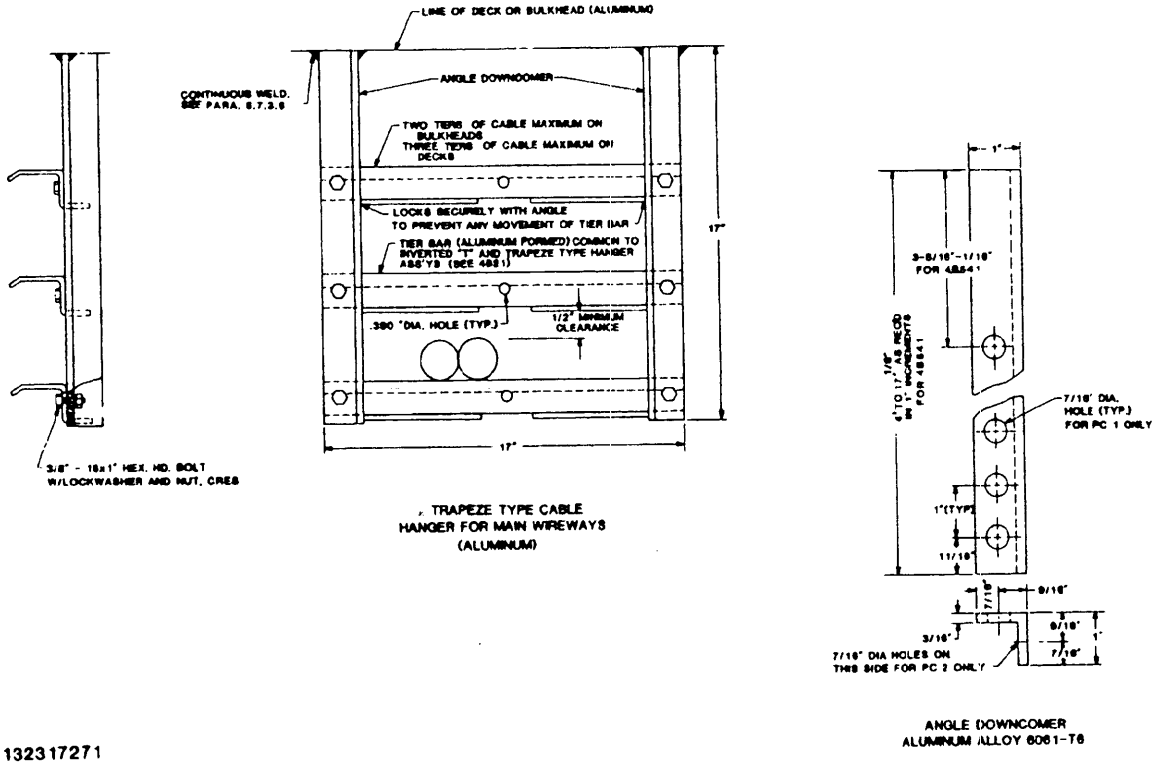
SECTION "A-A"

FIGURE 4B53. Cableways of main cable runs on small vessels.

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Check the source to verify that this is the current version before use.

4B541



TRAPEZE TYPE CABLE HANGER FOR MAIN WIREWAYS (ALUMINUM)

ANGLE DOWNCOMER ALUMINUM ALLOY 6061-T6

SH 132317271

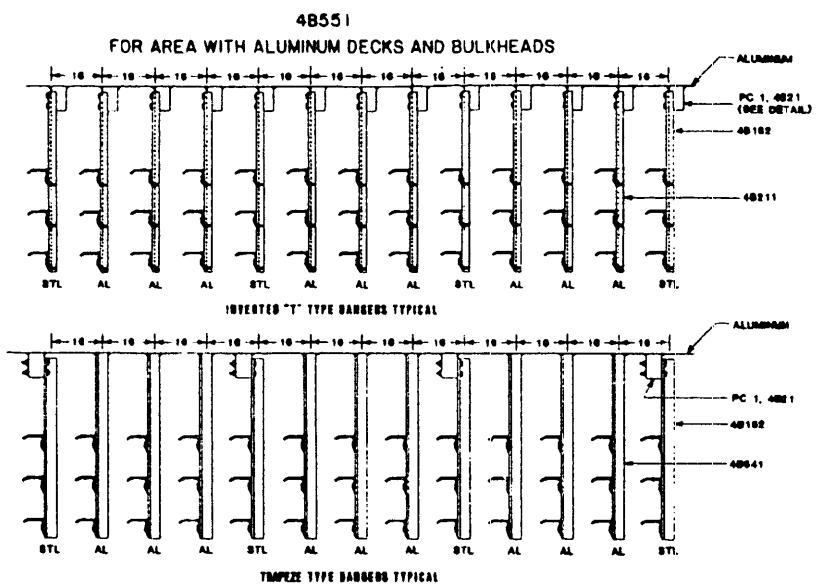
NOTES:

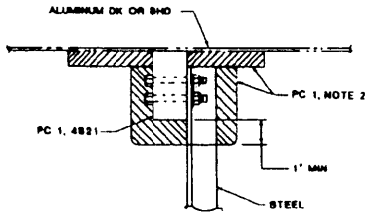
1. SPACING, NUMBER OF TIERS AND CABLE WEIGHT SHALL NOT EXCEED THE VALUES SPECIFIED BELOW. DOWNWARD EXTENSION TO THE LESSER CONDITION OF FEWER TIERS FOR HANGER ASSEMBLIES (WITH CABLE LOAD NOT EXCEEDING THE MAXIMUM PER TIER AS SPECIFIED BELOW) IS ACCEPTABLE.

CABLE RUN	TYPE ASS'Y.	CENTER TO CENTER SPACING (INCHES)	MAX. NO. OF TIERS	MAX. CABLE WT. (LBS.) SUPPORTED BY EA. TIER
HORIZONTAL	DECK MTD. 36" 17"	24	3	45.5
VERTICAL	BLK. HD MTD. 20" 17"	24	2	45.5

2. SHARP CORNERS SHALL BE SLIGHTLY ROUNDED. ALL PARTS SHALL HAVE SMOOTH SURFACES.
3. FOR SPACING OF HANGERS SEE FIGURE 4B55
4. SEE FIGURE 4C23 FOR INSTALLATION OF BONDING STRAPS
5. THIS FIGURE SUPERSEDES SHEET 4B54 OF DRAWING 803-5001027

FIGURE 4B54. Multiple cableway supports on aluminum or steel decks and bulkheads (surface ships only).





DETAIL OF FIRE PROTECTION INSTALLATION
(TYPICAL FOR ALUMINUM DECKS AND BULKHEADS, NOTE 2).

NOTES:

1. SEE FIGURES 4C22 & 4C23 FOR INSTALLATION OF BINDING STRAPS.
2. FIRE INSTALLATION DETAIL SHALL BE IN ACCORDANCE WITH NAVSEA DWG. 803-5184182.
3. THIS FIGURE SUPERSEDES SHEET 4B55 OF DRAWING 803-5001027.

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SH 132317272

FIGURE 4B55. Spacing of multiple cableway supports for aluminum and steel decks and bulkheads (surface ships only).

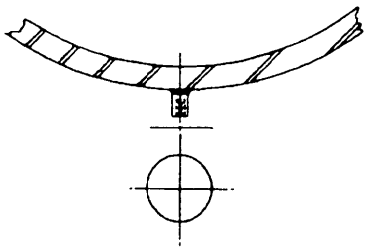
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. METHODS SHOWN ON THIS SHEET ARE FOR USE WITH CABLES UP TO 2 IN. IN DIA. FOR CABLES GREATER THAN 2 IN. IN DIA USE METHOD SHOWN ON FIGURE 4C2.
2. THIS FIGURE SUPERSEDES SHEET 4C1 OF DRAWING 803-5001027 AND SECTION 2, SHEET 195, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

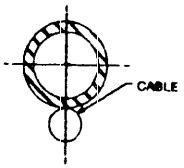
4C11

SINGLE CABLES ON STEEL STANCHIONS HANGER SECURED WITH STUD.



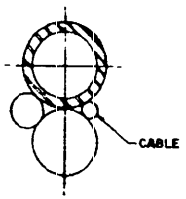
4C12

SINGLE CABLES ON SMALL STANCHIONS



4C13

TWO TO THREE CABLES ON SMALL STANCHIONS

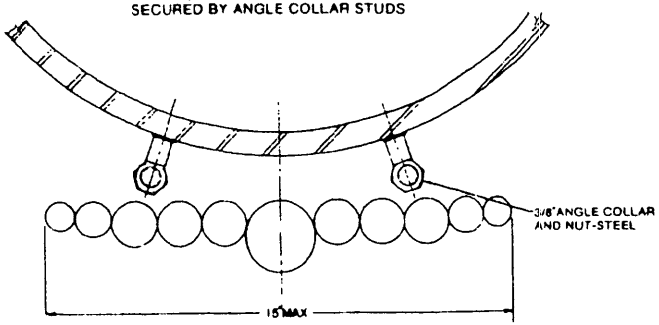


SH 132317273

FIGURE 4C1. Single cableway on steel stanchions.

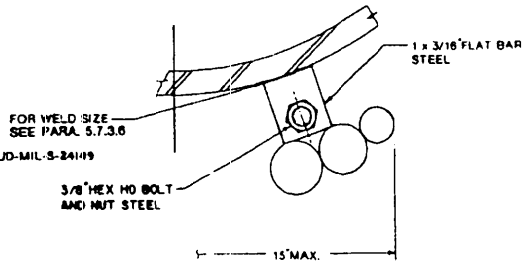
4C21

SINGLE TIER OF CABLES ON STEEL STANCHIONS
SECURED BY ANGLE COLLAR STUDS



4C22

SINGLE TIER OF CABLES ON STEEL STANCHIONS
SECURED BY FLAT BAR



NOTES

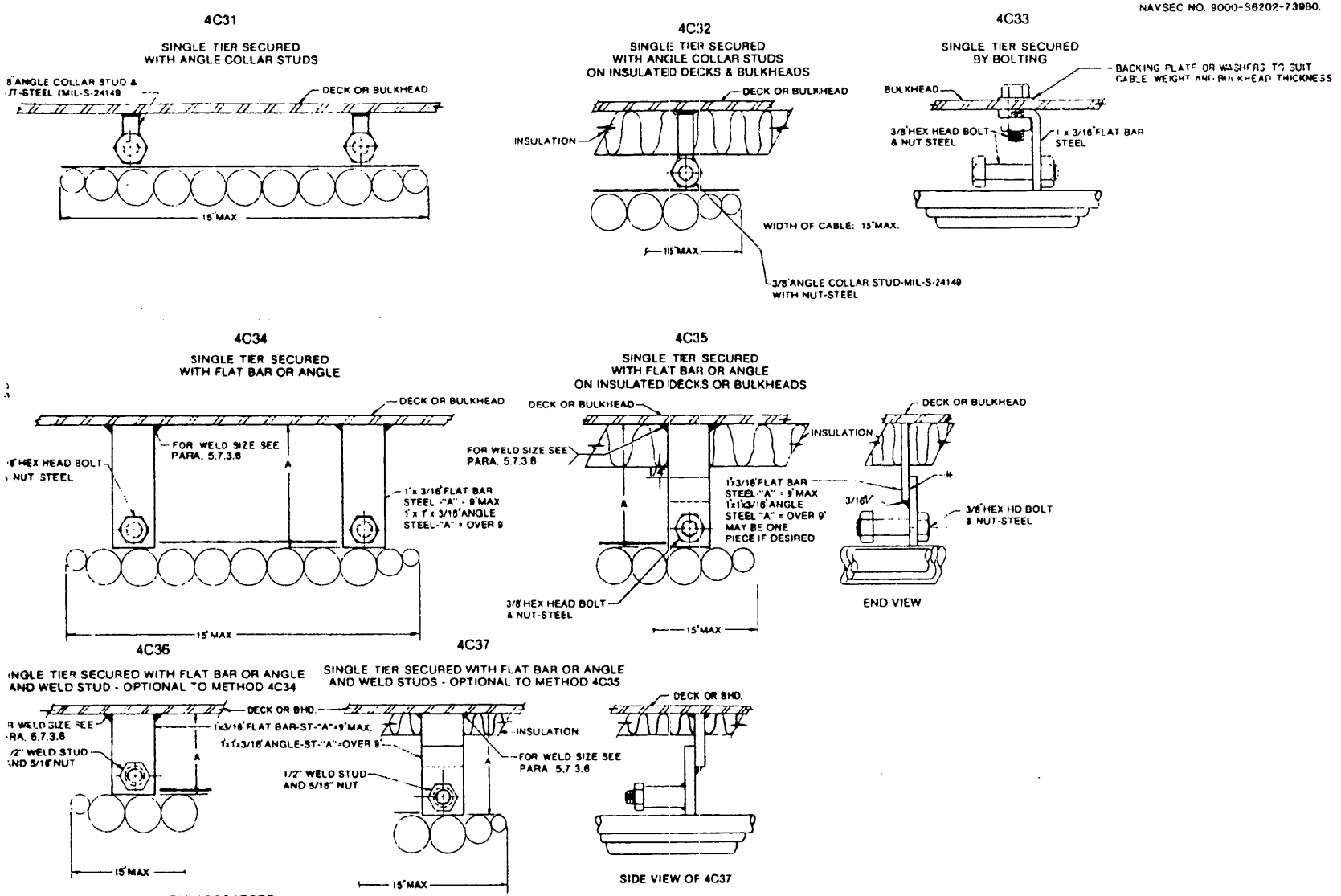
1. THIS FIGURE SUPERSEDES SHEET 4C2 OF DRAWING 803-5001027 AND SECTION 2, SHEET 204, OF DRAWING NAVSEC NO 9000-S6203-73980.

SH 132317274

FIGURE 4C2. Single cableway on steel stanchion.

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NOTES:
1 THIS FIGURE SUPERSEDES SHEET 4C3 OF DRAWING 803-5001027 AND SECTION 2, SHEET 202, OF DRAWING NAVSEC NO. 9000-S6202-73980.



SH 132317275

FIGURE 4C3. Single cableways on steel structure.

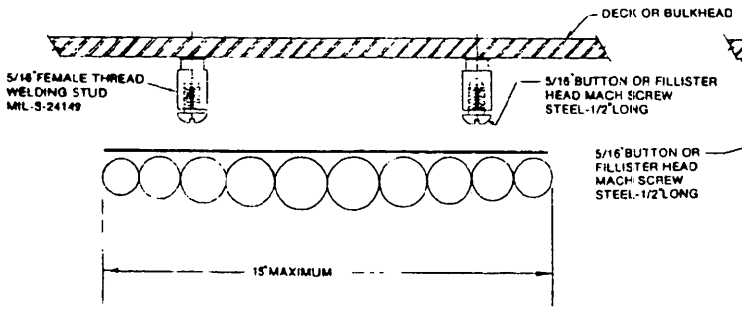
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NOTES:

1. METHODS SHOWN ON THIS FIGURE MAY BE USED IN LIEU OF THOSE SHOWN ON FIGURES 4C2, 4C3 & 4C14.
2. THIS FIGURE SUPERSEDES SHEET 4C4 OF DRAWING 803-5001027 AND SECTION 2, SHEET 203, OF DRAWING, NAVSEC NO. 9000-S6202-73950.

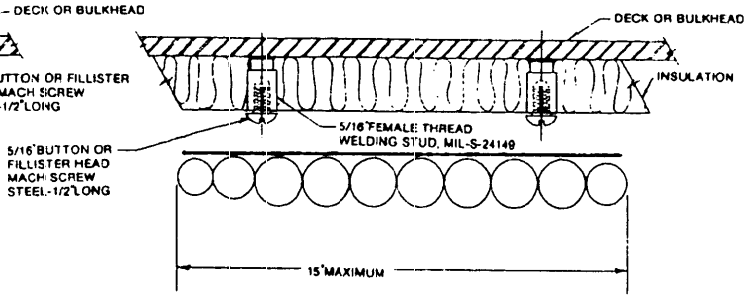
4C41

SINGLE TIER SECURED WITH
FEMALE THREAD WELDING STUDS
ON STEEL DECKS AND BULKHEADS



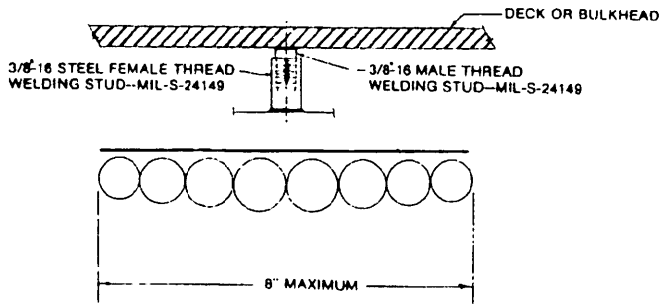
4C42

SINGLE TIER SECURED WITH
FEMALE THREAD WELDING STUDS ON
INSULATED DECKS AND BULKHEADS



4C43

SINGLE TIER SECURED WITH
WELD STUD ON STEEL DECKS AND BULKHEADS
MAXIMUM CABLE WEIGHT 10 POUNDS PER SUPPORT



SH 132317276

FIGURE 4C4. Single cableway on steel structure.

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Check the source to verify that this is the current version before use.

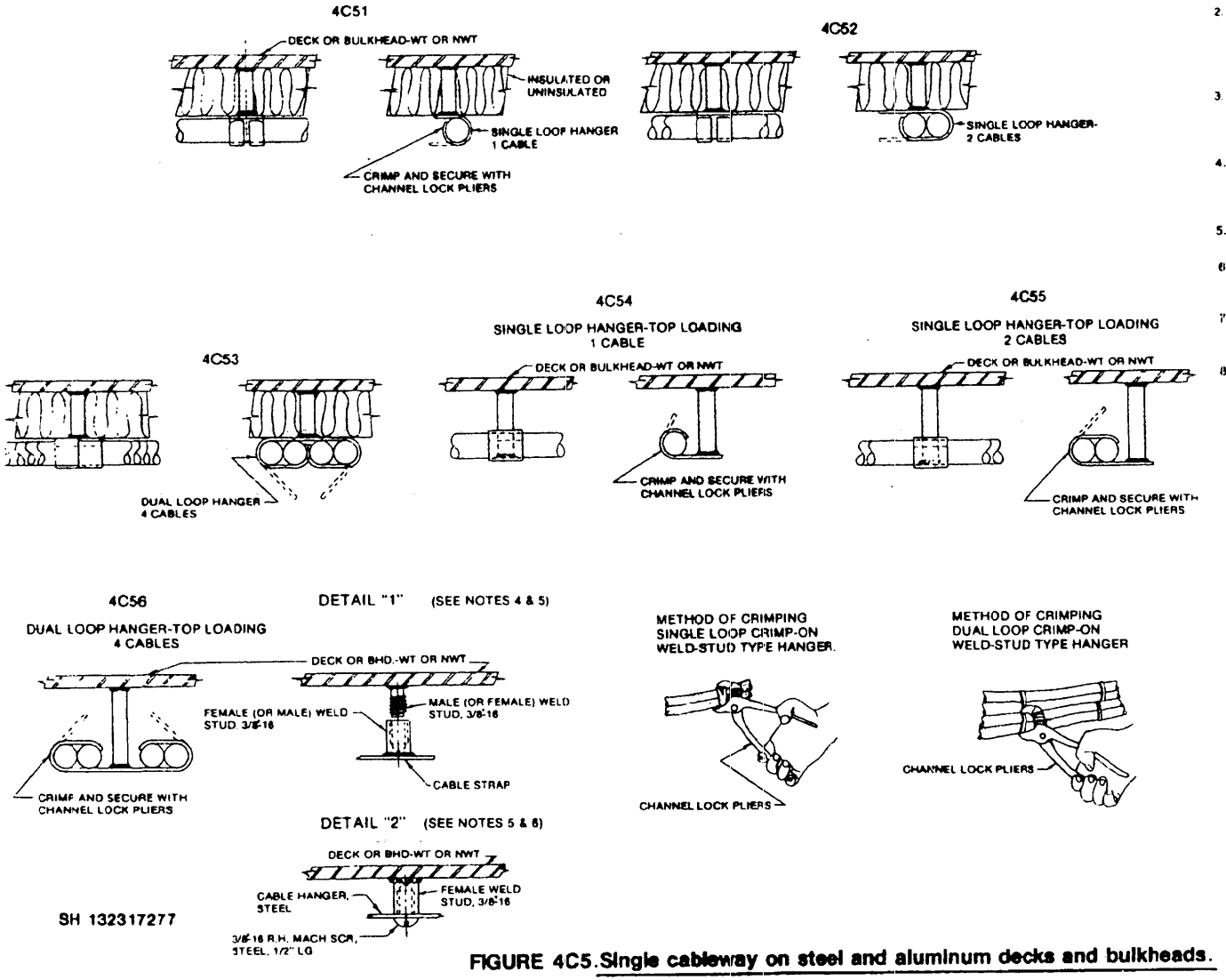


FIGURE 4C5. Single cableway on steel and aluminum decks and bulkheads.

NOTES

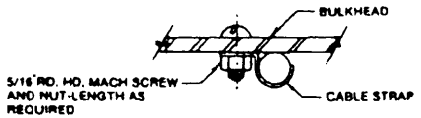
1. CRIMP-ON WELD-STUD TYPE CABLE HANGER SHOWN HEREON MAY BE USED AS ALTERNATES TO METHODS 4C72 AND 4C73.
2. WEIGHT OF CABLES SUPPORTED SHALL NOT EXCEED TWO POUNDS PER STUD FOR SINGLE-LOOP HANGER AND FOUR POUNDS PER STUD FOR A DUAL-LOOP HANGER SPACINGS BETWEEN STUDS SHALL NOT EXCEED 32 INCHES.
3. MAKE OF HANGERS PROCURED SHALL HAVE PASSED HIGH IMPACT SHOCK TEST, MIL-S-901, AND VIBRATION TEST, MIL-STD-167, USING THE MAXIMUM HANGER LOADINGS AND SPACING.
4. THE FEMALE WELD STUD METHOD SHOWN IN DETAIL "1" HEREON (ATTACHED TO EITHER THE CABLE STRAP OR SHIP'S STRUCTURE), MAY BE USED AS ALTERNATE TO METHODS 4C51 THROUGH 4C56.
5. THREADS WITH ALUMINUM MALE THREADS SHALL BE COATED.
6. THE CONTACT OF STEEL FEMALE THREADS WITH ALUMINUM MALE THREADS SHALL BE COATED WITH ANTI-SEIZE COMPOUND
7. THE FEMALE WELD STUD METHOD SHOWN IN DETAIL "2" MAY BE USED AS ALTERNATE TO METHODS 4C71 AND 4C75 and 4C51 THROUGH 4C56.
8. THIS FIGURE SUPERSEDES SHEET 4C5 OF DWG. 803-5001027 AND SECTION 2, SHEET 196, OF DRAWING, NAVSEC NO 9000-86202-73980.

NOTES

1. ALL HANGER MATERIAL SHALL BE THE SAME AS BULKHEAD MATERIAL
2. CABLE STRAP SHALL NOT DENT OR OTHERWISE EXERT PRESSURE ON THE SEMISOLID DIELECTRIC COAXIAL CABLES IN ANY OF THESE METHODS ADD SPACER WASHERS IF NECESSARY TO KEEP CABLE FREE OF PRESSURE.
3. THIS FIGURE SUPERSEDES SHEET 4C6 OF DWG 803-5001027 AND SECTION 2, SHEETS 193 & 194, OF DRAWING, NAVSEC NO. 6000-S6202-73660.

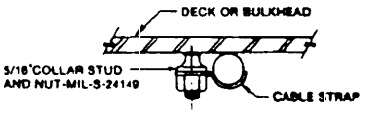
4C61

(SEE NOTE 1)
CABLES SECURED TO NON-WATERTIGHT BULKHEADS - CABLES 5/8"DIA MAXIMUM



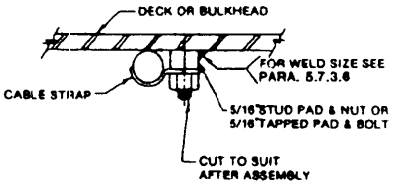
4C62

(SEE NOTE 1)
CABLE SECURED WITH COLLAR STUDS
CABLES 5/8"DIA MAXIMUM



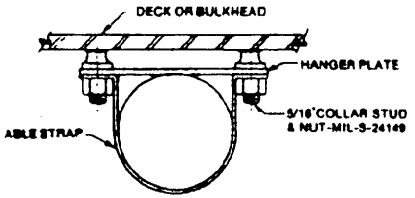
4C63

(SEE NOTE 1)
CABLES SECURED WITH PADS
CABLES 5/8"DIA MAXIMUM



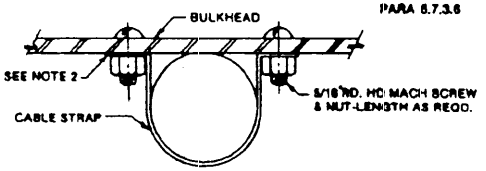
4C64

(SEE NOTE 1)
CABLES SECURED WITH COLLAR STUDS



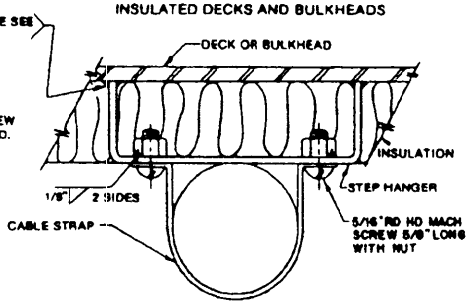
4C65

(SEE NOTE 1)
CABLES SECURED TO NON-WATERTIGHT BULKHEAD



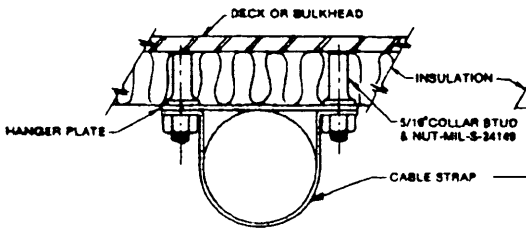
4C66

(SEE NOTE 1)
CABLE SECURED WITH STEP-HANGER ON INSULATED DECKS AND BULKHEADS



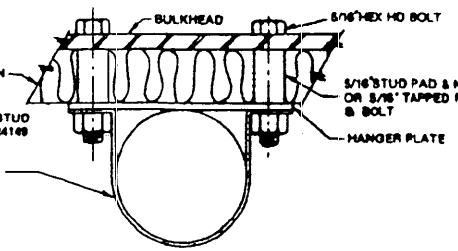
4C67

(SEE NOTE 1)
CABLES SECURED WITH COLLAR STUDS ON INSULATED DECKS OR BULKHEADS



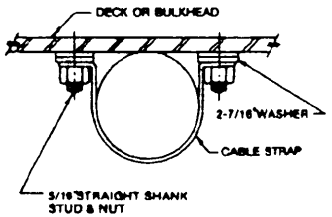
4C68

(SEE NOTE 1)
CABLES SECURED TO NON-WATERTIGHT INSULATED BULKHEADS



4C69

(SEE NOTE 1)
CABLES SECURED WITH STRAIGHT SHANK STUDS



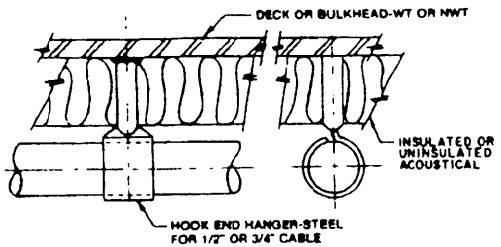
SH 132317278

FIGURE 4C6. Single cableway for coaxial cable.

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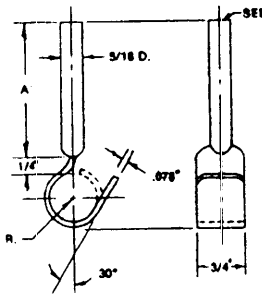
4C71

ALTERNATE METHODS 4C51, 4C54



4C72

3/4 INCH CABLE CLAMP (HOOK END)



SIZE	"A"	"R"
1/2"	5/8"	1/4"
3/4"		3/8"
1/2"	3/8"	1/4"
	1"	
	2"	
	3"	
3/4"	3/8"	3/8"
	1"	
	2"	
	3"	

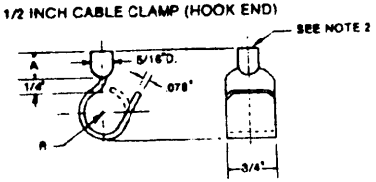
(SEE NOTE 3.)

NOTES:

1. PIECES 4C72 THRU 4C74 SHALL BE UNPLATED UNLESS OTHERWISE SPECIFIED.
2. HANGERS 4C72 THRU 4C74 SHALL BE SUITABLE FOR WELDING BY THE METAL ARC STUD WELDING PROCESS.
3. THE APPLICABLE PC NO. AND SIZE SHALL BE STAMPED ON EACH HOOK END HANGER IN 1/8 INCH FIGURES.
4. THIS FIGURE SUPERSEDES SHEET 4C7 OF DWG. 803-5001027 AND SECTION 1, SHEET 18, AND SECTION 2, SHEET 191 OF DRAWING, NAVSEC NO. 9000-96202-73980.

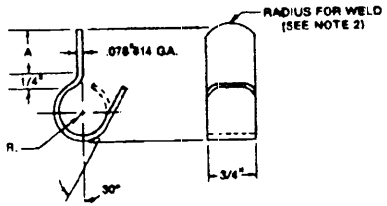
4C73

1/2 INCH CABLE CLAMP (HOOK END)

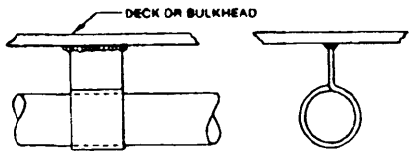


4C74

1/2 & 3/4 INCH CABLE CLAMP (HOOK END)



4C75



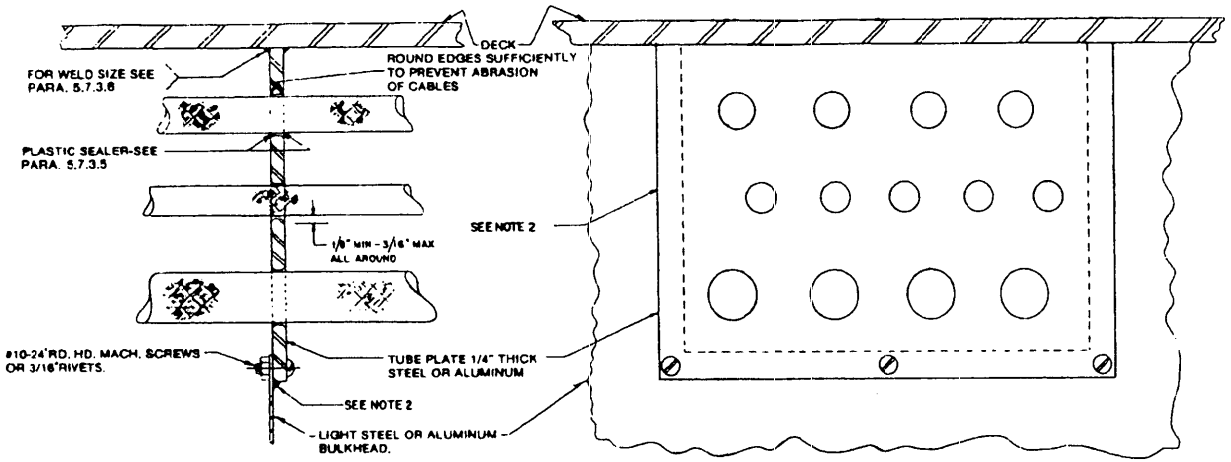
SH 132317279

FIGURE 4C7. Single cableway on steel and aluminum decks and bulkheads.

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4C81

MULTIPLE TIERS OF CABLES THROUGH LIGHT BULKHEAD WITH AIR AND FUME TIGHTNESS REQUIREMENTS



NOTES:

1. MINIMUM SPACING OF TUBES TO BE SUFFICIENT TO PERMIT SATISFACTORY WELDING.
2. PLACE BEAD OF RTV SEALER OR EQUIVALENT ALL AROUND EDGE.
3. THIS FIGURE SUPERSEDES SHEET 4C8 OF DRAWING 803-5001027 AND SECTION 4, SHEET 21, OF DRAWING NAVSEC NO. 9000-S8202-73980.

SH 132317280

FIGURE 4C8. Cableway through non-watertight bulkheads & beams without firestops.

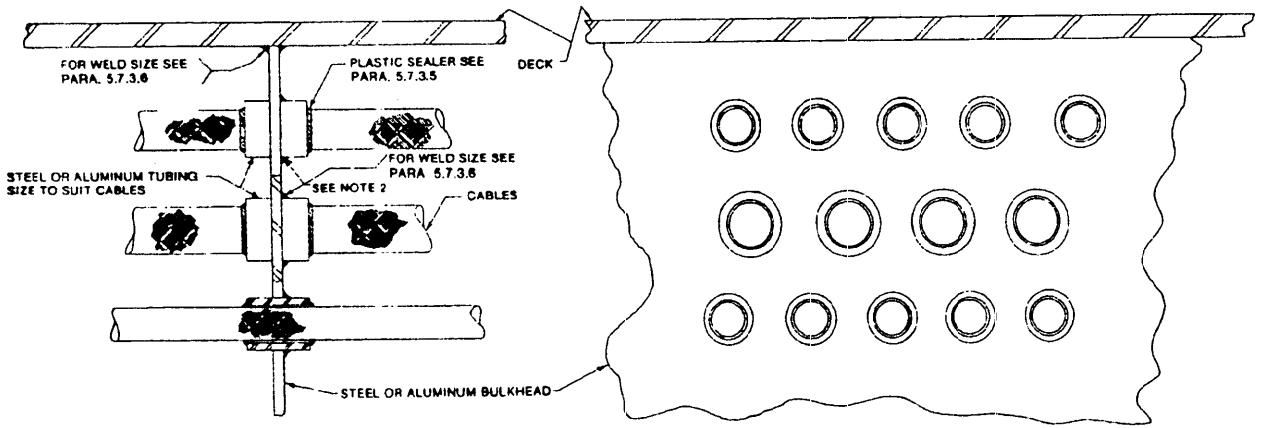
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NOTES:

1. MINIMUM SPACING OF TUBES TO BE SUFFICIENT TO PERMIT SATISFACTORY WELDING.
2. CHAFING NIPPLES USED TO PROTECT CABLES PASSING THRU BULKHEADS SHALL BE WELDED ON ONE SIDE OF THE BULKHEAD PENETRATED, EXCEPT FOR THE FOLLOWING TYPES OF BULKHEADS WHERE WELDS ON BOTH SIDES ARE REQUIRED
   A. BALLISTIC BULKHEADS
   B. SHELL PLATING
   C. TORPEDO DEFENSE BULKHEADS
3. THIS FIGURE SUPERSEDES SHEET 4C9 OF DRAWING 803-5001027 AND SECTION 4, SHEET 20, OF DRAWING NAVSEC NO. 9000-S5202-73980.

4C91

MULTIPLE TIERS OF CABLES THROUGH BULKHEADS
WITH AIR AND FUME TIGHTNESS REQUIRMENTS
CABLES THROUGH TUBING



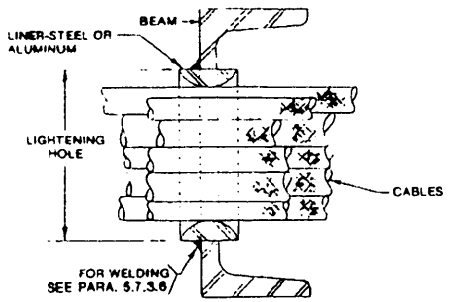
SH 132317281

FIGURE 4C9. Cableway through non-watertight bulkheads & beams without firestops.

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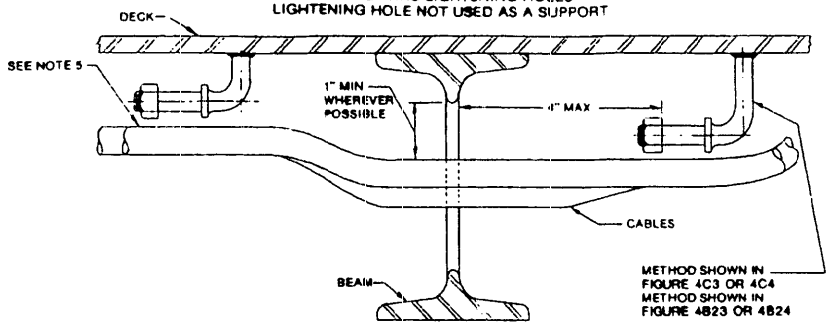
4C101

CABLES THROUGH LIGHTENING HOLES
OR CABLEWAYS IN STRUCTURE MEMBERS
(NON-LIGHT TIGHT)



4C102

A GROUP OF CABLES
PASSING THRU LIGHTENING HOLES
LIGHTENING HOLE NOT USED AS A SUPPORT

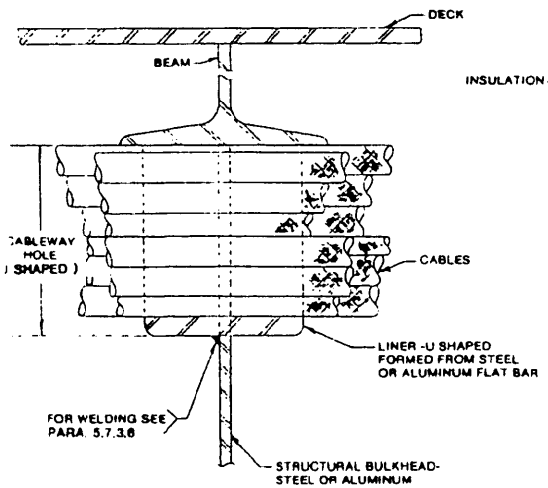


NOTES:

1. WHEN HOLES ARE CUT IN SHIPS STRUCTURE SOLELY FOR THE PASSAGE OF CABLES, THE LINER SHALL BE OF SUFFIC NT WEIGHT TO COMPENSATE FOR THE LOSS OF STRENG TH DUE TO THE METAL REMOVED.
2. LINERS 1/16" THICK CAN BE USED AS AN ALTERNATE TO HALF ROUND BAR.
3. INSTALLATION OF LINER IN BEAMS MAY BE MADE IN SHOP OR IN THE SHIP.
4. WHERE NO LIGHTENING HOLES ARE PROVIDED, CABLES MAY BE RUN OUTSIDE OF BEAM WHEN NO INTERFERENCE WITH EITHER EQUIPMENT OR NORMAL HEADROOM RESULTS.
5. THIS FIGURE SUPERSEDES SHEET 4C10 OF DRAWING 803-5001027 AND SECTION 4, SHEET 19, OF DRAWING, NAVSEC NO 9000-S6202-73980.

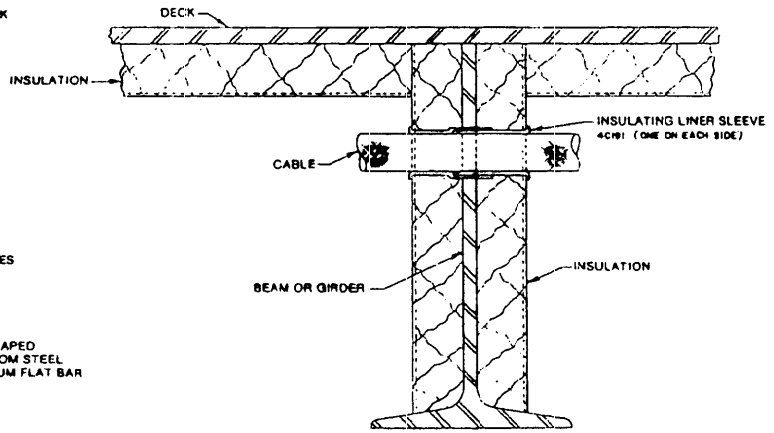
4C103

CABLEWAY IN STRUCTURAL BULKHEADS
(NON-LIGHT TIGHT)
ALTERNATE TO METHOD 4C101



4C104

SINGLE CABLE THROUGH BEAMS AND GIRDERS
WITH INSULATION, WITHOUT METAL SHEATHING.
ALTERNATE TO METHOD 4C112.



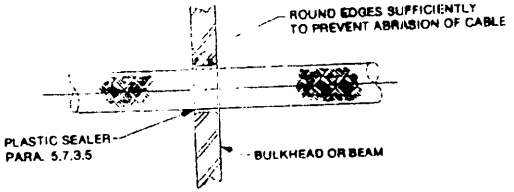
SH 132317282

FIGURE 4C10. Cableway through non-watertight bulkheads & beams without firestops.

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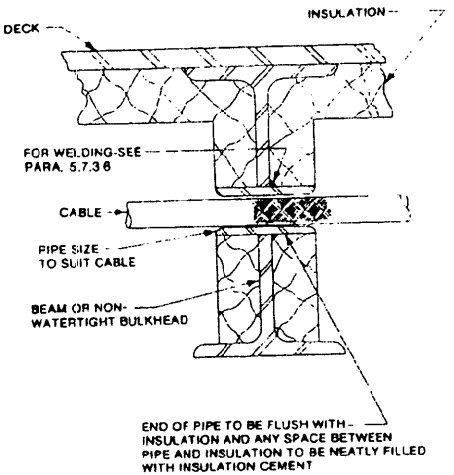
4C111

CABLES THROUGH BULKHEADS AND BEAMS
1/4" THICKNESS AND OVER



4C112

INSULATED BULKHEAD OR BEAM
CABLE THROUGH TUBING



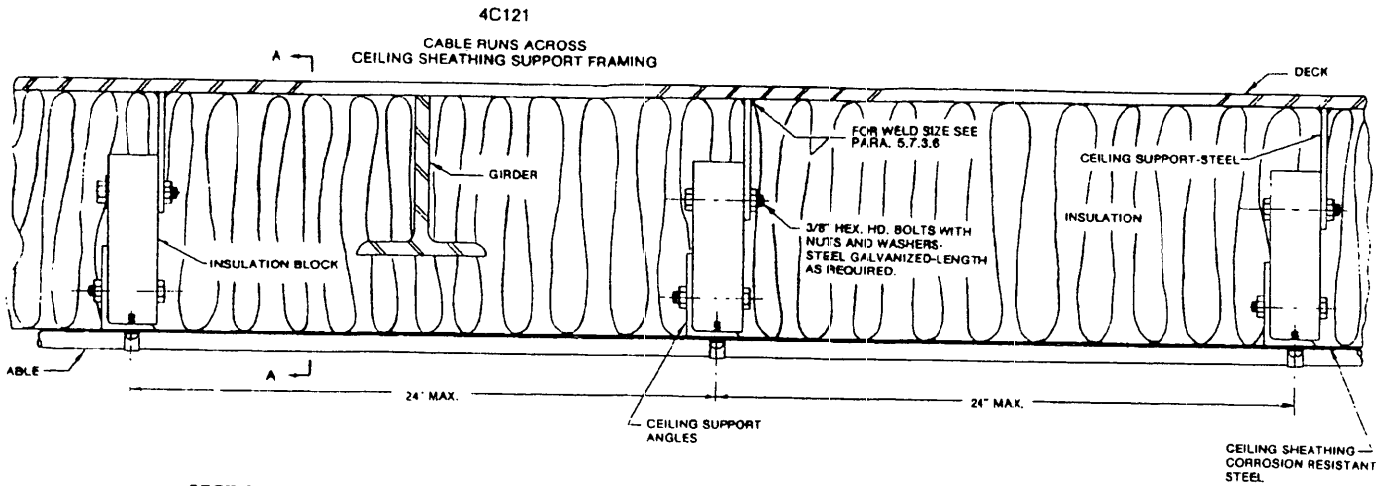
NOTES:

1. WHEN CABLES PIERCE NON-WATERTIGHT BULKHEADS REQUIRED TO BE LIGHT - TIGHT, THE SPACE BETWEEN THE CABLE AND INSIDE OF THE BUSHING TUBING OR DRILLED HOLE SHALL BE THOROUGHLY PACKED WITH PLASTIC SEALER SPECIFICATION MIL-I-3064, TYPE HF TO FORM A PERMANENT LIGHT-TIGHT SEAL.
2. INSTALLATION OF LINER IN BEAMS MAY BE MADE IN SHOP OR IN THE SHIP.
3. WHERE NO LIGHTENING HOLES ARE PROVIDED, CABLES MAY BE RUN OUTSIDE OF BEAM WHEN NO INTERFERENCE WITH EITHER EQUIPMENT OR NORMAL HEADROOM RESULTS.
4. DELETED
5. DELETED
6. THIS FIGURE SUPERSEDES SHEET 4C11 OF DRAWING 803-5001027 AND SECTION 4, SHEET 18, OF DRAWING, NAVSEC NO 9000-S6202-739180.

SH 132317283

FIGURE 4C11. Cableway through non-watertight bulkheads and beams without firestops.

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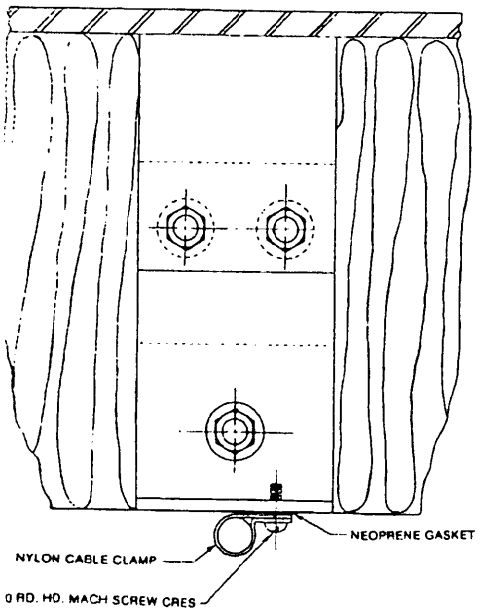


NOTES

1. THIS FIGURE SUPERSEDES SHEET 4C12 OF DWG. 803-5001027 AND SECTION 2 SHEET 119 OF DRAWING NAVSEC NO. 9000-S6202-73980.

SECTION A-A

INTERMEDIATE SUPPORT



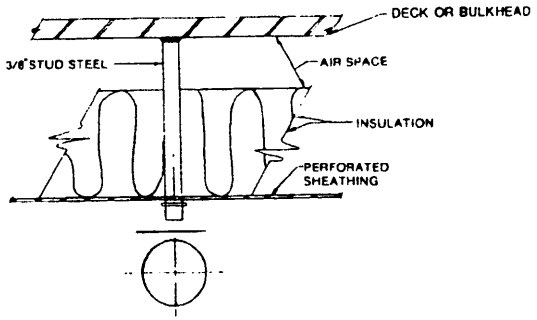
SH 132317284

FIGURE 4C12. Single cableway in refrigerated spaces.

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4C131

WATERTIGHT DECK OR BULKHEAD
WITH INSULATION AND SHEATING
HANGER SECURED WITH COLLAR STUD



NOTES:

1. METHODS SHOWN ON THIS FIGURE ARE FOR USE WITH CABLE UP TO 2 IN. IN DIA. FOR CABLES GREATER THAN 2 IN. IN DIA, USE METHODS SHOWN ON FIGURE 4C14.
2. THIS FIGURE SUPERSEDES SHEET 4C13 OF DRAWING 803-5001027 AND SECTION 2, SHEET 103, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

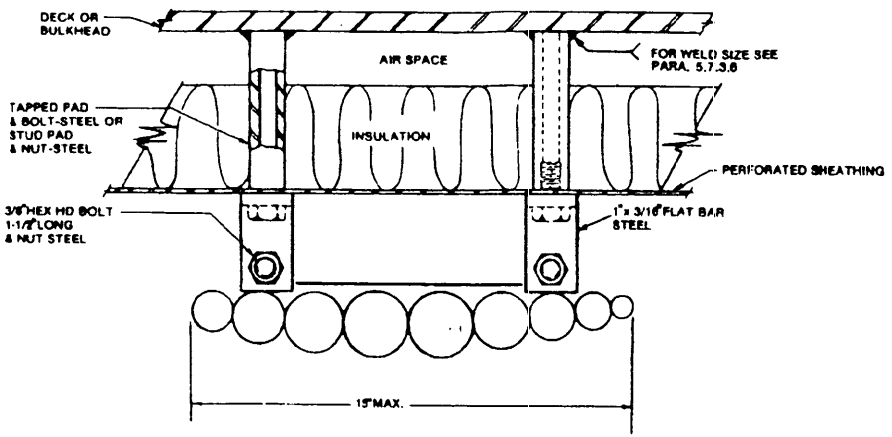
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SH 132317285

FIGURE 4C13. Single cableways in acoustical spaces.

4C141

DECKS OR BULKHEADS WITH INSULATION AND SHEATHING - CABLES SECURED TO STUD PADS OR TAPPED PADS



NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4C14 OF DRAWING 803-5001027 AND SECTION 2, SHEET 105, OF DRAWING NAVSEC NO. 9000-S6202-73980.

SH 132317286

FIGURE 4C14. Single cableways in acoustical spaces.

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NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4C15 OF DRAWING 803-5001027 AND SECTION 3, SHEET 19, OF DRAWING NAVSEC NO. 9000-98202-73880.

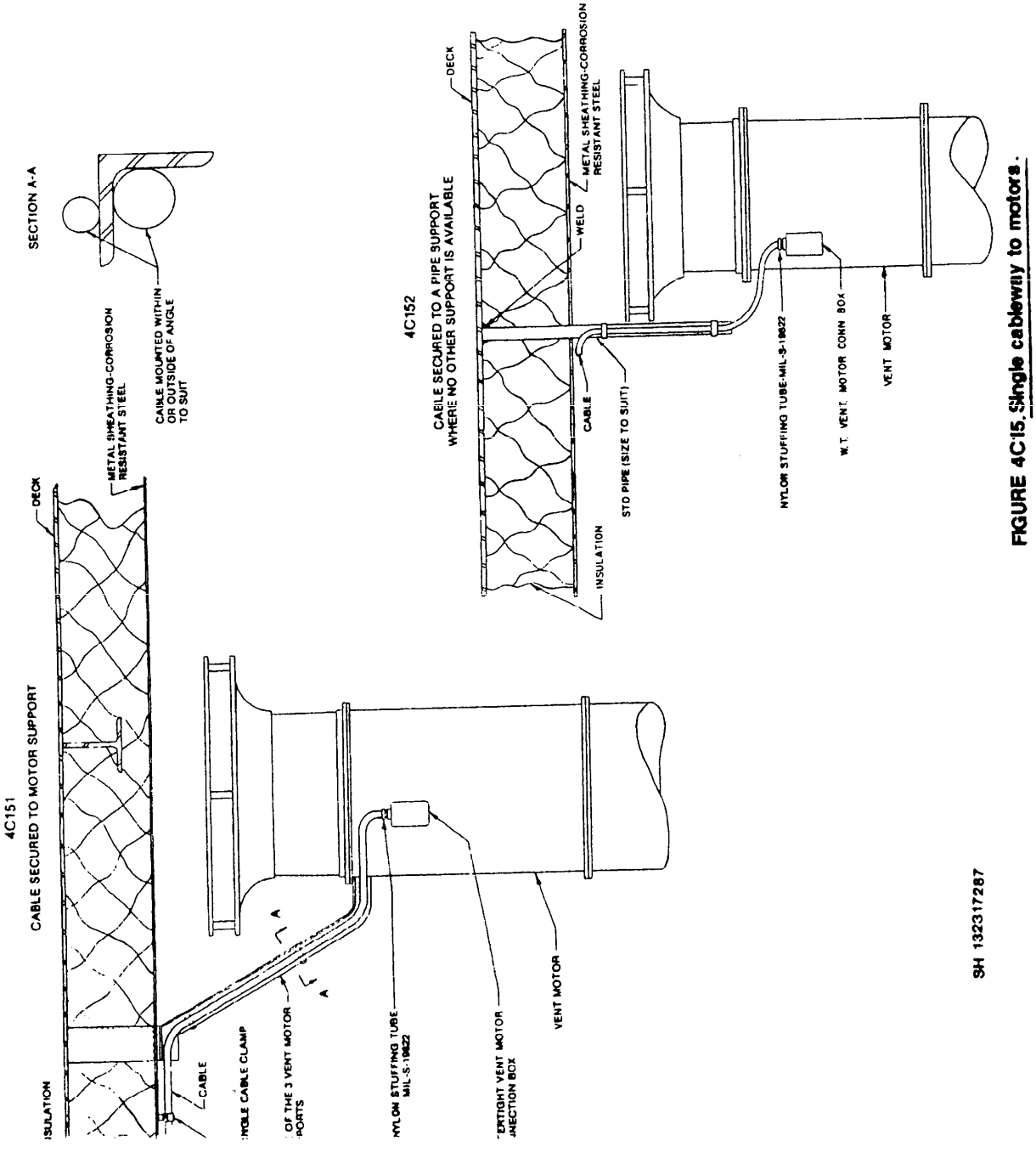


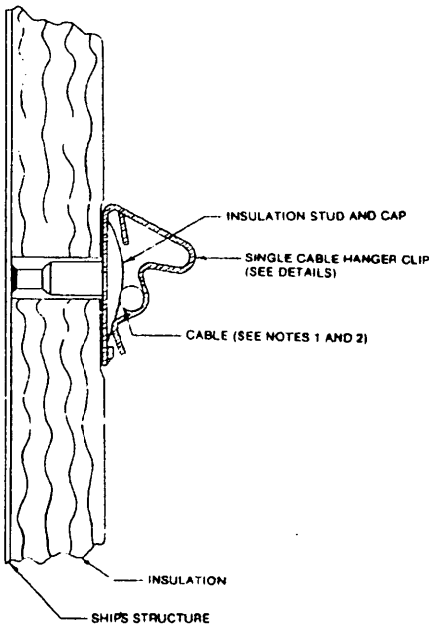
FIGURE 4C15. Single cableway to motors.

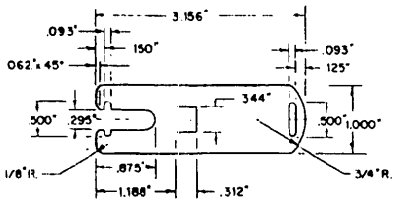
SH 132317287

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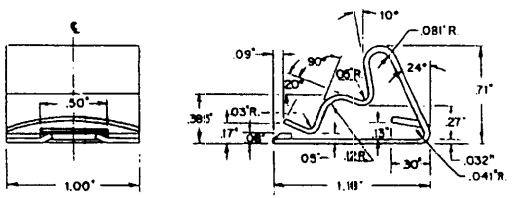
METHOD 4C161

SINGLE CABLE HANGER CLIP USED IN CONJUNCTION WITH INSULATION STUD CAP





BLANK CLIP (FULL SCALE)



FORMED CLIP (2 x SIZE)

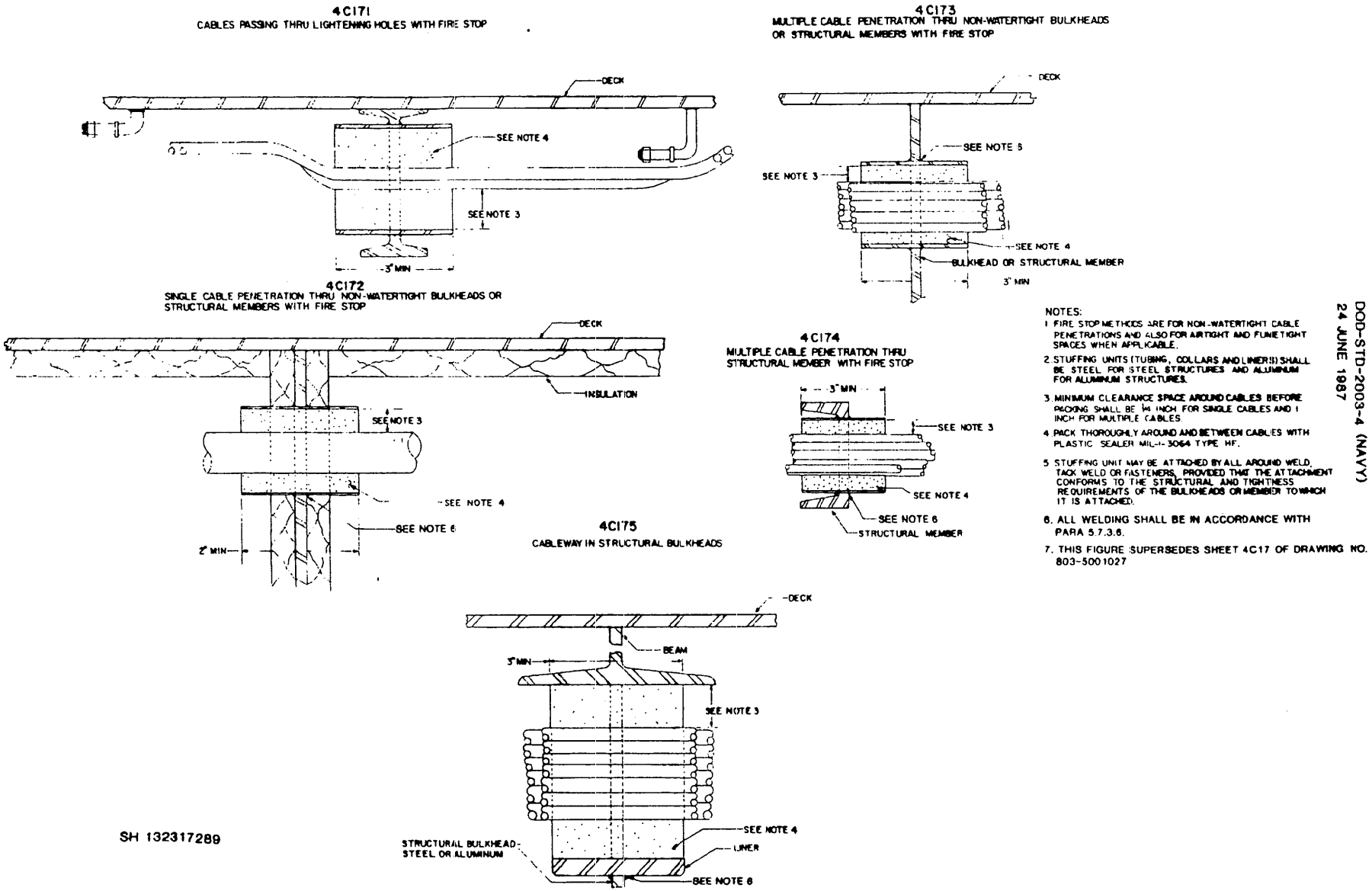
NOTES:

1. THIS HANGER CLIP IS FOR CABLE 1/2" IN DIAMETER OR LESS AND WEIGHING 1/4 POUND PER FOOT OR LESS.
2. WHERE A MAXIMUM LENGTH OF THE CABLE RUN IS INSTALLED ALONG THE SURFACE OF THE HULL BOARD INSULATION AND IT IS NECESSARY TO ORIGINATE OR TERMINATE THE RUN OUTSIDE THE INSULATION BOUNDARY THIS HANGER MAY BE USED FOR CONTINUATION OF THE RUN IF ADVANTAGEOUS TO DO SO.
3. MATERIAL: SPRING STEEL, SPHEROIDIZED, AISI C-1070, ROCKWELL HARDNESS 45-55, ZINC ELECTRO-PLATED, SPEC ASTM-B-633, TYPE I, CLASS SC2.
4. THIS FIGURE SUPERSEDES SHEET 4C16 OF DRAWING 803-5001027 AND SECTION 2, SHEET 197, OF DRAWING NAVSEC NO. 9000-86202-73980.

SH 132317288

FIGURE 4C16. Single cableway on hullboard insulation.

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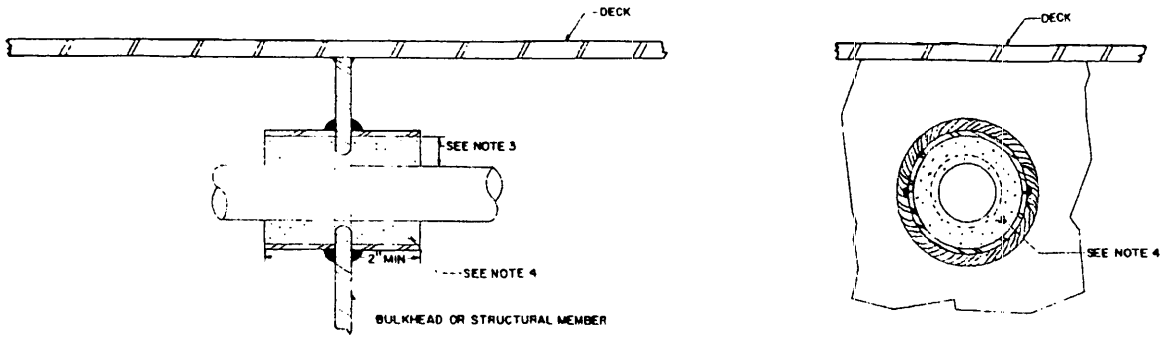
NOTES:
1. FIRE STOP METHODS ARE FOR NON-WATERTIGHT CABLE PENETRATIONS AND ALSO FOR AIRTIGHT AND FUMETIGHT SPACES WHEN APPLICABLE.
2. STUFFING UNITS (TUBING, COLLARS AND LINERS) SHALL BE STEEL FOR STEEL STRUCTURES AND ALUMINUM FOR ALUMINUM STRUCTURES.
3. MINIMUM CLEARANCE SPACE AROUND CABLES BEFORE PACKING SHALL BE ¼ INCH FOR SINGLE CABLES AND 1 INCH FOR MULTIPLE CABLES.
4. PACK THOROUGHLY AROUND AND BETWEEN CABLES WITH PLASTIC SEALER MIL-A-3064 TYPE HF.
5. STUFFING UNIT MAY BE ATTACHED BY ALL AROUND WELD, TACK WELD OR FASTENERS, PROVIDED THAT THE ATTACHMENT CONFORMS TO THE STRUCTURAL AND TIGHTNESS REQUIREMENTS OF THE BULKHEADS OR MEMBER TO WHICH IT IS ATTACHED.
6. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA 5.7.3.6.
7. THIS FIGURE SUPERSEDES SHEET 4C17 OF DRAWING NO. 803-5001027

SH 132317289

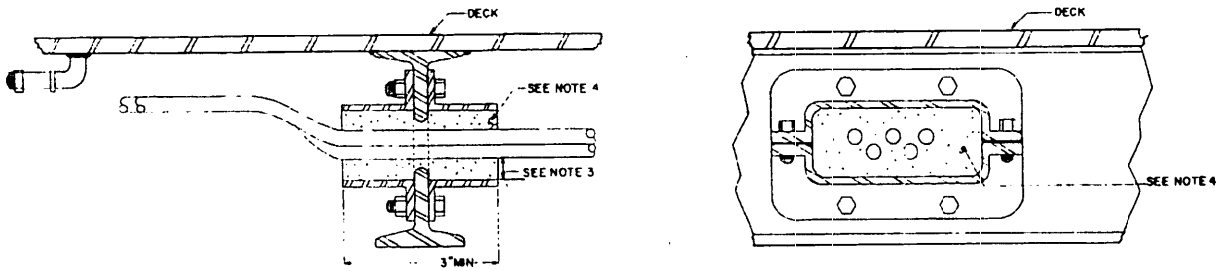
FIGURE 4C17. Cableway through non-watertight bulkheads and beams with fire stops.

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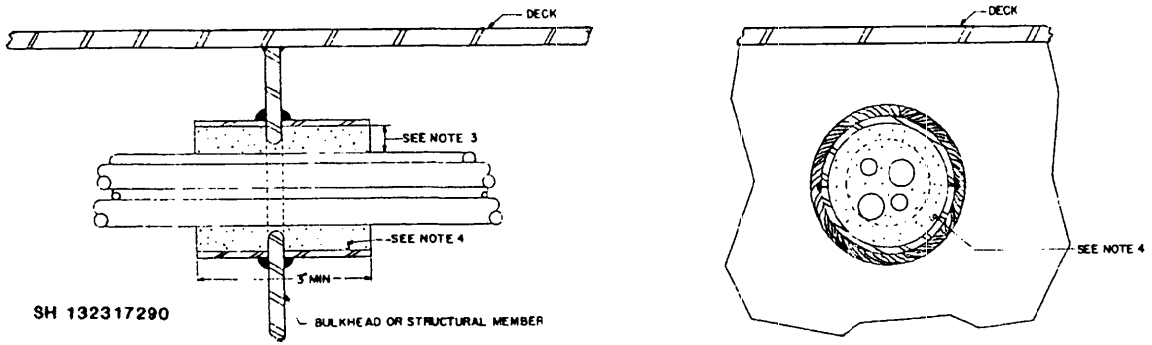
4C181
SINGLE CABLE PENETRATION THRU NON-WATERTIGHT BULKHEADS OR STRUCTURAL MEMBERS WITH FIRE STOPS.



4C182
CABLE PASSING THRU LIGHTENING HOLES WITH FIRE STOPS
MULTIPLE CABLE PENETRATION



4C183
MULTIPLE CABLE PENETRATION THRU NON-WATERTIGHT BULKHEADS OR STRUCTURAL MEMBERS WITH FIRE STOPS



SH 132317290

NOTES:

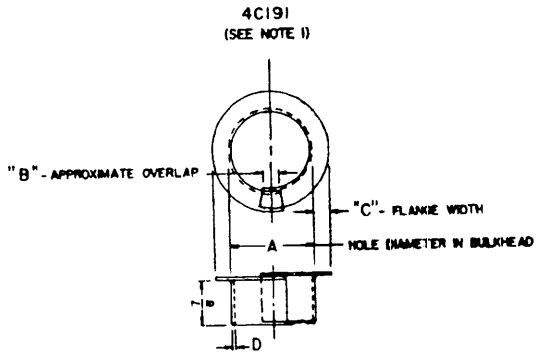
1. FIRE STOP METHODS ARE FOR NON-WATER TIGHT CABLE PENETRATIONS AND ALSO FOR AIR TIGHT AND FUMETIGHT
2. STUFFING UNITS (TUBING, COLLARS, AND LINERS) SHALL BE STEEL FOR STEEL STRUCTURES AND ALUMINUM FOR ALUMINUM STRUCTURES.
3. MINIMUM CLEARANCE AND SPACE AROUND CABLES BEFORE PACKING SHALL BE 1/4" INCH FOR SINGLE CABLES AND 1" INCH FOR MULTIPLE CABLES.
4. PACK THOROUGHLY AROUND AND BETWEEN CABLES WITH PLASTIC SEALER MIL-I-3064, TYPE HF.
5. STUFFING UNIT MAY BE ATTACHED BY ALL AROUND WELD, TACK WELD, OR FASTENERS PROVIDED THAT THE ATTACHMENT CONFORMS TO THE STRUCTURAL AND TIGHTNESS REQUIREMENTS OF THE BULKHEADS OR MEMBERS TO WHICH IT IS ATTACHED.
6. FOR EXISTING INSTALLATIONS STUFFING UNITS MAY BE SPLIT LENGTHWISE FOR INSTALLING OVER CABLE.
7. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA 6.7.3.6.
8. THIS FIGURE SUPERSEDES SHEET 4C18 OF DRAWING NO. 803-5001027.

FIGURE 4C18. Cableways through non-watertight bulkheads with fire stops (existing cable penetrations).

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NOTES:

1. THESE LINERS MAY BE OF COMERCIAL SIZES: THE MATERIAL SHALL BE OF VULCANIZED FIBER OR OTHER PLASTIC MATERIAL OF NON-TOXIC TYPE.
2. REFER TO METHOD 4C04.
3. THIS FIGURE SUPERSEDES SHEET 4C19 OF DRAWING NO. 803-5001027 AND SECTION 1, SHEET 85 OF DRAWING, NAVSEC NO 9000-S6202-73980.



A	B	C	D
1-5/32	3/16	1/8	.025
1-7/16	3/16	5/32	.025
1-13/16	1/4	5/32	.030
2-1/16	1/4	5/32	.030
2-9/32	5/16	5/32	.030
2-5/16	3/8	7/32	.040
3-3/8	3/8	7/32	.040
4-1/4	7/16	7/32	.040
3/4	3/16	1/8	.025

SH 132317291

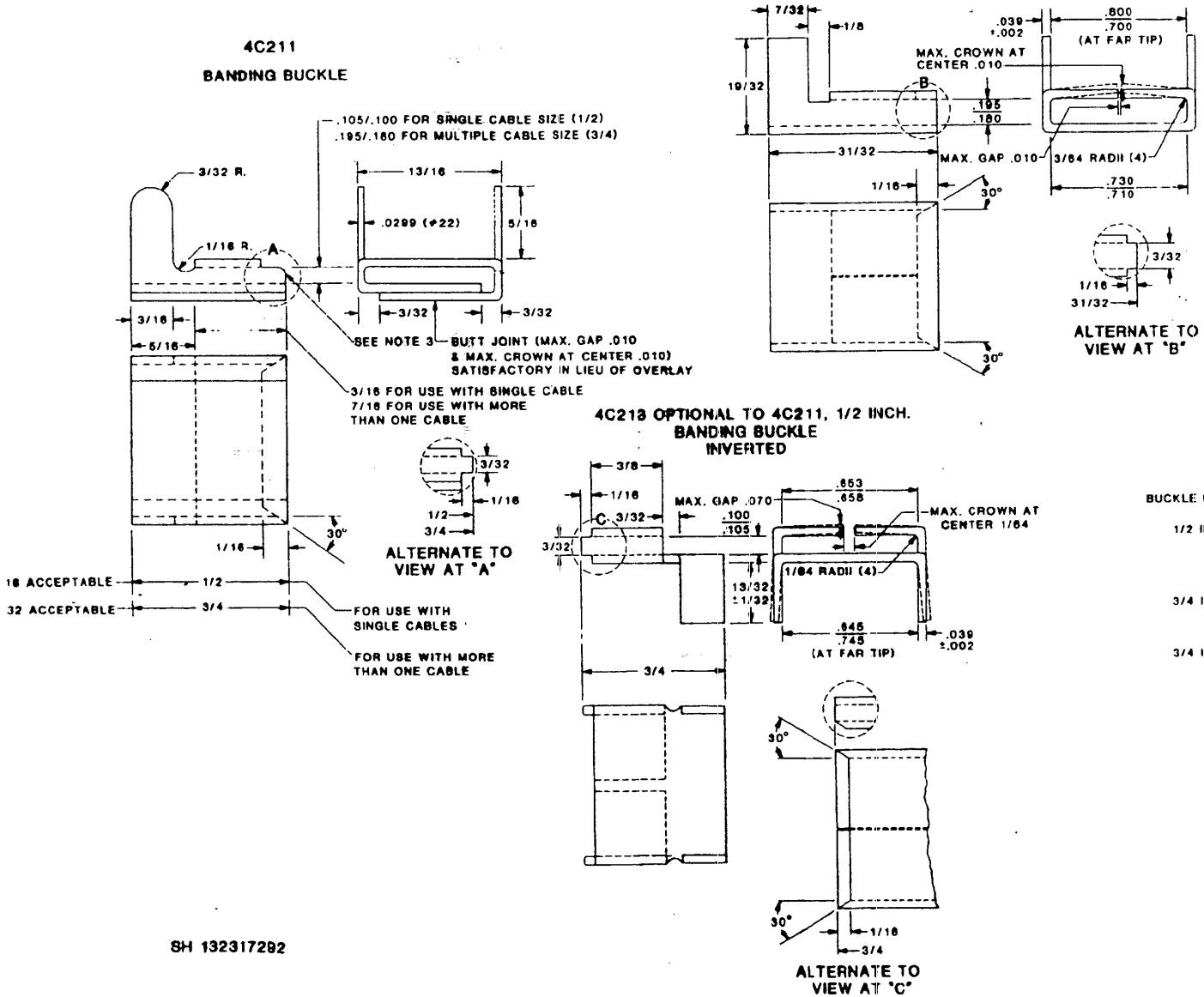
FIGURE 4C19. Insulating liners for cable holes in beams & girders.

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SH 132317292

NOTES

1. BANDING BUCKLES SHALL BE MADE OF CORROSION-RESISTANT STEEL, SPEC QQ-S-766, CL 304, COND A, FIN 1.
2. EDGES OF BANDING BUCKLE IN CONTACT WITH BANDING MATERIAL SHALL BE ROUNDED SLIGHTLY.
3. BANDING BUCKLES ARE AVAILABLE FROM THE FOLLOWING COMPANIES:

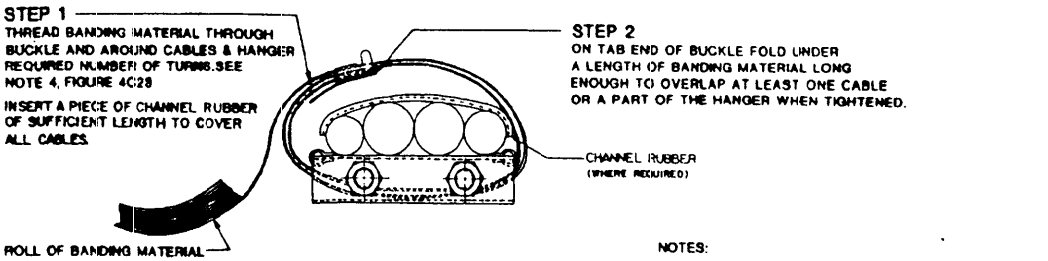
BUCKLE LENGTH	COMMERCIAL
1/2 INCH	A.J.GERRARD CO. MELROSE PARK, ILL. "INVERTED WING SEAL" (STAINLESS STEEL) NO. 202SS
3/4 INCH	A.J.GERRARD CO. NO. 204SS "WING SEAL" (STAINLESS STEEL).
3/4 INCH	INDEPENDENT METAL STRAP CO., INC. BROOKLYN, NEW YORK, "ROYAL WING SEAL" (STAINLESS STEEL).

4. THIS FIGURE SUPERSEDES SHEET 4C21 OF DRAWING 803-5001027.

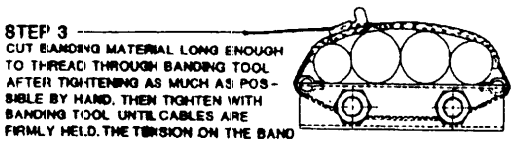
FIGURE 4C21. Cable banding buckle & misc. details.

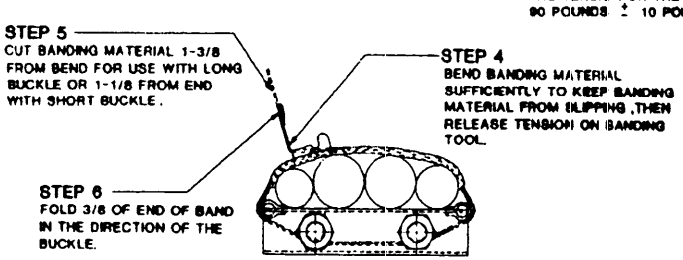
STRAPPING MATERIAL					
PIECE NUMBER	MATERIAL	WIDTH	THICKNESS	SPECIFICATIONS	FINISH
4C221	MILD STEEL	5/8	.015	QQ-S-781,TYPE 1 CLASS B,GRADE 1	ZINC ELECTRO-PLATED
4C222	CRES.	5/8	.015	MIL-S-17996 CLASS 1	2D
4C223	ALUMINUM	5/8	.025	QQ-A-327 TEMPER T6	NONE
4C224	MILD STEEL	5/8	.020	QQ-S-781,TYPE 1 CLASS B,GRADE 1	ZINC ELECTRO-PLATED
4C225	CRES.	5/8	.020	MIL-S-17996 CLASS 1	2D

CHANNEL RUBBER			
PIECE NUMBER	MATERIAL	SIZE	SPECIFICATIONS
4C226	RUBBER	7/8 W x 5/8 CH x 5/16 H	MIL R 6855 CL 2, GR 60



NOTES:
1. THIS FIGURE SUPERSEDES SHEET 4C22 OF DRAWING 803-5001027.





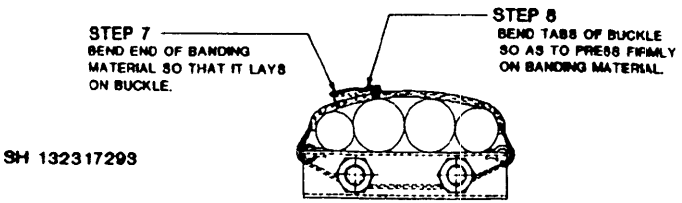
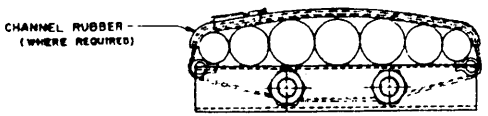


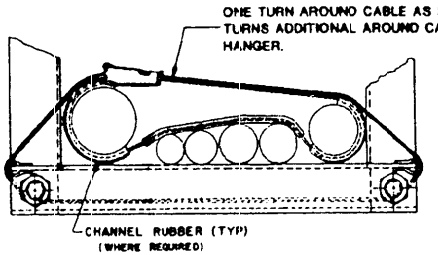
FIGURE 4C22. Applying banding material.

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Check the source to verify that this is the current version before use.

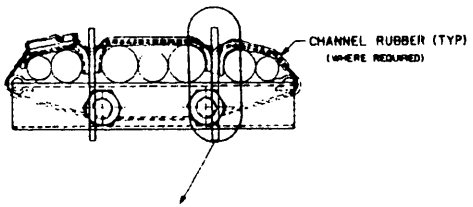
FULLY LOADED HANGERS 4C231

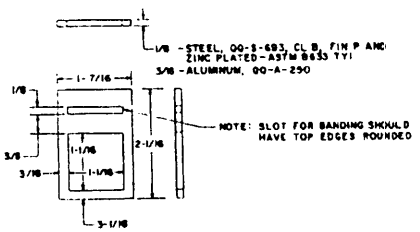


HANGERS WITH LARGE & SMALL CABLES INTERMIXED 4C233



PARTIALLY LOADED HANGERS 4C232

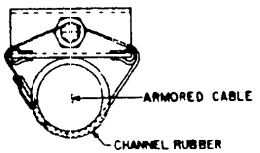




4C234

USE OF CHANNEL RUBBER FOR SOUND ISOLATION

(SEE NOTE 10)



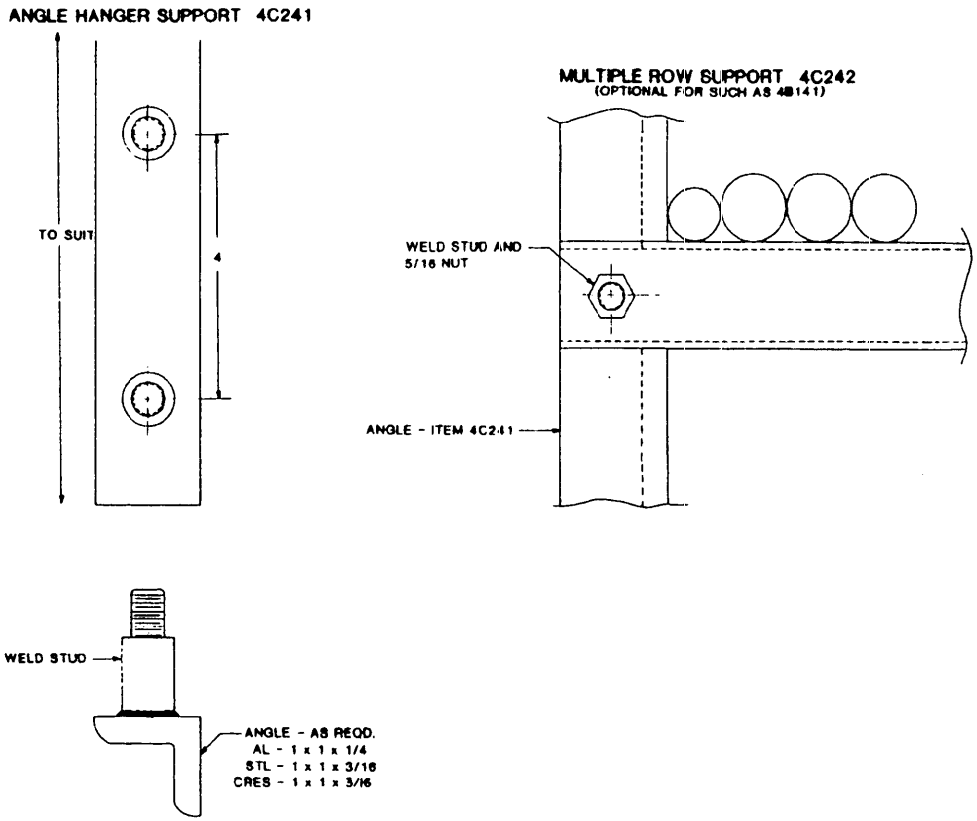
SH 132317294

FIGURE 4C23. Details of partially and full loaded banding hangers.

NOTES

1. FOR FULLY LOADED HANGERS (WHERE HANGER WIDTH DOES NOT EXCEED THE WIDTH OF A SINGLE CABLE, OR A SINGLE ROW OF CABLES BY MORE THAN 1/2 INCH) USE METHOD 4C231.
2. FOR PARTIALLY LOADED HANGERS (WHERE HANGER WIDTH EXCEEDS THE WIDTH OF A SINGLE CABLE OR A SINGLE ROW OF CABLES BY MORE THAN 1/2 INCH) USE METHOD 4C232.
3. THE LENGTH OF BANDING HANGER SHALL NOT BE LESS THAN THE OVERALL WIDTH OF CABLES FOR A SINGLE CABLE OR A ROW OF CABLES.
4. BAND REQUIREMENTS ARE AS FOLLOWS: SINGLE CABLES 1 IN. DIA.AND UNDER - .015 THICK - 1 TURN. SINGLE CABLES OVER 1 IN. DIA. - .015 THICK - TWO TURNS MULTIPLE CABLES 2 IN. DIA. AND UNDER - .015 THICK - 2 TURNS. MULTIPLE CABLES OVER 2 IN. DIA. .015 THICK, 3 TURNS OR .020 IN. THICK, 2 TURNS.
5. CABLES SHOULD BE ARRANGED SO THAT THE LARGER CABLES ARE IN THE CENTER OF THE HANGER. IF POSSIBLE.
6. IN NONMAGNETIC MINESWEEPERS USE CRES BAND EXCEPT THAT ALUMINUM BAND ITEM 4C223 (.025 THICK) MAY BE USED WHERE CABLES ARE UNDER 2 IN. IN DIA. AND HANGER SPACING IS 21 INCHES ON CENTER, MAXIMUM.
7. FOR WEATHER DECKS USE CRES BAND, SIZES AS PER NOTE 4.
8. THE BANDING METHOD SHALL BE USED TO SUPPORT COAXIAL CABLES EXCEPT COAXIAL CABLES OF THE SEMI-SOLID DIELECTRIC TYPE.
9. SEMI-SOLID DIELECTRIC COAXIAL CABLES SHALL BE SUPPORTED BY THE CONTOUR STRAP METHODS, SUCH AS METHODS ON FIGURE 4C6.
10. CHANNEL RUBBER SHALL BE INSTALLED AROUND ARMORED CABLES TERMINATING AT SOUND ISOLATED EQUIPMENT.
11. THIS FIGURE SUPERSEDES SHEET 4C23 OF DRAWING 803-5001027.

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SH 132317295

FIGURE 4C24. Angle hangers support details.

NOTES:

1. ALUMINUM AND STEEL ANGLES MAY BE PURCHASED IN COMMERCIAL LENGTHS AND CUT TO THE DIMENSIONS REQUIRED FOR THE COMPONENT PARTS OF AN INSTALLATION.
2. THIS FIGURE SUPERSEDES SHEET 4C24 OF DRAWING 803-5001027.

MATERIAL:

ANGLE HANGER SUPPORTS SHALL BE ALUMINUM ALLOY 6061-T6 OR STEEL COMMERCIAL GRADE.

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NOTES:
1. SQUARE TUBING MAY BE PURCHASED IN COMMERCIAL LENGTHS.
2. CUT TUBULAR HANGERS TO THE LENGTHS REQUIRED IN TABULATIONS
3. TUBULAR HANGERS SHOWN HEREON MAY BE USED AS ALTERNATE TO TUBULAR HANGERS FERRULES NOT REQUIRED.
4. THIS FIGURE SUPERSEDES SHEET 4C25 OF DWG 803-5001027

4C251 - ALUMINUM
4C252 - STEEL
4C253 - CRES

SIDE MOUNTED TUBULAR HANGER (SEE NOTE 3)

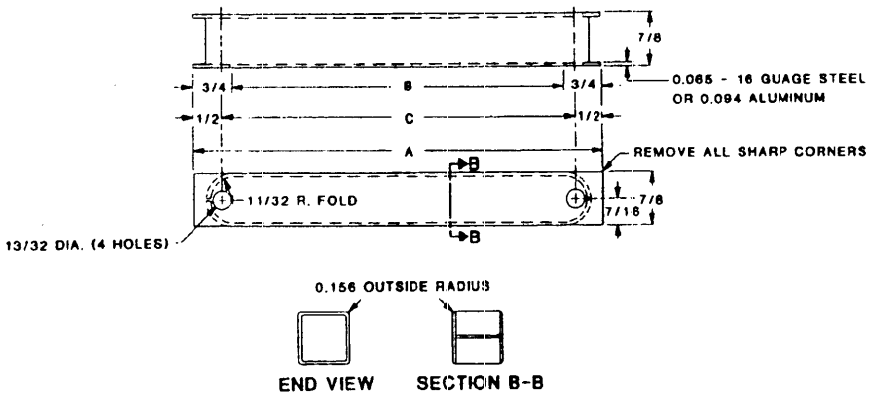
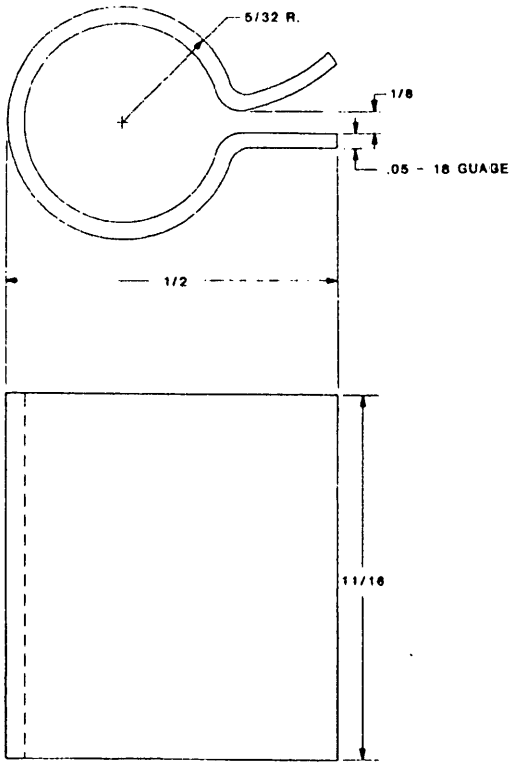


TABLE OF DIMENSIONS			
ITEM NO.	LENGTH A	LENGTH B	MOUNTING DIM. C
4C251 4C252 4C253	3	1-1/2	2
	4	2-1/2	3
	5	3-1/2	4
	6	4-1/2	5
	7	5-1/2	6
	8	6-1/2	7
	9	7-1/2	8
	10	8-1/2	9
	11	9-1/2	10
	12	10-1/2	11
	13	11-1/2	12
	14	12-1/2	13
	15	13-1/2	14
	16	14-1/2	15
	17	15-1/2	16

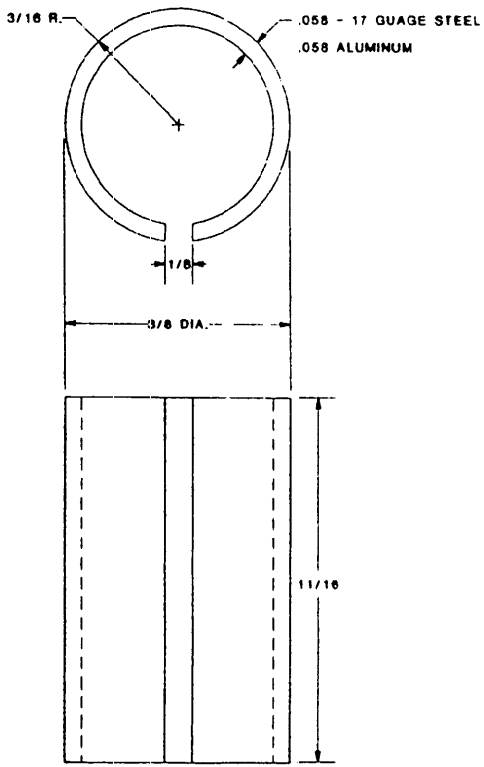
FIGURE 4C25. Details of tubular hanger.

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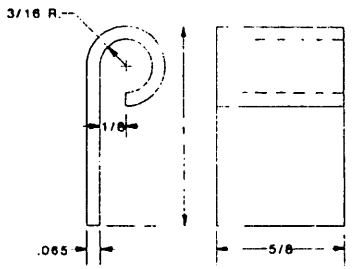
4C261 - ALUMINUM
4C262 - MILD STEEL
4C263 - CRES
FERRULE
(SEE NOTES 2 & 3)



4C264 - STEEL
4C465 - ALUMINUM
FERRULE
(SEE NOTES 2 & 3)



4C266 - STEEL
FERRULE
(SEE NOTE 2)



NOTES:

1. FERRULES SHALL BE FORMED OR EXTRUDED STEEL, ALUMINUM OR NYLON, COMMERCIAL GRADE.
2. FERRULES, ITEMS 4C264 OR 4C266, MAY BE USED AS ALTERNATES TO ITEM 4C262.
3. FERRULE, ITEM 4C265 MAY BE USED AS AN ALTERNATE TO ITEM 4C261.
4. THIS FIGURE SUPERSEDES SHEET 4C26 OF DWG. 803-5001027.

FIGURE 4C26. Tubular banding hangers ferrule details.

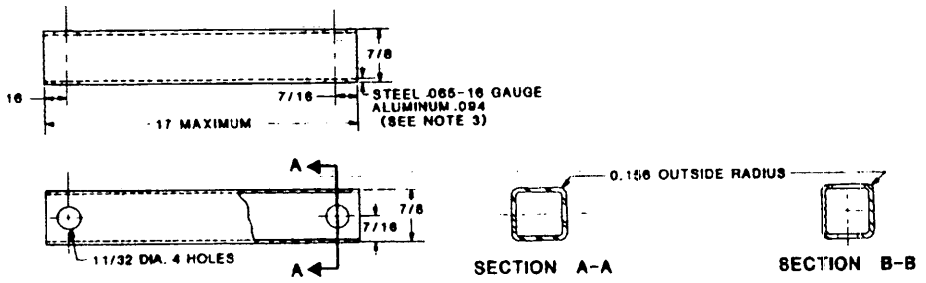
SH 132317297

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NOTES:

1. SQUARE TUBING MAY BE PURCHASED IN COMMERCIAL LENGTHS.
2. CUT TUBULAR HANGERS TO THE LENGTH REQUIRED IN THE TABULATIONS.
3. ALL TUBING MAY BE ONE INCH SQUARE IN LIEU OF 7/8 AS ALTERNATE.
4. TUBULAR HANGERS, ITEM 4C251,4C252,& 4C253 TO BE USED AS ALTERNATE TO 4C271,4C272 & 4C273.
5. THIS FIGURE SUPERSEDES SHEET 4C27 OF DWG 803-5001027.

4C271 - ALUMINUM
4C272 - STEEL
4C273 - CRES
SIDE MOUNTED TUBULAR HANGER (SEE NOTE 4)



4C274 - ALUMINUM
4C275 - STEEL
4C276 - CRES
BOTTOM MOUNTED TUBULAR HANGER

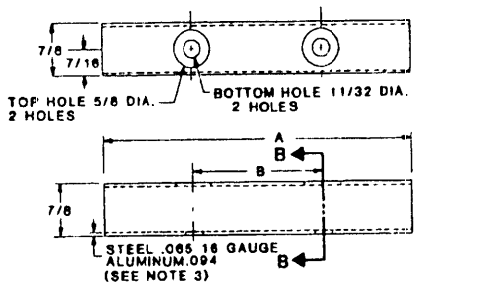


TABLE OF DIMENSIONS									
HANGER LENGTH	A	3	4	5	6	7	8	9	TO 17 MAXIMUM
HOLE SPACING	B	2	2	2	2	6-1/2	6-1/2	5-1/2	DIM. A MINUS 7/16

MATERIAL:

ALL TUBING SHALL BE SEAMLESS DRAWN & IN ACCORDANCE WITH SPEC. QQ-A-270 TEMPER 6

STEEL TUBING SHALL BE HOT ROLLED WELD SEAMFLASH IN COMMERCIAL GRADE

FIGURE 4C27. Details of tubular hanger.

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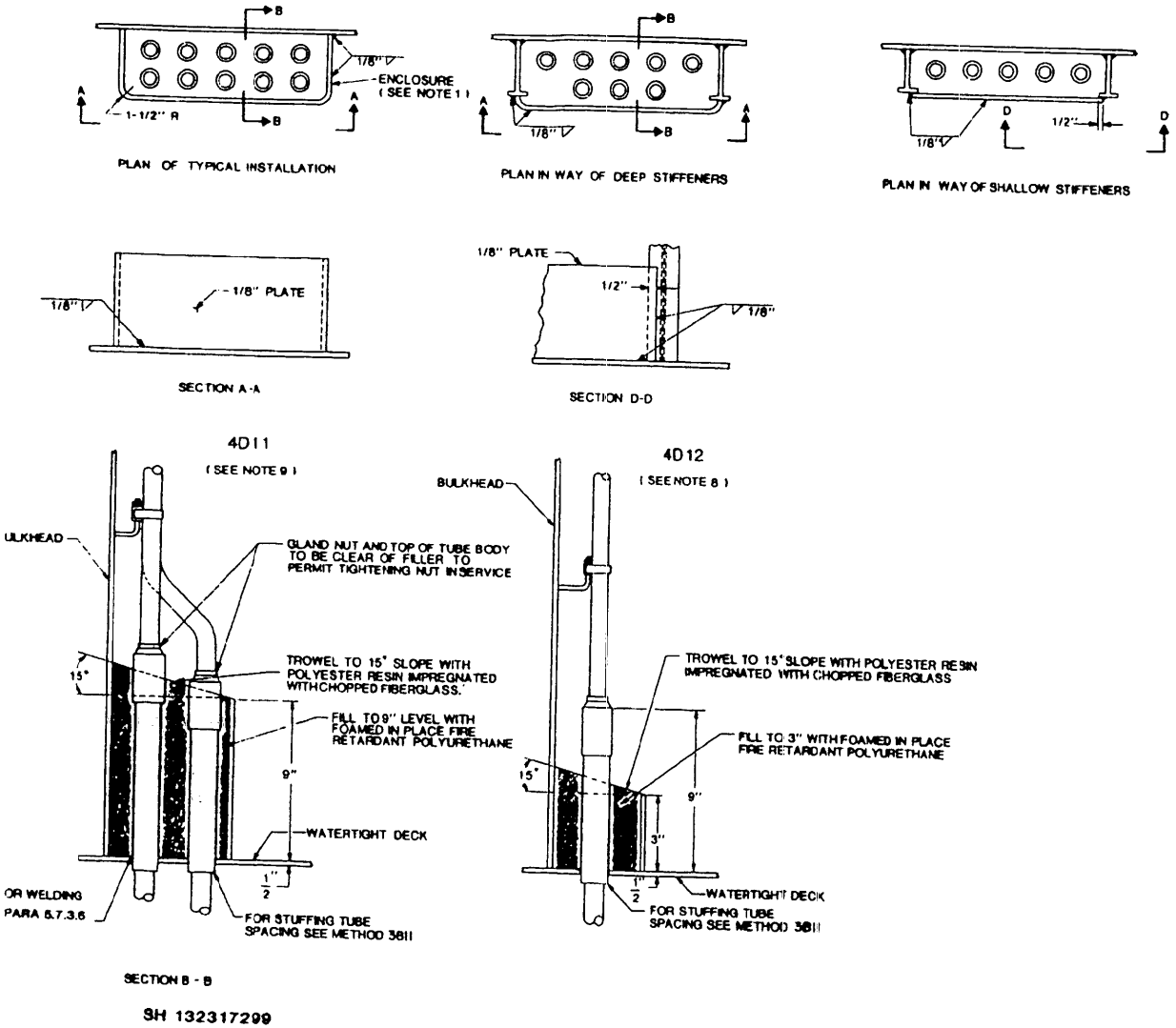
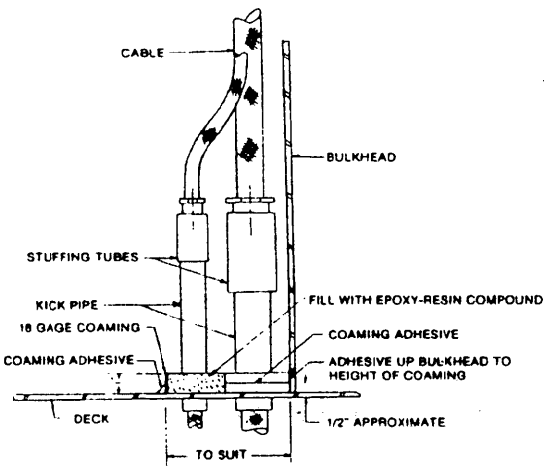


FIGURE 4D1. Protection of cables penetrating watertight decks.

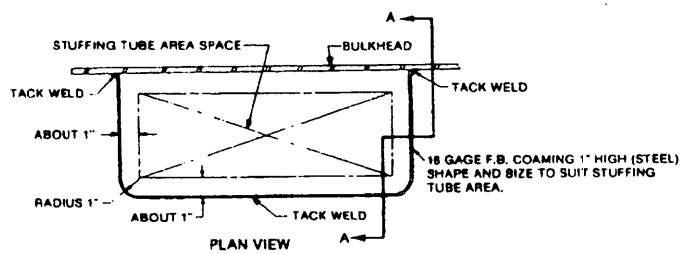
NOTES

1. LENGTH AND WIDTH OF ENCLOSURE TO SUIT CABLE GROUP. THE SIZE OF THE STUFFING TUBE GROUP IN DECKS AND BULKHEADS SHALL BE LIMITED SUCH AS TO PERMIT RETIGHTENING OF ALL GLAND NUTS IN THE GROUP BY SHIPS FORCE USING STUFFING TUBE WRENCH SET FEDERAL SPEC. GGG-W-646, TYPE II, CLASS I, STYLE A, FORM B, TABLE VI (NSN 5120-00-293-0013)
2. SURFACES IN CONTACT WITH PLASTICS TO BE THOROUGHLY CLEANED AND FREE OF OIL OR GREASE.
3. THE FINISHED 15° SURFACE SHALL BE SMOOTH.
4. POLYESTER RESIN IN ACCORDANCE WITH MIL-R-21607 COMMERCIAL SOURCE - OWENS CORNING OR SIMILAR. CHOPPED FIBERGLASS MIL-Y-1140 COMMERCIAL SOURCE OWENS CORNING OR SIMILAR. FIRE RETARDANT POLYURETHANE FOAM MIL-P-21929 SOURCE - GOLD BOND ZERO -O - CFL NATIONAL GYPSUM CO. OR SIMILAR.
5. THESE ENCLOSURES MAY BE USED ON STEEL OR ALUMINUM ALLOY STRUCTURES USING 1/8" STEEL OR 1/8" ALUMINUM TO SUIT.
6. THESE ENCLOSURES SHALL NOT BE INSTALLED UNTIL AFTER ALL CABLES HAVE BEEN INSTALLED AND COMPARTMENTS CHECKED FOR TIGHTNESS.
7. THESE ENCLOSURES SHALL NOT BE INSTALLED ON THE WEATHER DECKS.
8. COLLAR HEIGHT SHALL BE 3" IN DRY AREAS BELOW DECKS.
9. COLLAR HEIGHT SHALL BE 9" IN WET DECK SPACES BELOW DECKS SUCH AS WASHROOMS AND SCULLERIES.
10. A GLASS REINFORCED PLASTIC COLLAR MAY BE USED IN LIEU OF METAL COLLAR.
11. SEE ALTERNATE METHOD FIGURE 4D2.
12. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.6.
13. THIS FIGURE SUPERSEDES SHEET 4D1 OF DRAWING 803-5001027 AND SECTION 4, SHEET 158, OF DRAWING, NAVSEC NO. 9000-S6202-739810.

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TYPICAL SECTION A-A



TYPICAL DECK STUFFING TUBE AREA

SH 132317300

METHOD 4D21
(ALTERNATE METHOD 4D11)

1. FOR STUFFING TUBE AREAS OTHER THAN IN THE WEATHER, A METHOD USING AN EPOXY-COMPOUND RESIN ON THE DECK AROUND THE KICK-PIPES SHALL BE ACCOMPLISHED AS FOLLOWS:

A. AFTER THE KICK-PIPE AND STUFFING TUBES HAVE BEEN INSTALLED AND THE CABLES PULLED IN, THE 16 GAGE STEEL COAMING SHALL BE FABRICATED TO SUIT AND TACK-WELDED. WHERE DECKS HAVE CAMBER/SHEAR THE COAMING SHALL BE FABRICATED TO MATCH THE SLOPE OF THE DECK SO THAT THE EPOXY COMPOUND WILL BE LEVEL WITH THE TOP EDGE OF THE ENTIRE COAMING. THIS MAY NECESSITATE THE COAMING BE MADE AND TACK WELDED IN THREE SEPARATE PIECES IN LIEU OF ONE CONTINUOUS FLAT BAR AS SHOWN IN PLAN VIEW. THE HIGH SIDE OF THIS TYPE OF COAMING SHALL NOT EXCEED 1-1/2", AND THE CORNERS SHALL BE ROUNDED.

B. THE DECK AROUND THE KICK-PIPES WITHIN THE COAMING ENCLOSURE AND THE DECK AROUND THE OUTSIDE OF THE COAMING SHALL BE THOROUGHLY CLEANED AS FOLLOWS (IT IS NOT NECESSARY TO REMOVE PAINT):

(1) BLOW OUT ALL DIRT AND LOOSE MATERIAL.
(2) SCRUB WITH A DETERGENT SOLUTION.
(3) RINSE WITH FRESH WATER.
(4) BLOW OUT EXCESS WATER.

C. PUT A SMALL AMOUNT OF PHILADELPHIA RESIN CO. ADHESIVE RESIN (OR EQUIVALENT) INTO A SHALLOW CONTAINER OR ON A FLAT PLATE. ADD AN EQUAL AMOUNT OF PHILADELPHIA RESIN CO. ADHESIVE HARDNER (OR EQUIVALENT) AND MIX THOROUGHLY UNTIL THE MIXTURE IS UNIFORM IN COLOR. (3 TABLESPOONS EACH OF RESIN AND HARDNER WILL PROVIDE ENOUGH ADHESIVE FOR ABOUT 3 LINEAL FEET OF COAMING.)

D. APPLY THIS ADHESIVE "PUTTY" AROUND THE OUTSIDE OF THE COAMING WITH A ROUNDED END 3/4" WIDE SPATULA (OR EQUIVALENT) AND SHAPE IT TO A FILLET WITH THE END OF THE SPATULA. DIP THE SPATULA IN WATER TO PROVIDE A SMOOTH SURFACE.

E. POUR THE PHILADELPHIA RESIN CO. (OR EQUIVALENT) PRS10FR HARDENER (BLACK LABELED CAN) INTO THE WHITE LABELED CAN CONTAINING PRS10FR RESIN AND MIX THOROUGHLY FOR ABOUT 5 MINUTES UNTIL THE MIXTURE IS A UNIFORM LIGHT GRAY COLOR AND ANY SOLID MATERIAL IN THE BOTTOM OF THE CAN HAS DISPERSED. DO NOT THIN THE MIXTURE. (MIXING CAN BE ACCOMPLISHED BY HAND OR WITH A SMALL ELECTRIC DRILL WITH A PAINT MIXER ATTACHMENT)

F. POUR THIS MIXTURE INTO THE KICK-PIPE AREA ENCLOSURE UNTIL IT BEGINS TO OVERFLOW THE COAMING. WIPE EXCESS OFF OUTSIDE OF COAMING. ALLOW TO CURE HARD BEFORE PAINTING.

2. THE PRS10FR MATERIAL IS PACKAGED IN 6 LB. UNITS (1/2 GALLON SIZE) AND 12 LB. UNITS (1 GALLON SIZE). THE FOLLOWING IS A GUIDE FOR INSTALLATION:

6 LB. UNIT WILL COVER APPROX. A 9" x 12" TUBE AREA
12 LB. UNIT WILL COVER APPROX. A 12"x18" TUBE AREA
6 LB. UNIT CONSISTS OF ½ GAL. RESIN AND 1 PINT HARDENER
12 LB. UNIT CONSISTS OF 1 GAL. RESIN AND 1 QUART HARDENER

THE OV ADHESIVE MATERIAL IS PACKAGED IN SIZE UNITS (CONSISTING OF 1 QUART SIZE RESIN AND 1 QUART SIZE HARDNER).

3. CURE TIME IS 4 HOURS AT 85° F, 8 HOURS AT 75° F AND 12 HOURS AT 65° F. USE HEAT LAMPS FOR TEMPERATURES BELOW 60° F. FOR CURING.

POT LIFE IS 45 MINUTES WHICH MEANS THAT THE EPOXY-RESIN COMPOUND SHOULD BE USED WITHIN 45 MINUTES AFTER MIXING. SHELF LIFE OF THE MATERIAL BEFORE MIXING IS 2 YEARS.

4. LOCAL EXHAUST VENTILATION IS NOT NECESSARY WHEN USING THESE EPOXY MATERIALS, HOWEVER, GOOD PERSONAL HYGIENE HABITS SHOULD BE USED AND CONTACT WITH THE SKIN SHOULD BE AVOIDED. AFTER USING THE EPOXY MATERIAL, WASH WITH BORAX SOAP OR WATERLESS SKIN CLEANER. SOLVENTS SHOULD BE AVOIDED AS A SKIN CLEANER BUT IF NECESSARY, ALCOHOL SHOULD BE USED (DENATURED ETHYL OR ISOPROPYL ARE PREFERRED). IN NO CASE SHOULD PETROLEUM SOLVENTS OR CHLORINATED HYDROCARBONS SUCH AS GASOLINE, KEROSENE, BENZENE, DEGREASING SOLVENTS OR DRY CLEANING FLUIDS BE USED FOR CLEANING THE SKIN.

NOTE:

1. THIS FIGURE SUPERSEDES SHEET 4D2 OF DWG. 803-5001027 AND SECTION 4, SHEET 157, OF DRAWING, NAVSEC NO. 9000-98202-73980.

FIGURE 4D2. Protection of cables penetrating watertight decks.

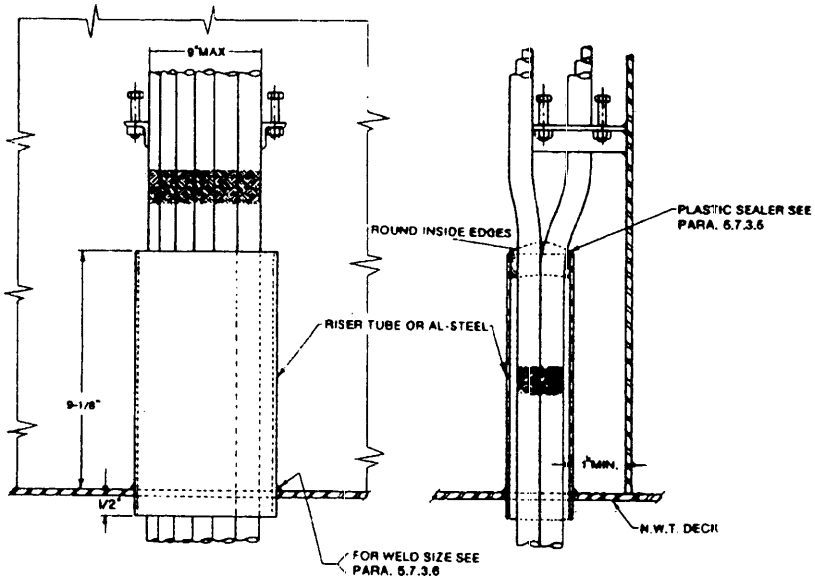
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NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4D3 OF DRAWING 803-5001027 AND SECTION 4, SHEET 37, OF DRAWING NAVSEC NO. 9000-S6202-73960.

4D31

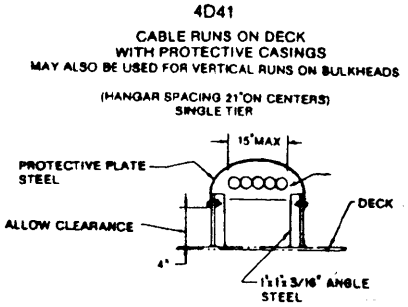
FOR PROTECTING THREE OR MORE CABLES PASSING THROUGH NON-WATERTIGHT DECKS.

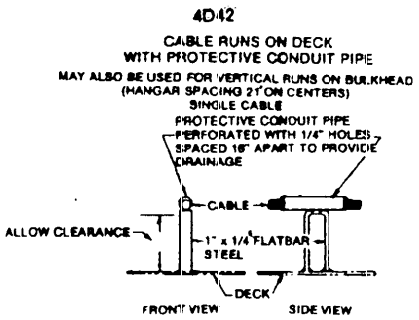


SH 132317301

FIGURE 4D3. Protection of cables penetrating NTW deck.

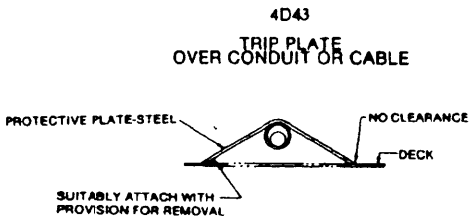
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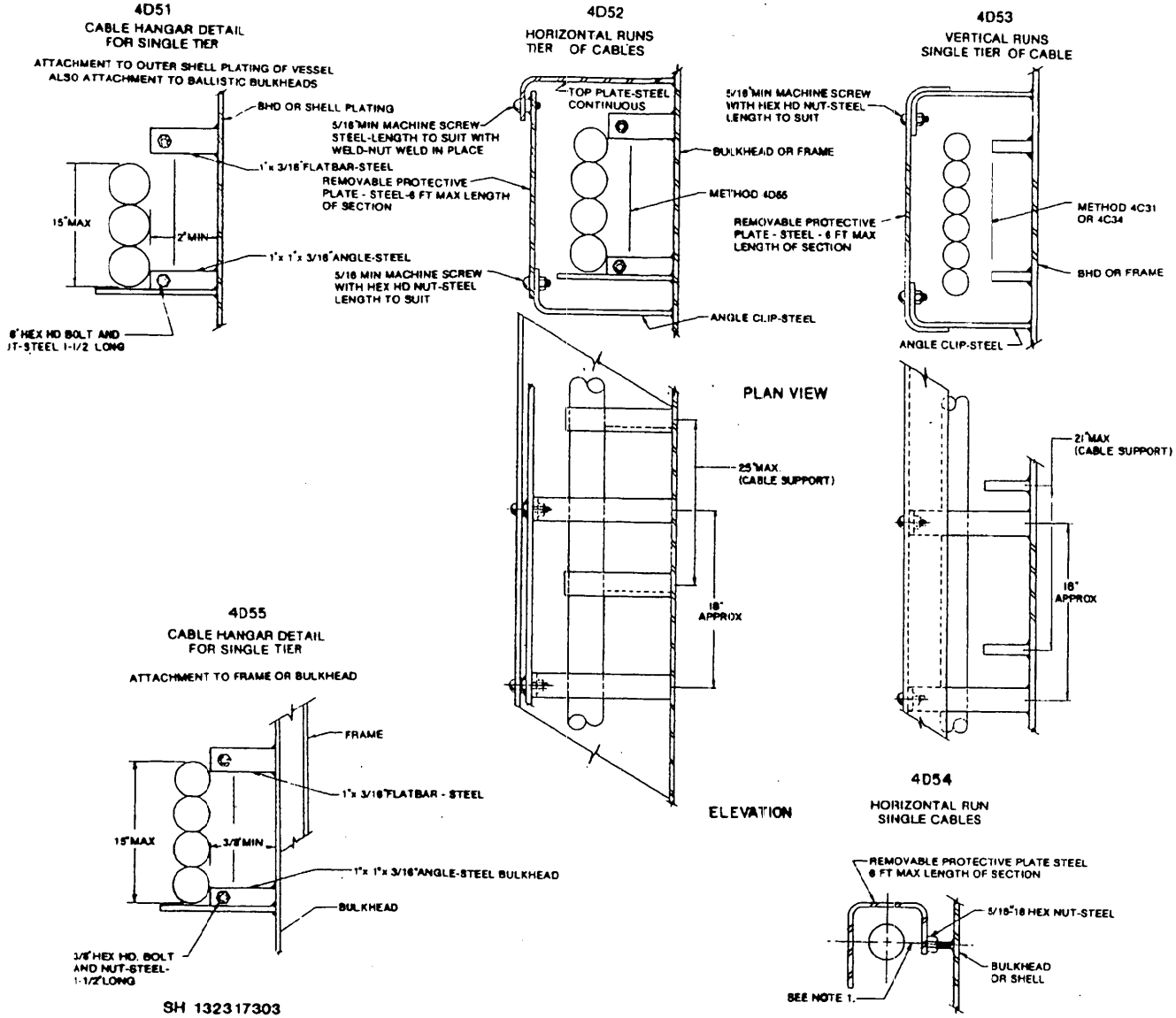
1. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.6.
2. THIS FIGURE SUPERSEDES SHEET 4D4 OF DRAWING 803-5001027 AND SECTION 2, SHEET 156, OF DRAWING, NAVSEC NO. 9000-S6202-73980.



SH 132317302

FIGURE 4D4. Protection of cables from mechanical injury.

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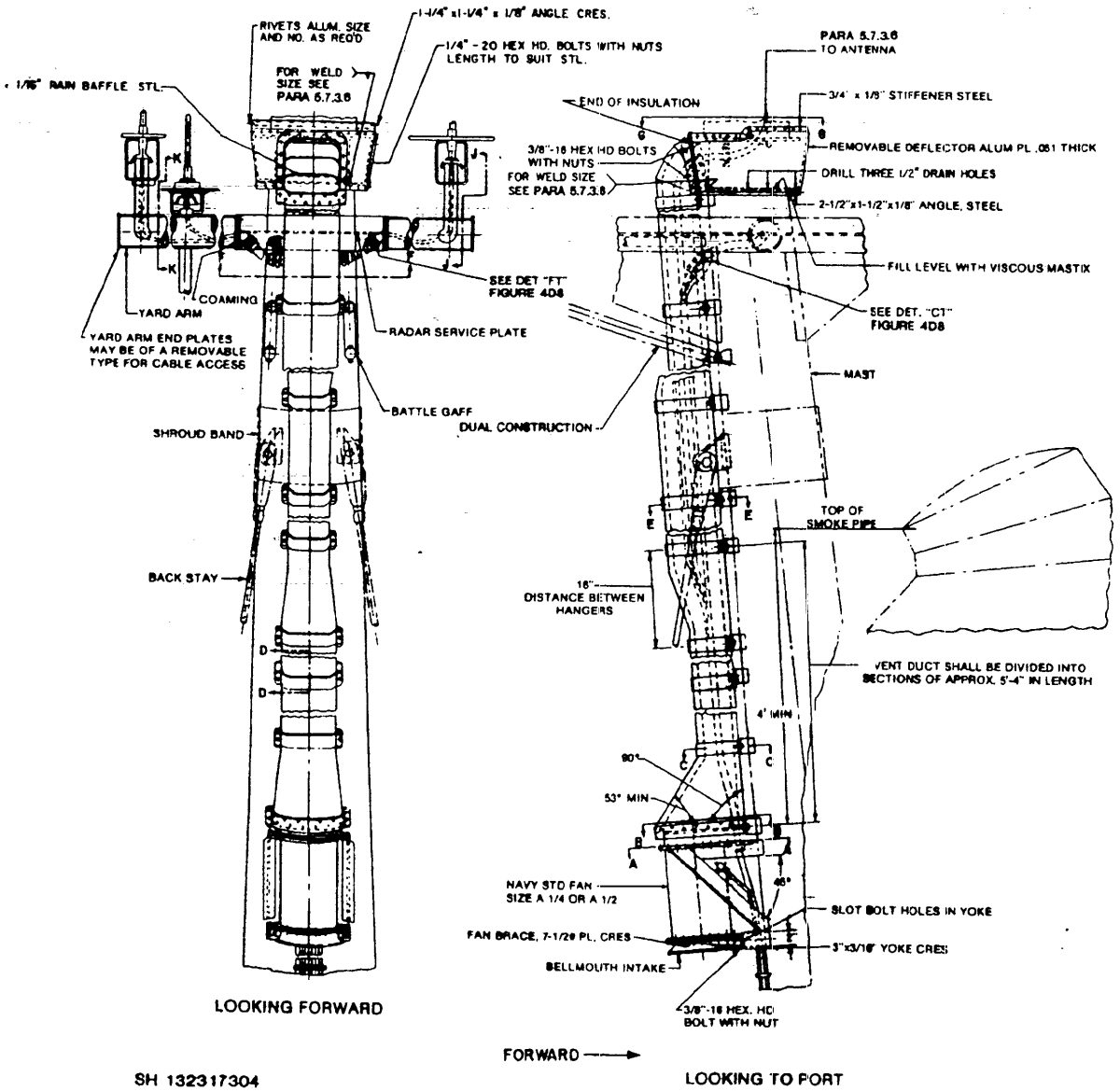


NOTES:

1. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.6.
2. THIS FIGURE SUPERSEDES SHEET 4D5 OF DRAWING 803-5001027 AND SECTION 2, SHEET 157, OF DRAWING, NAVSEC NO. 9000-S9202-73980.

FIGURE 4D5. Protection of degaussing cables from mechanical injury.

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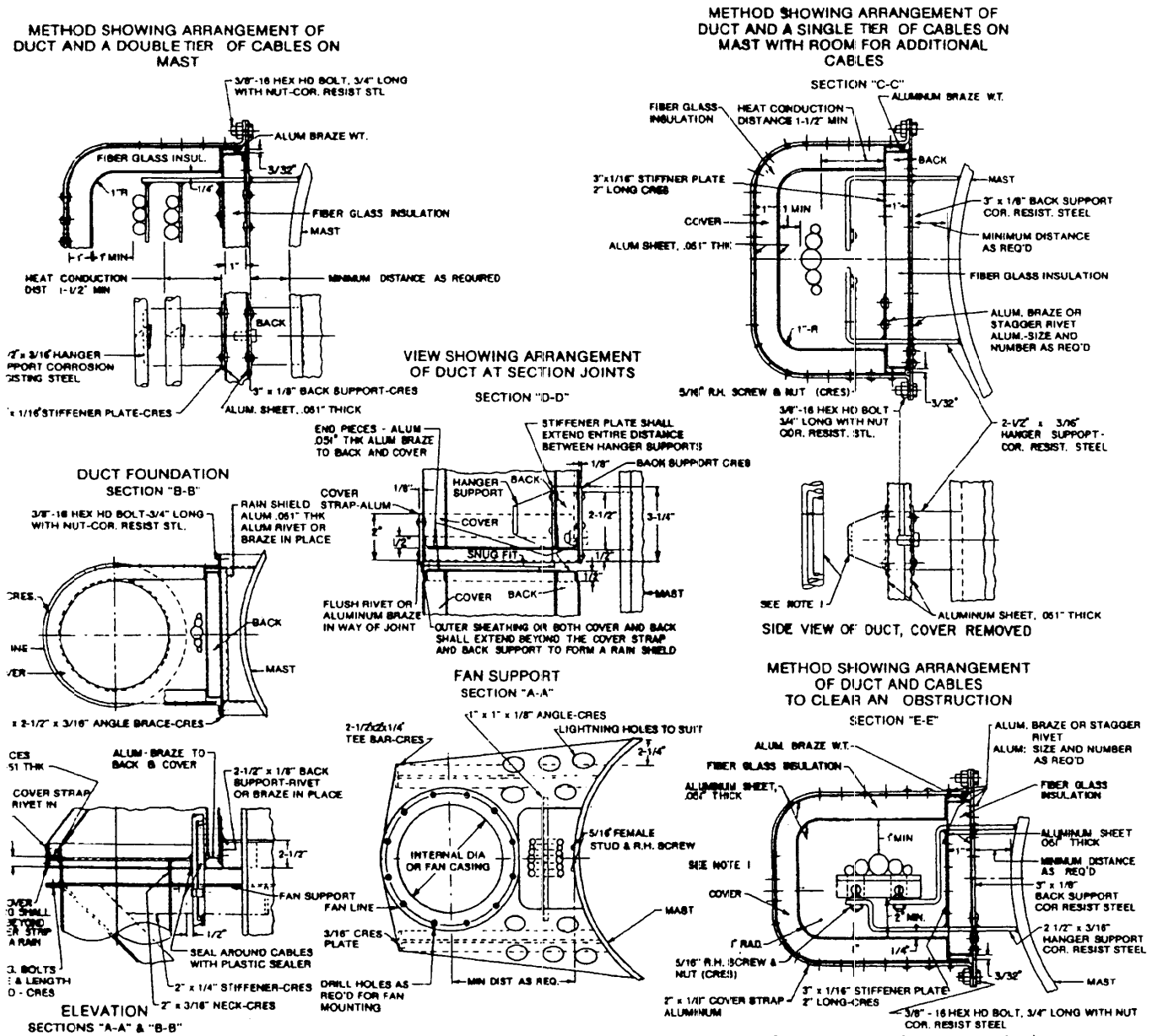
SH 132317304

FIGURE 4D6. Protection of cables on mast (surface ships).

NOTES:

1. THE AREA OF THE VENTILATION DUCT SHALL BE CALCULATED TO GIVE A MINIMUM FREE FLOW AREA OF 30 SQUARE INCHES WITH THE CLOSEST APPROACH OF DUCT WALL TO THE CABLE OF 1-INCH.
2. THE MINIMUM DISTANCE OF HANGER SUPPORT FROM THE INSULATION TO THE HANGER CHANNEL SHALL BE 1 1/2" TO REDUCE CONDUCTION ALONG HANGER.
3. ALL SEAMS, RIVETS AND OPENINGS IN DUCT WALL AND INSULATING MATERIAL SHALL BE MADE WEATHER TIGHT.
4. PROVIDE DOUBLER PLATES AS NECESSARY, WHERE BRACKETS, COAMING, YARD ARM, ETC., ARE WEAKENED BY ENTRANCE OF THE VENTILATION DUCT.
5. FAYING SURFACES OF STEEL AND ALUMINUM PARTS SHALL RECEIVE ONE COAT OF ZINC CHROMATE PRIMER.
6. AFTER INSTALLATION, ALL SURFACES OF DUCT SHALL RECEIVE NECESSARY COATS OF HEAT RESISTING PAINT TO MATCH THE SURROUNDING SURFACES.
7. IN DETERMINING THE SIZE OF DUCT FOR NEW CONSTRUCTION, ALLOWANCE SHALL BE MADE FOR SEVERAL ADDITIONAL CABLES TO ACCOMMODATE FUTURE RADAR AND ELECTRICAL REQUIREMENTS.
8. FOR VESSELS ON WHICH THE SCREENED SPEED LIGHT IS LOCATED ON THE MAINMAST, THE SERVICE CABLE SHALL BE BRANCHED FROM THE VENTILATION DUCT, THROUGH FLEXIBLE TUBING.
9. FREE FLOW AREAS WITHIN RAIN CAPS AND AIR DEFLECTORS SHALL HAVE 1 1/2 TO 2 TIMES THE AREA OF THE FEEDING PIPE.
10. FLEXIBLE TUBING SHALL BE INSTALLED WITHOUT TWIST, SHALL HAVE SUFFICIENT SLACK TO PREVENT THE TUBING FROM BEING STRAINED UPON WORKING OF MAST AND YARD ARMS, AND WITH THE RUN PITCHED TO AVOID WATER POCKETS.
11. FLEXIBLE TUBING SHALL BE SUPPORTED AT 16" INTERVALS WHERE PRACTICABLE, OTHERWISE AT SUCH SPACING AND LOCATION THAT TUBING SHALL TAKE EASY BENDS. SUPPORTS SHALL NOT BE SO PLACED THAT THE FLEXING OF THE TUBING WILL BE RESTRAINED.
12. THIS FIGURE SUPERSEDES SHEET 4D6 OF DRAWING 803-5001027 AND SECTION 2, SHEET 220, OF DRAWING NAVSEC NO. 9000-S6202-73980.

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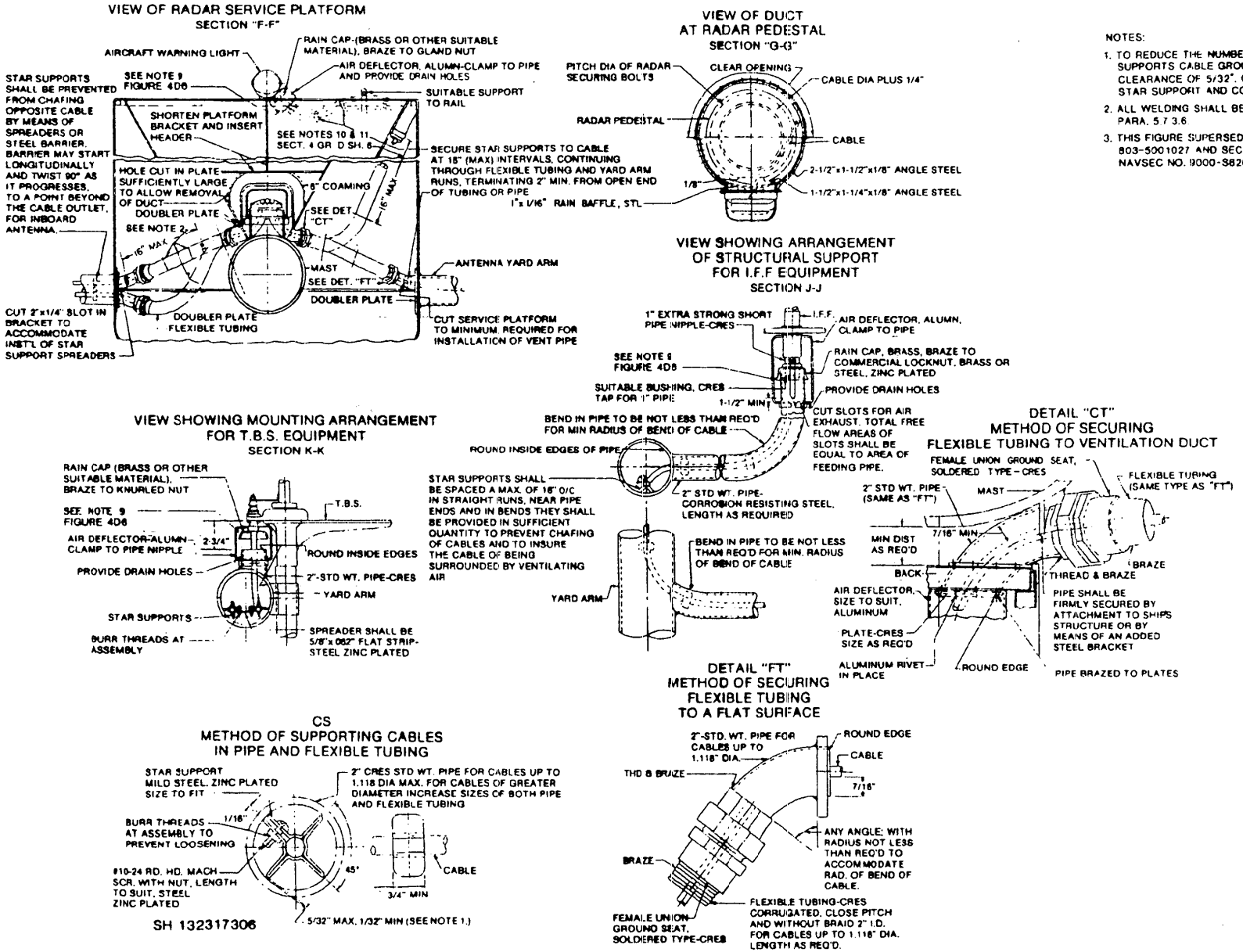
NOTES:

1 ALL WELDING REQUIRED TO FABRICATE SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.6.

2 THIS FIGURE SUPERSEDES SHEET 4D7 OF DRAWING 803-5001027 AND SECTION 2, SHEET 221, OF DRAWING, NAVSEC NO. 9000-S6202-73980.

FIGURE 4D7. Protection of cables on masts (surface ships).

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NOTES:

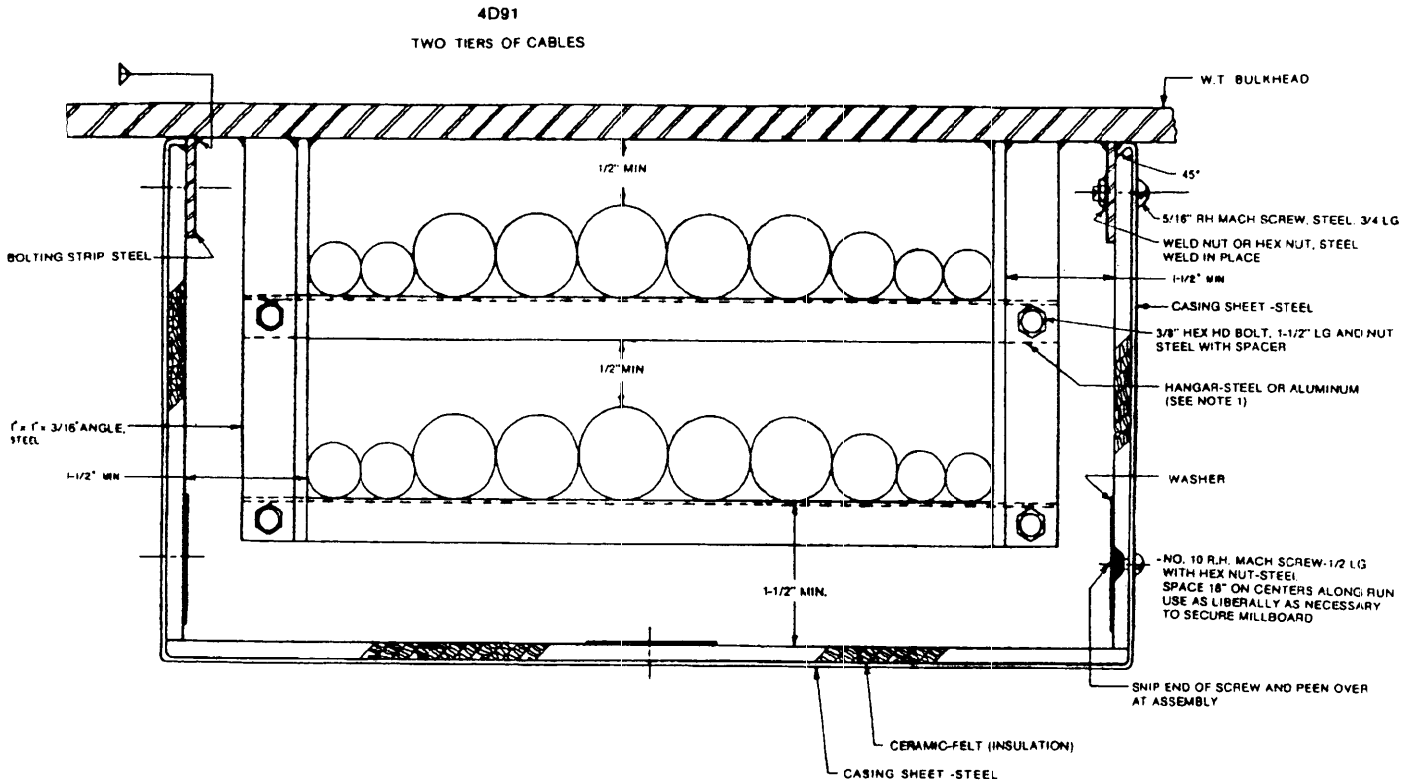
1. TO REDUCE THE NUMBER OF SIZES REQUIRED FOR STAR SUPPORTS CABLE GROUPING MAY BE USED WITH A MAX CLEARANCE OF 5/32". ONLY AS NECESSARY BETWEEN STAR SUPPORT AND CORROSION RESISTING STEEL PIPE.
2. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.6.
3. THIS FIGURE SUPERSEDES SHEET 4D8 OF DRAWING 803-5001027 AND SECTION 2, SHEET 222, OF DRAWING, NAVSEC NO. 9000-S8202-73980.

FIGURE 4D8. Protection of cables on mast (surface ships).

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NOTES:

1. THIS FIGURE SUPERSEDES SHEET 4D9 OF DRAWING 803-5001027 AND SECTION 2, SHEET 140, OF DRAWING, NAVSEC NO. 9000-S6202-73980.



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FIGURE 4D9. Protection of cables in hangar spaces (surface ships).

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NOTES:

1. ALL WELDING SHALL BE IN ACCORDANCE WITH PARA. 5.7.3.6.
2. THIS FIGURE SUPERSEDES SHEET 4D10 OF DRAWING 803-5001027 AND SECTION 2, SHEET 1432, OF DRAWING NAVSEC NO 9000-86202-73980.

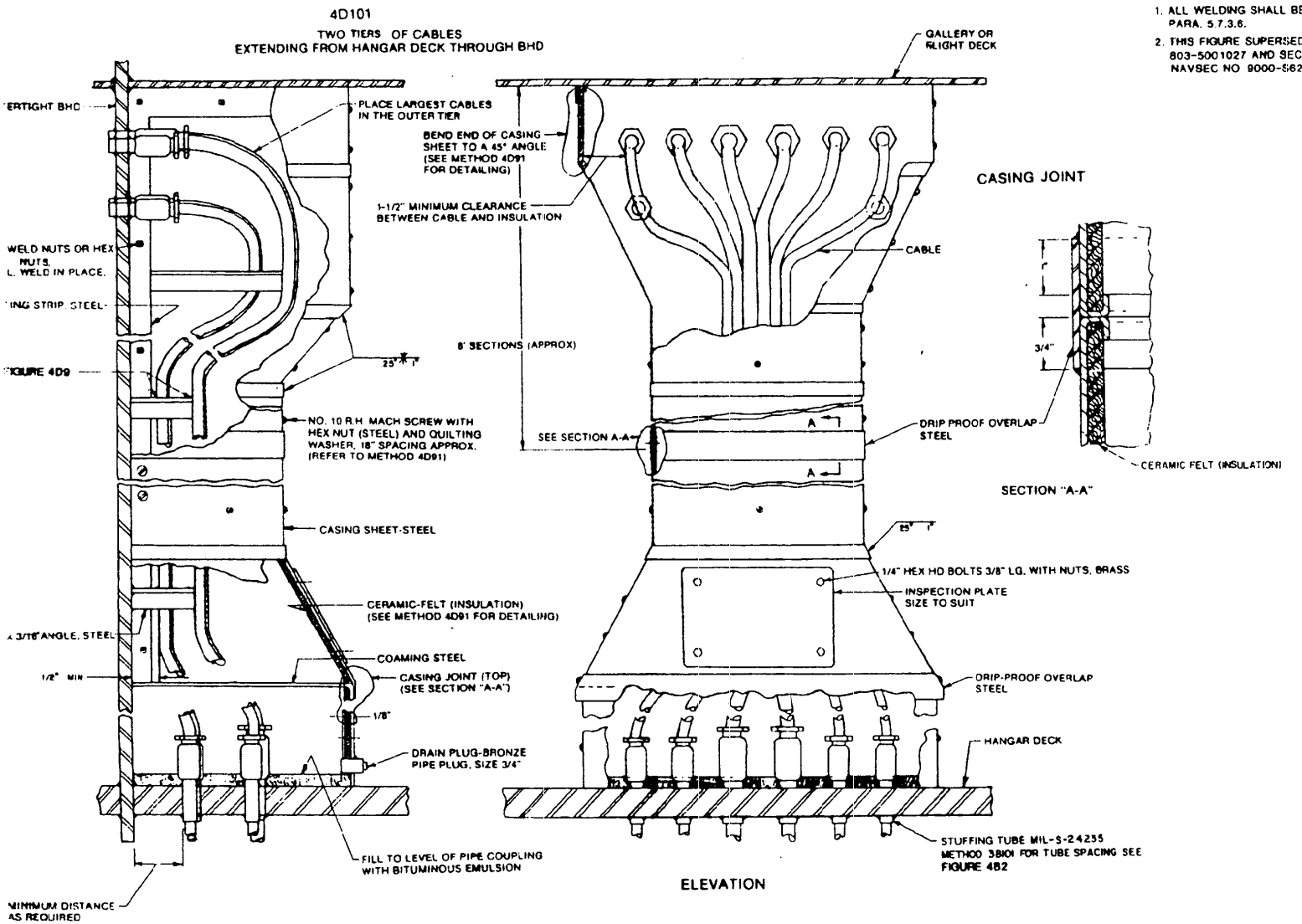


FIGURE 4D10. Protection of cables in hangar spaces (surface ships).

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-STD-2003-4(NAVY)	ELECTRIC PLANT INSTALLATION STANDARD METHODS FOR SURFACE SHIPS AND SUBMARINES (CABLEWAYS) SECTION 4 OF 5

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
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a. Paragraph Number and Wording:

b. Recommended Wording:

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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